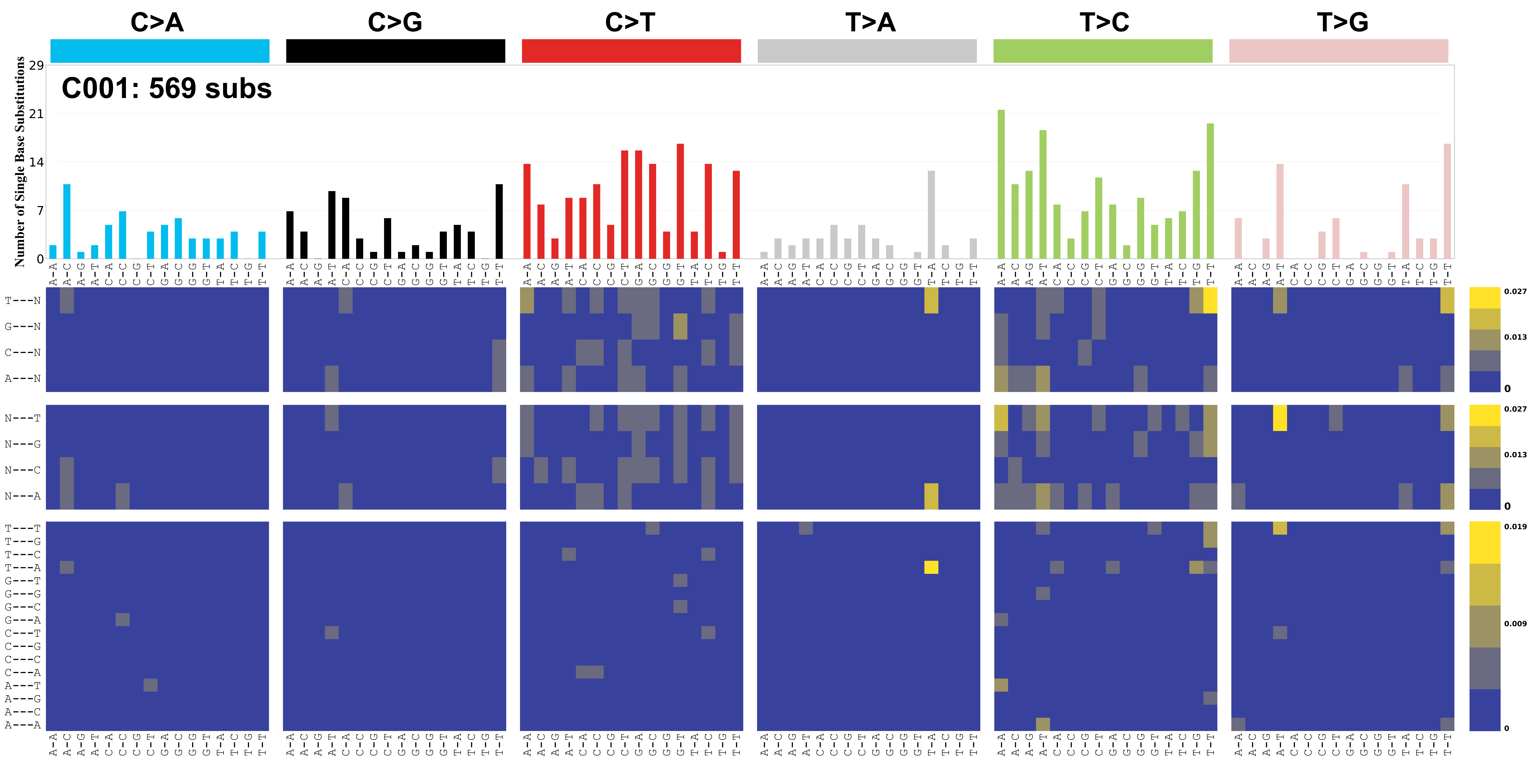
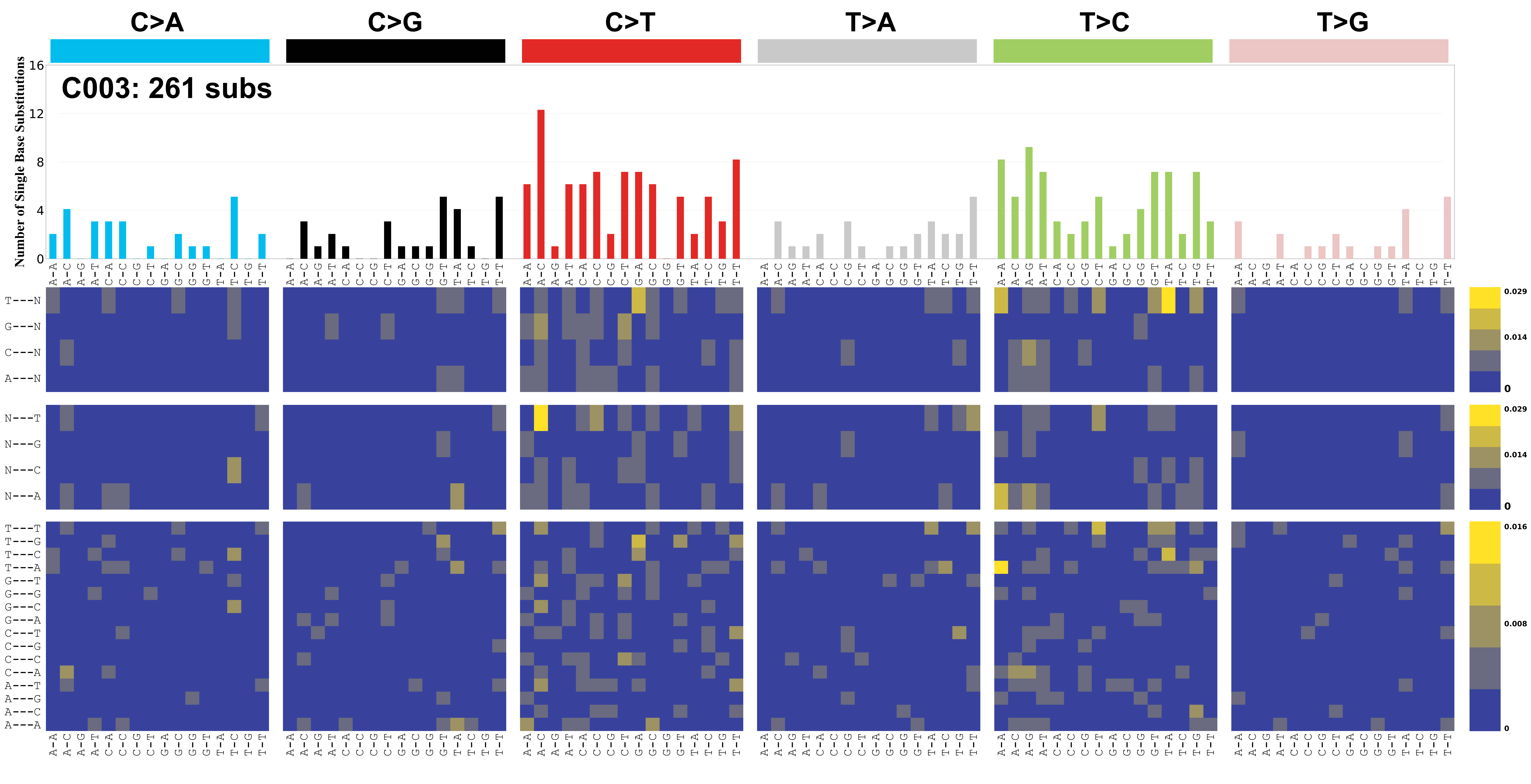
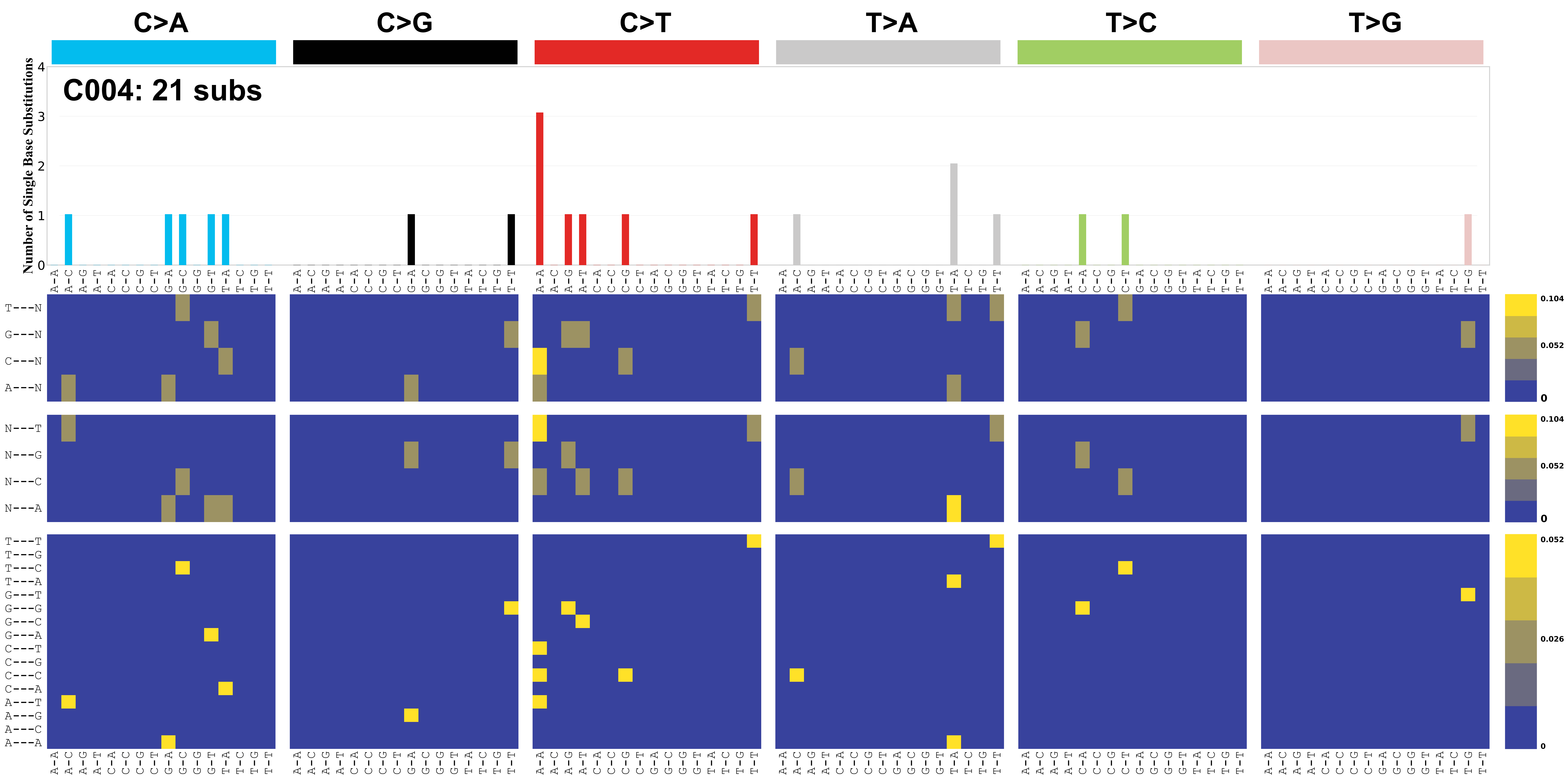
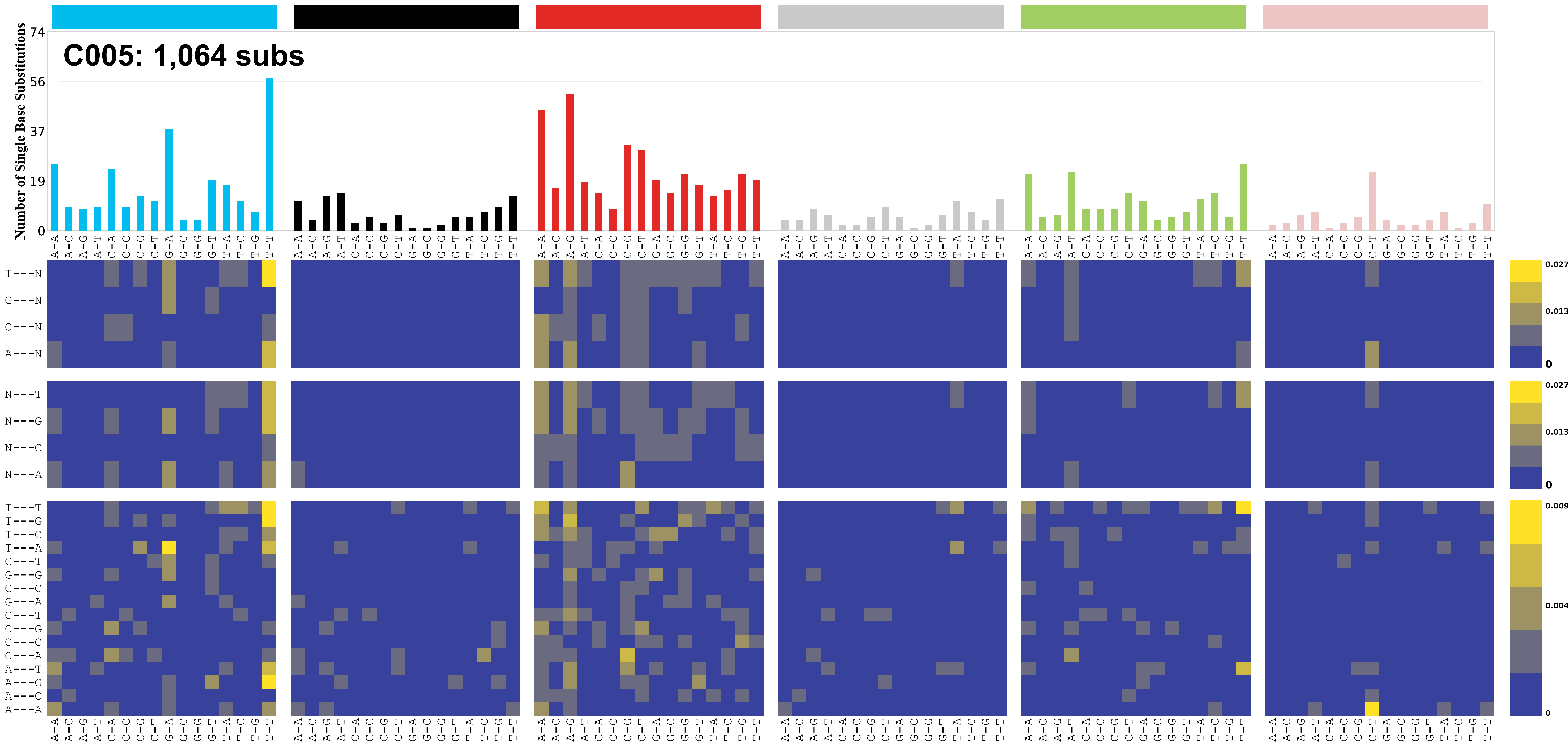


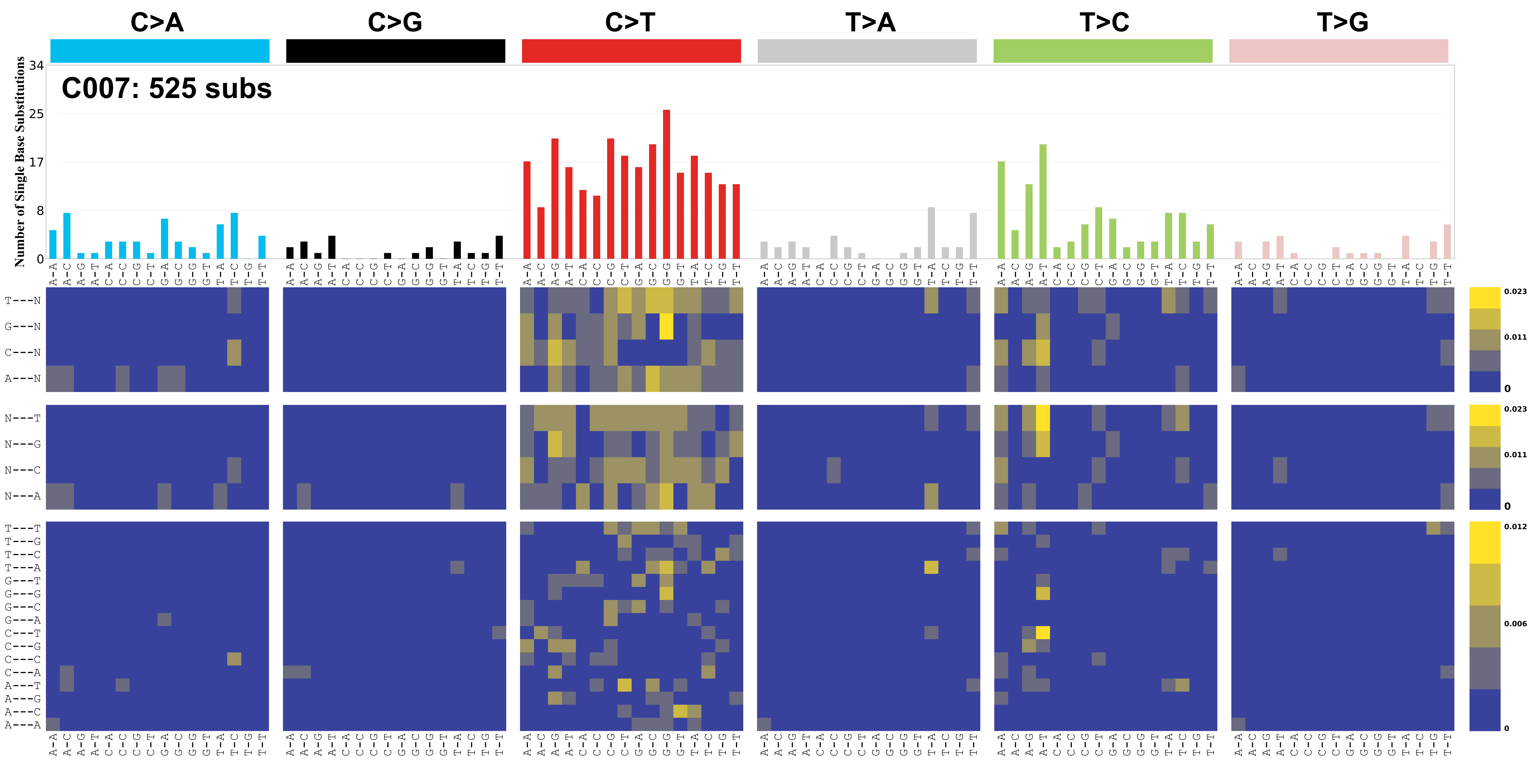


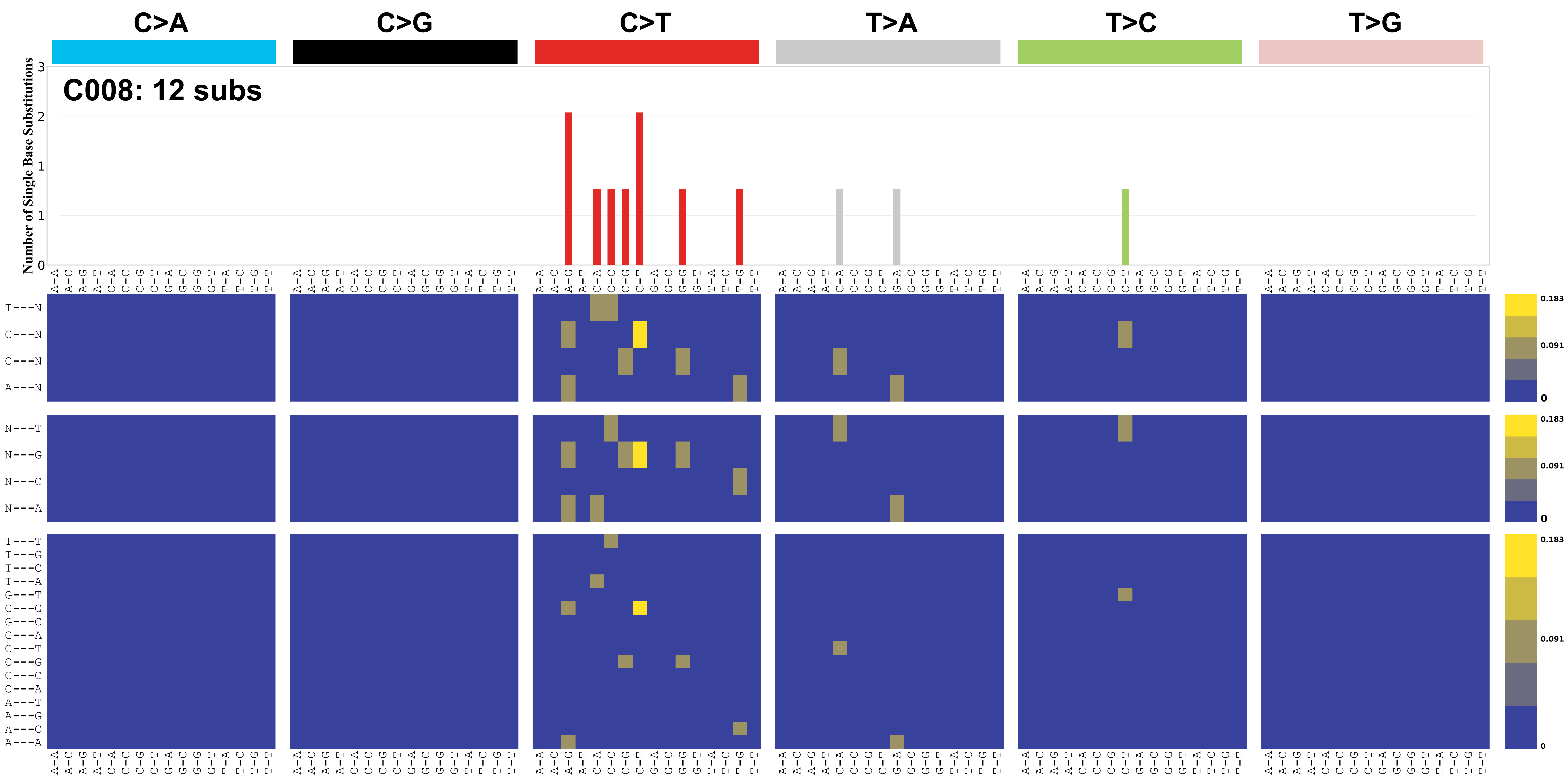




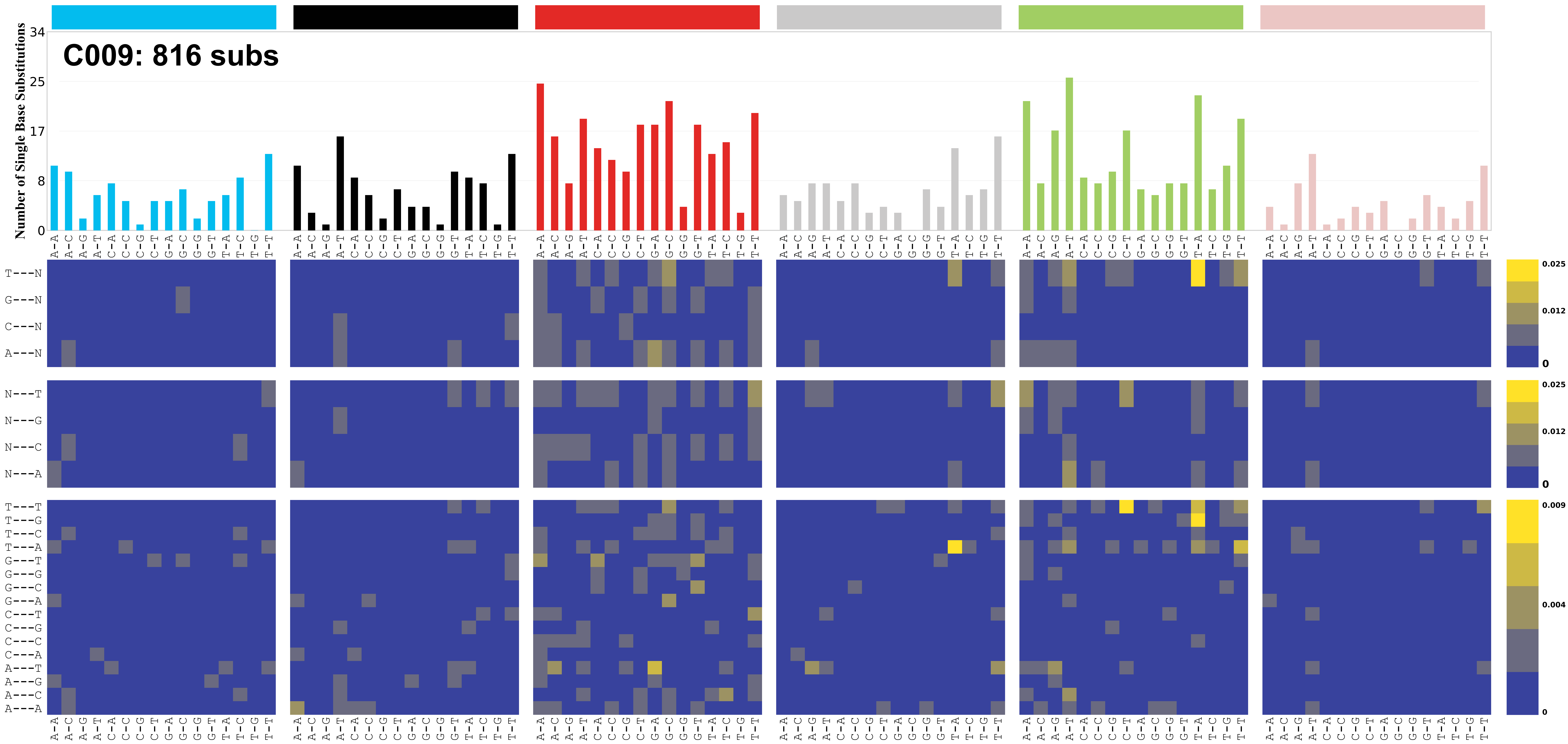
T>G



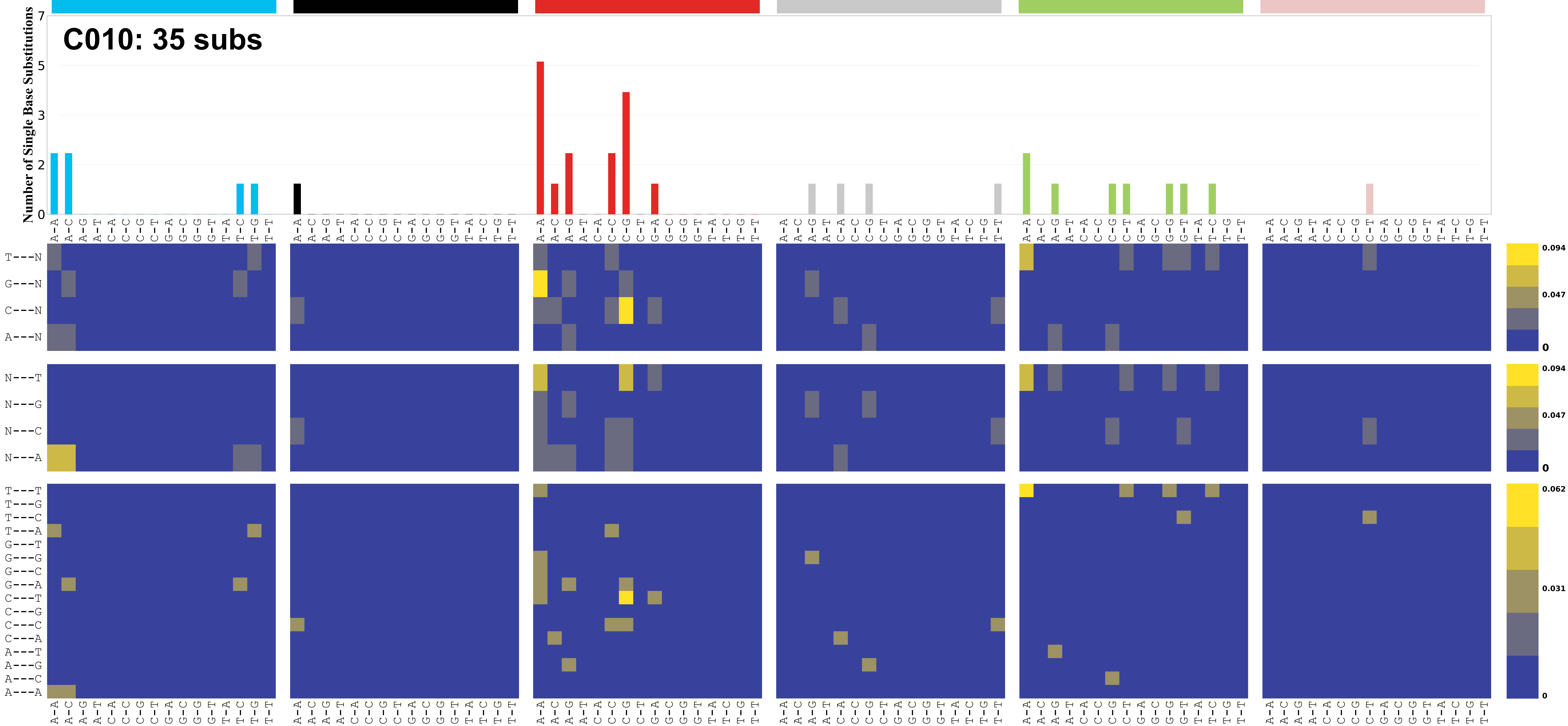


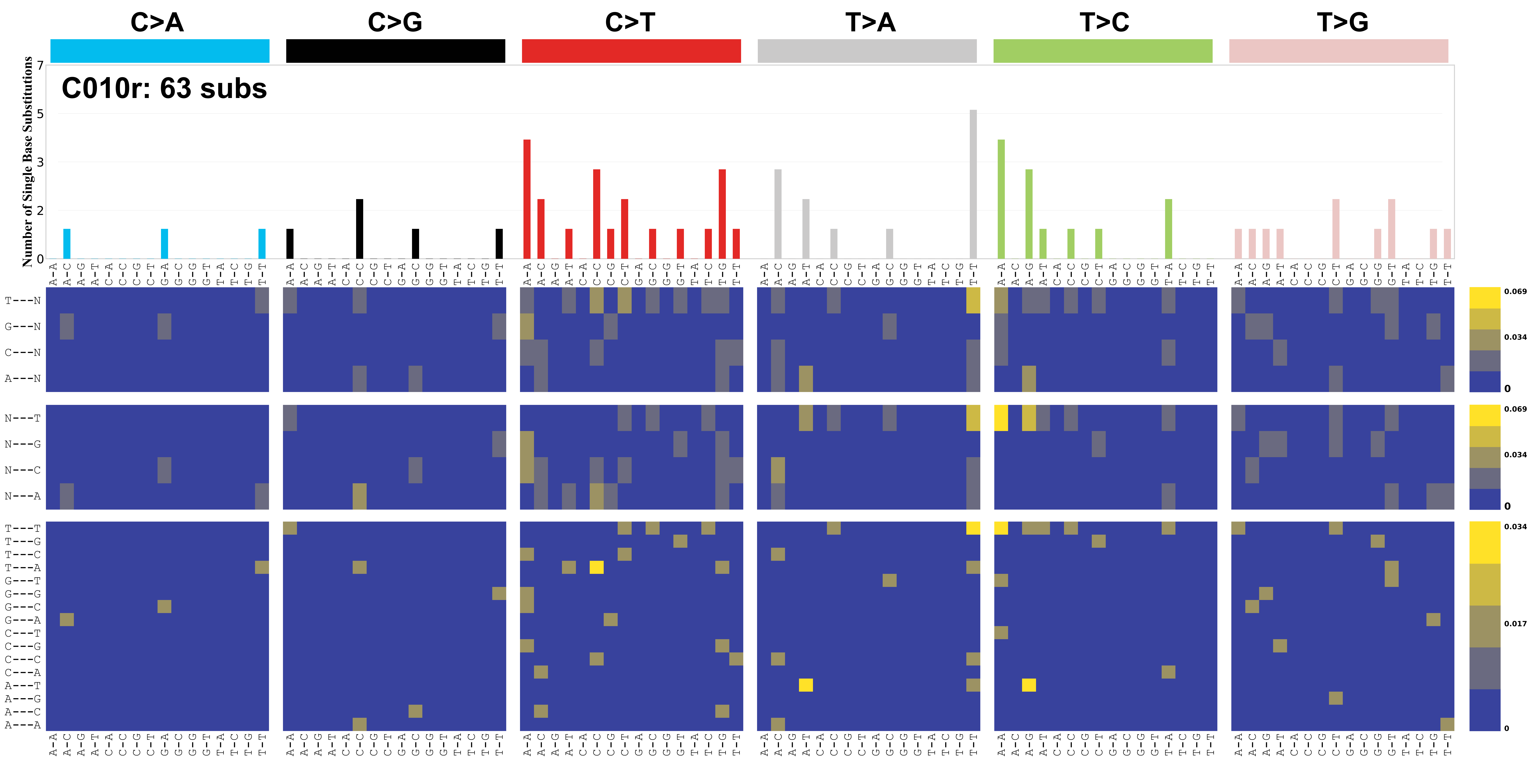


T>G



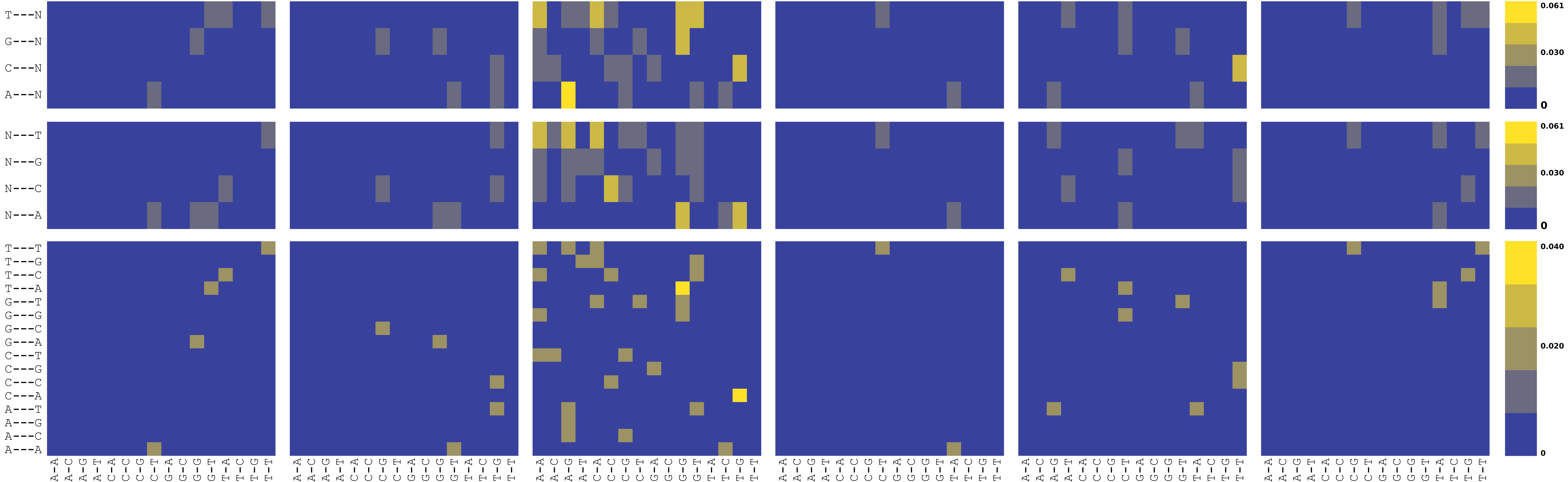
T>G



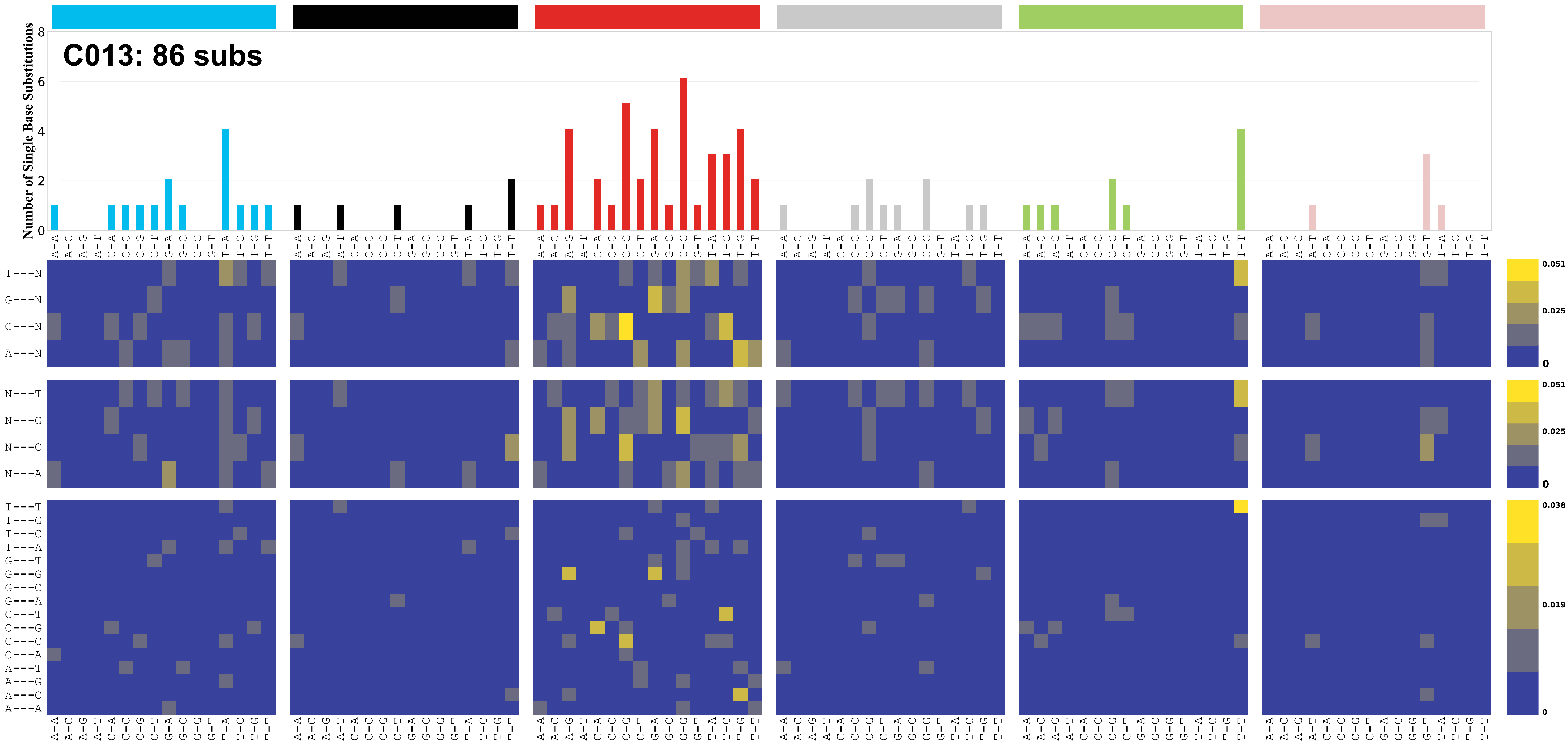


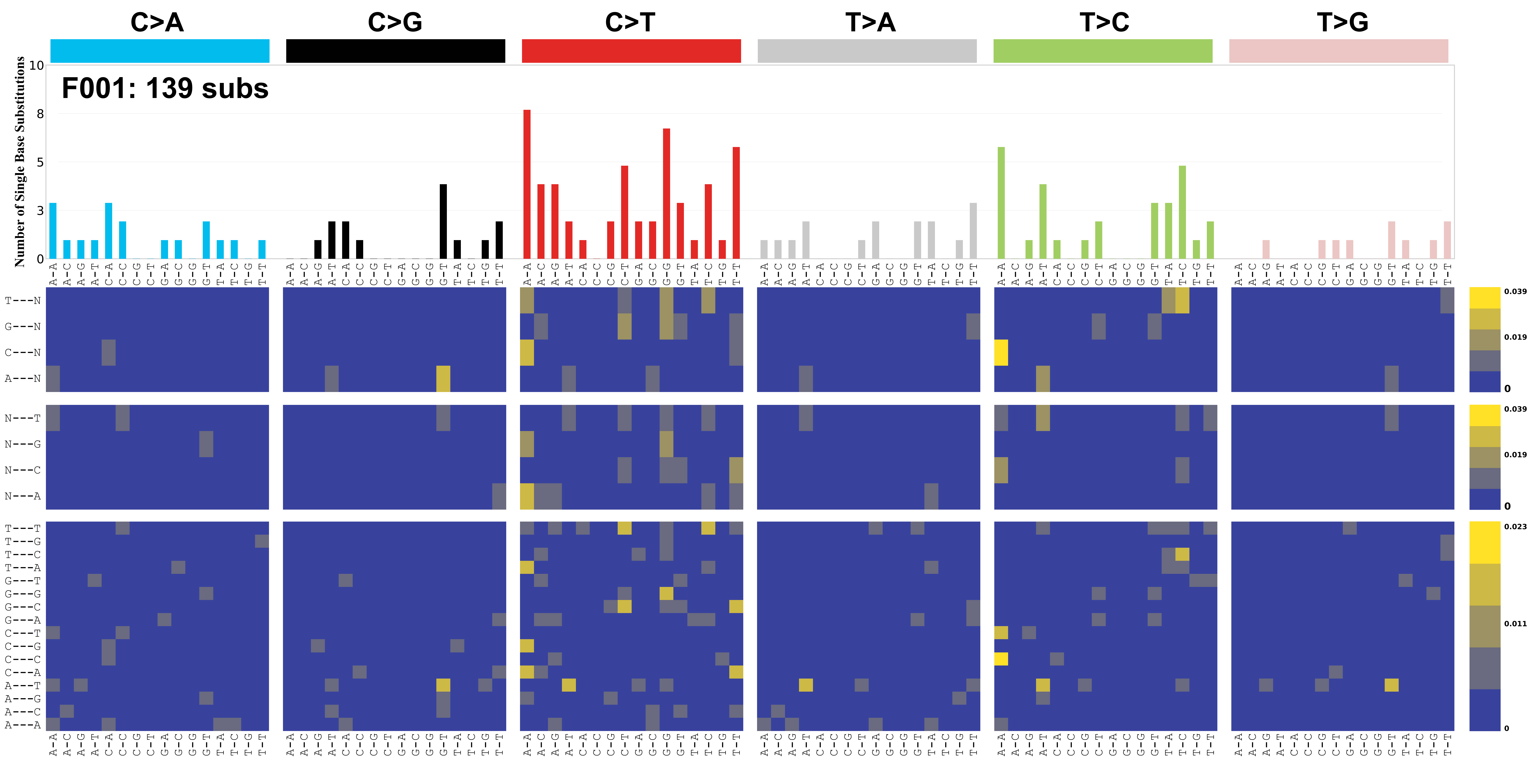
T>G

Number of Single Base Substitutions

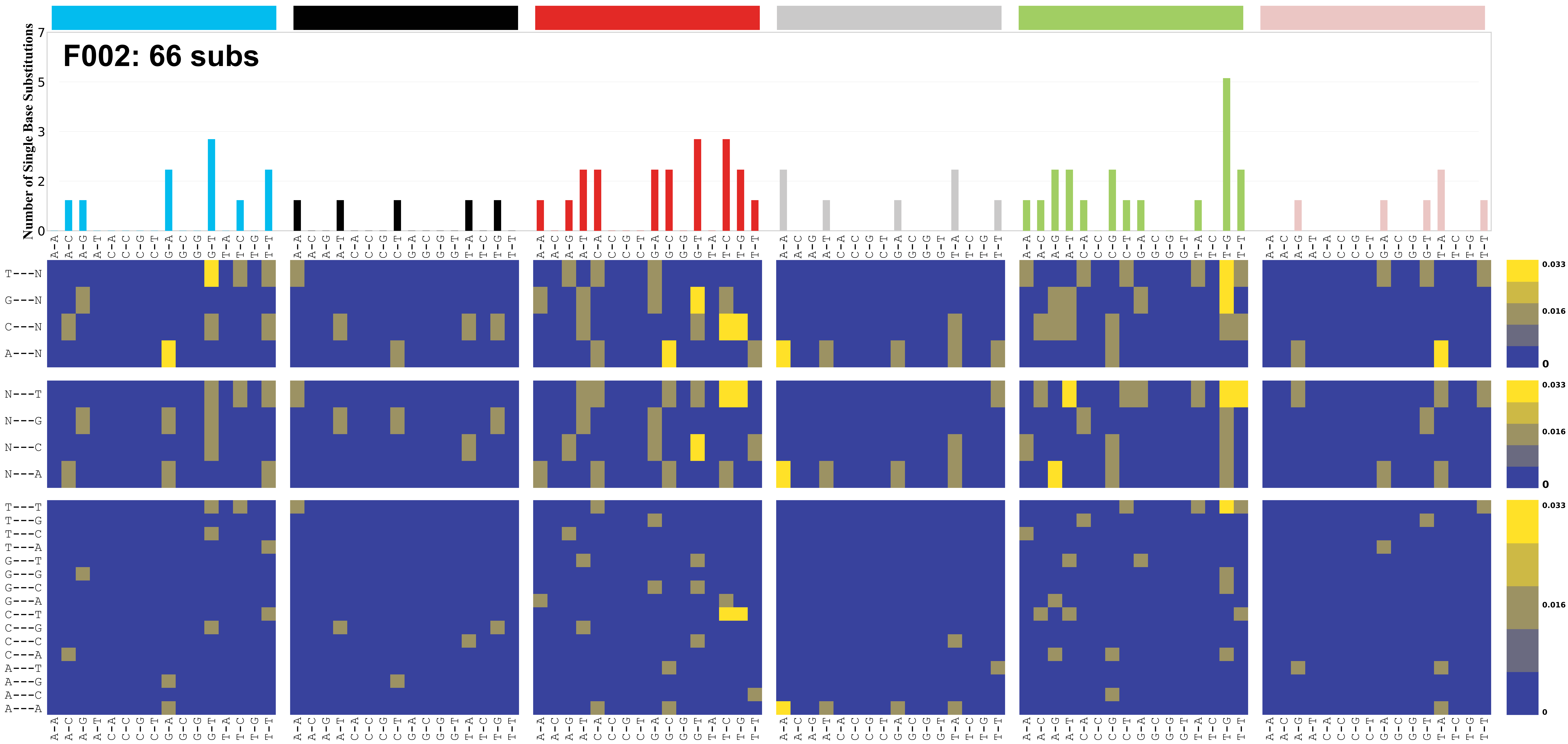


T>G



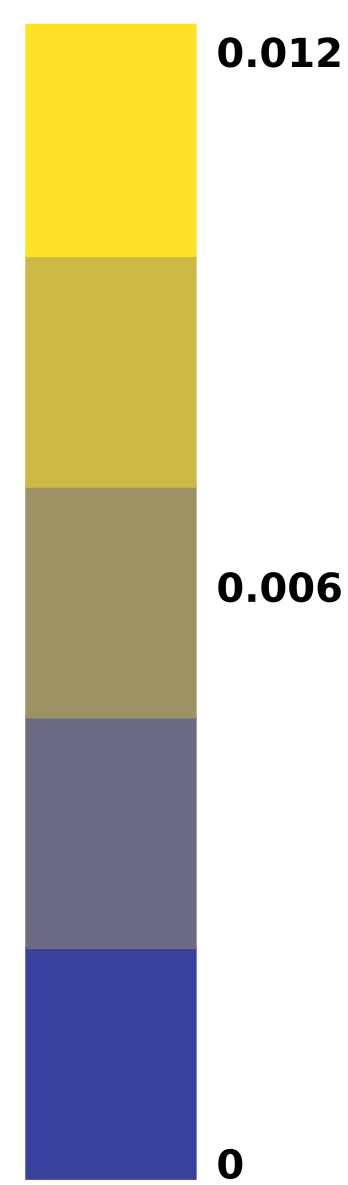
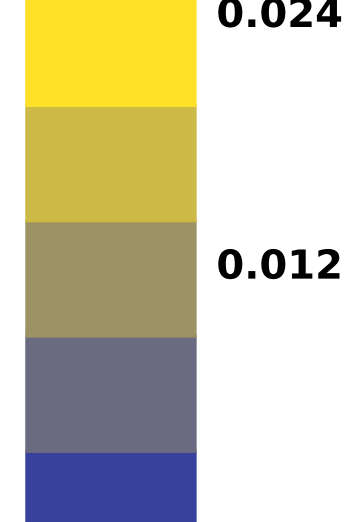
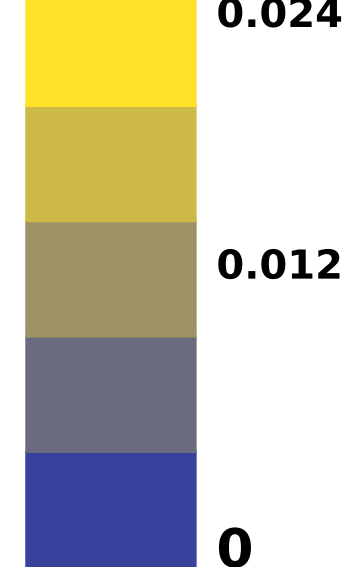
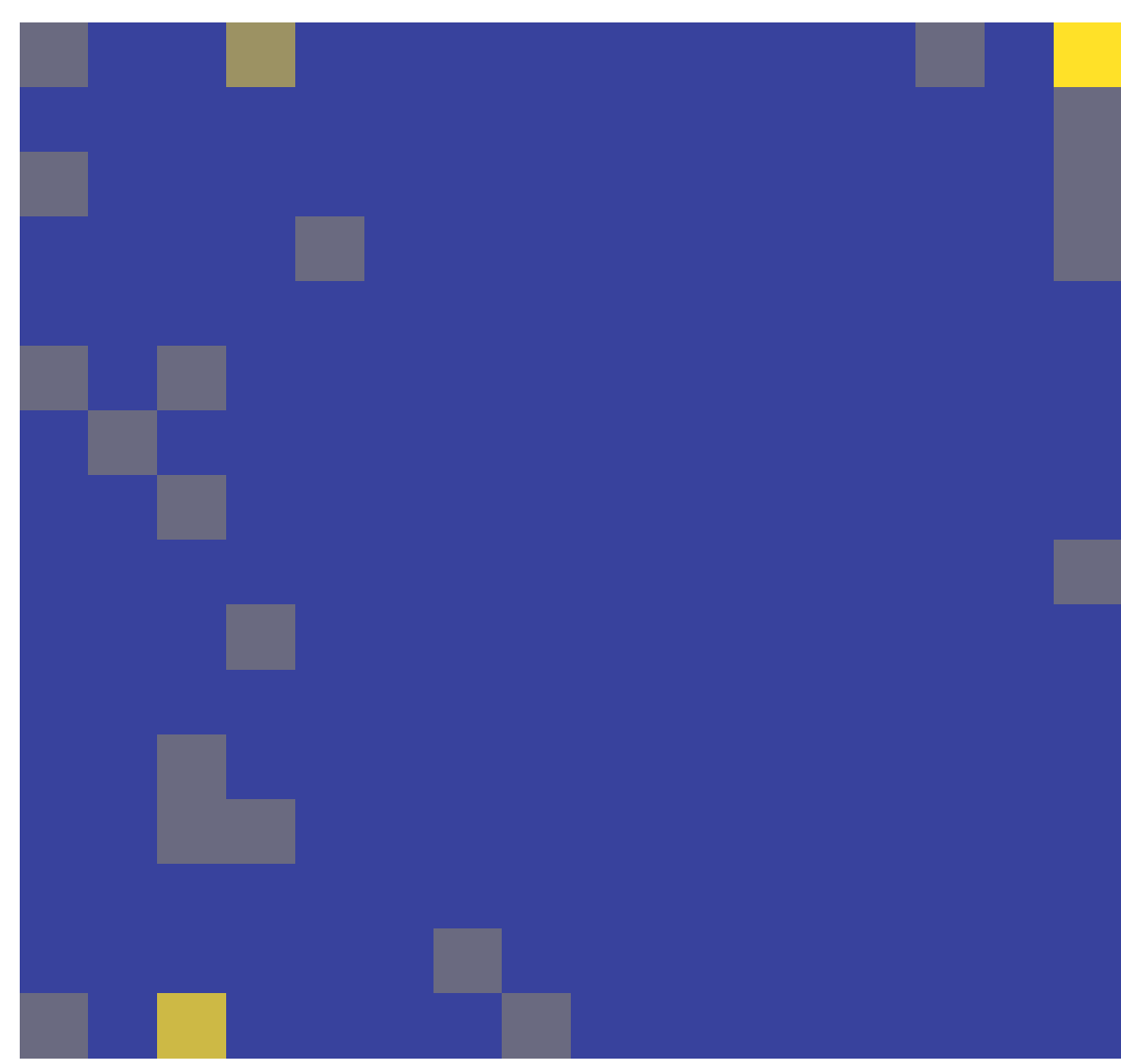
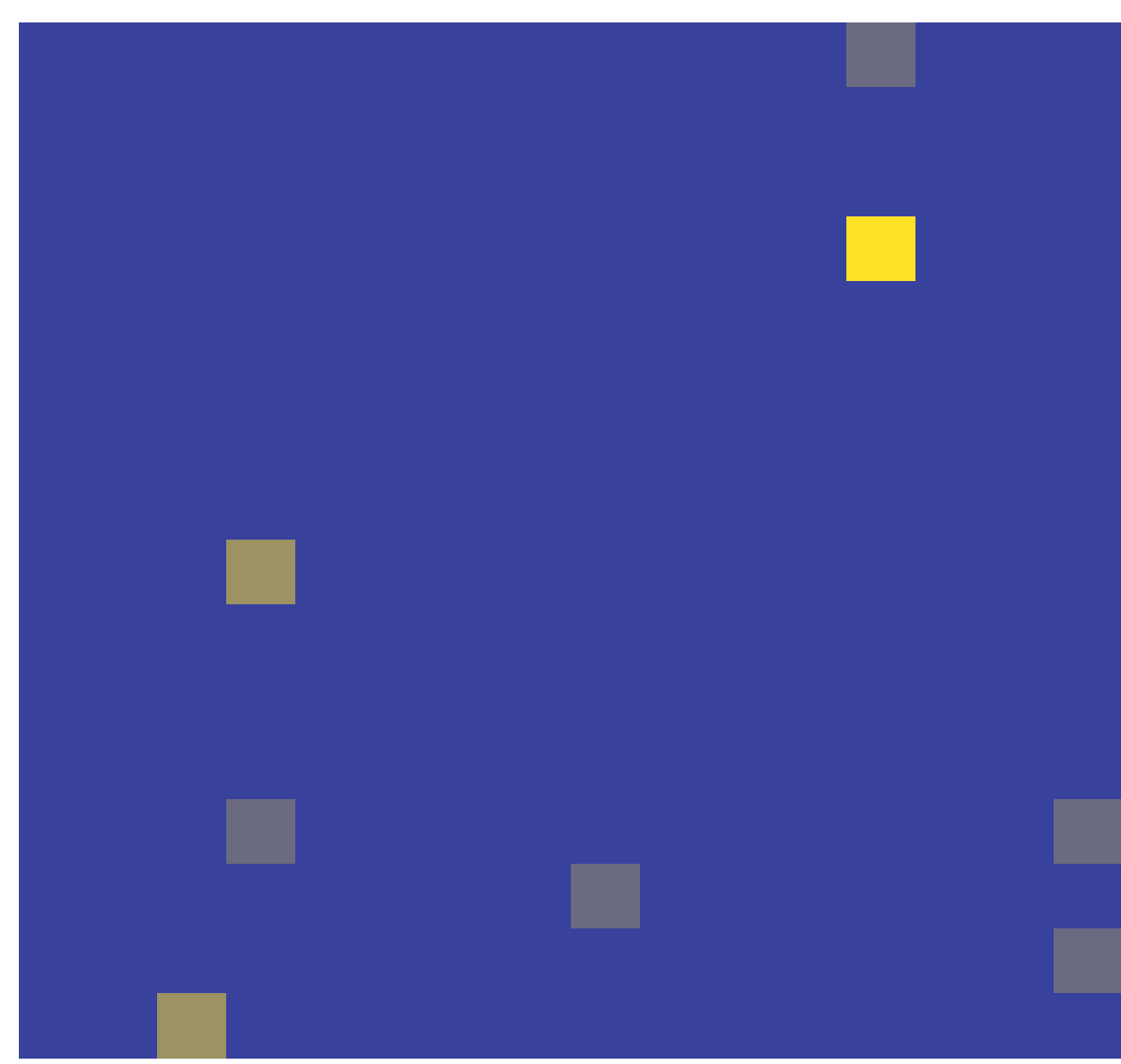
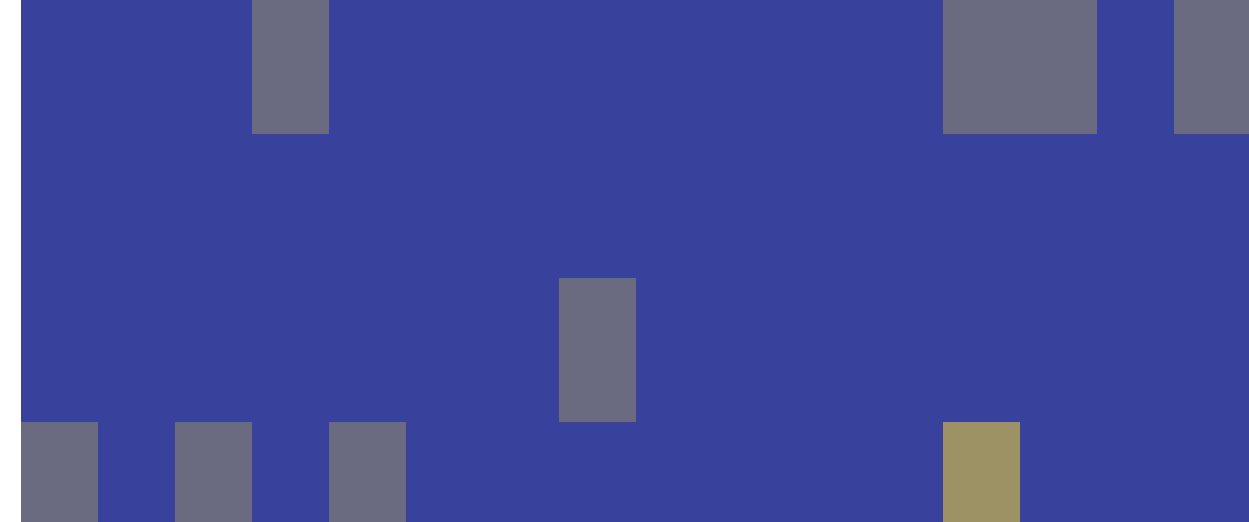
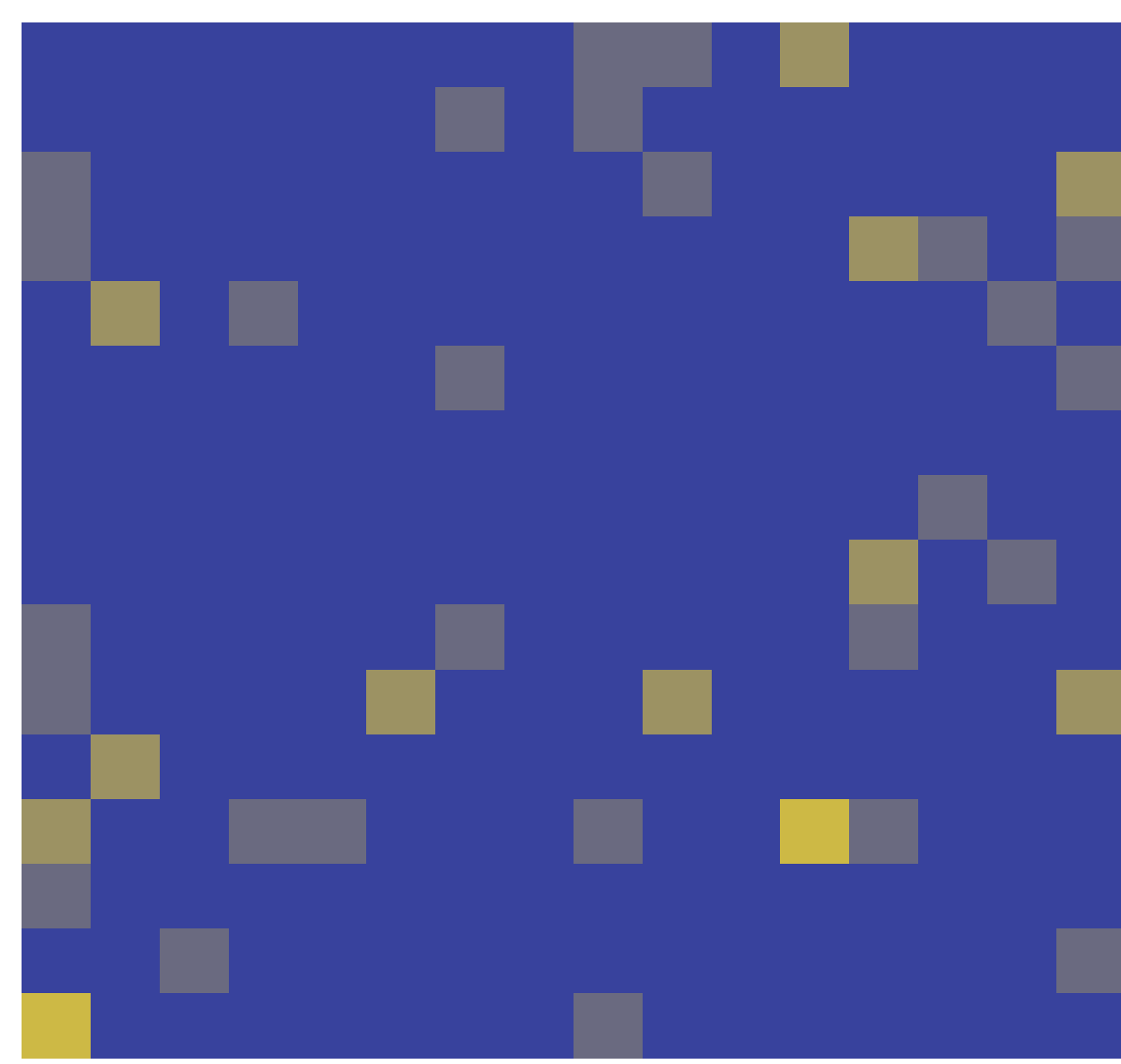
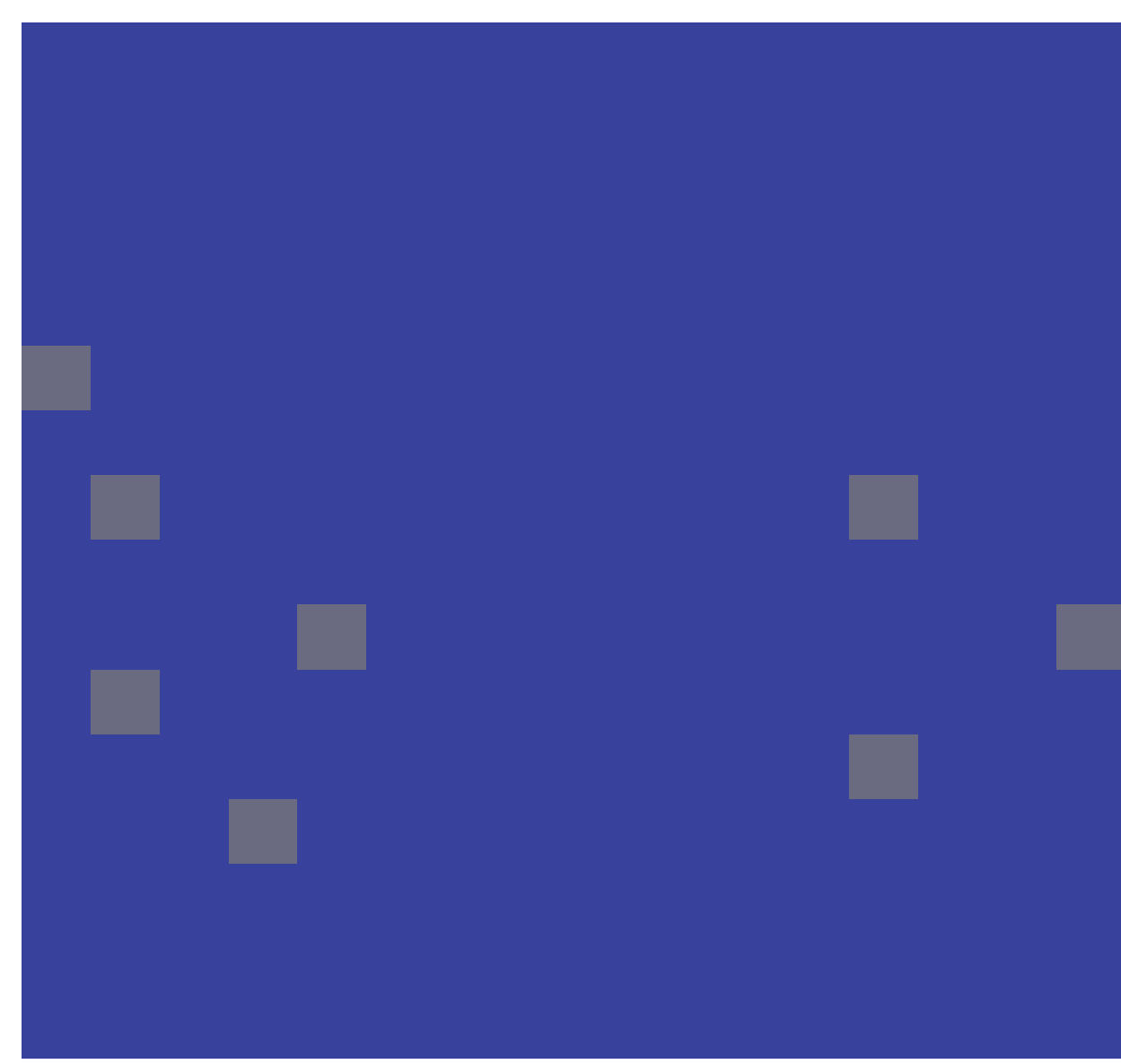
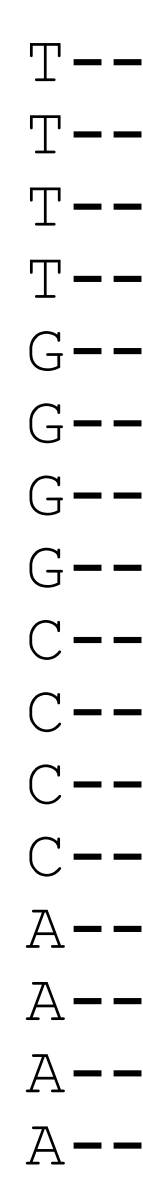
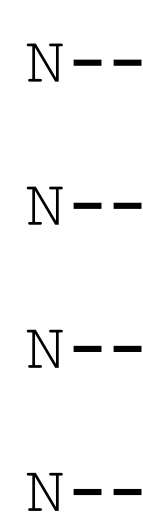
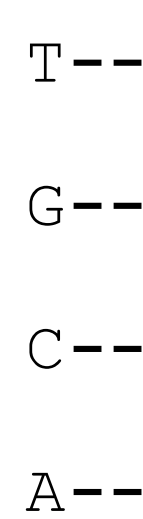


T>G



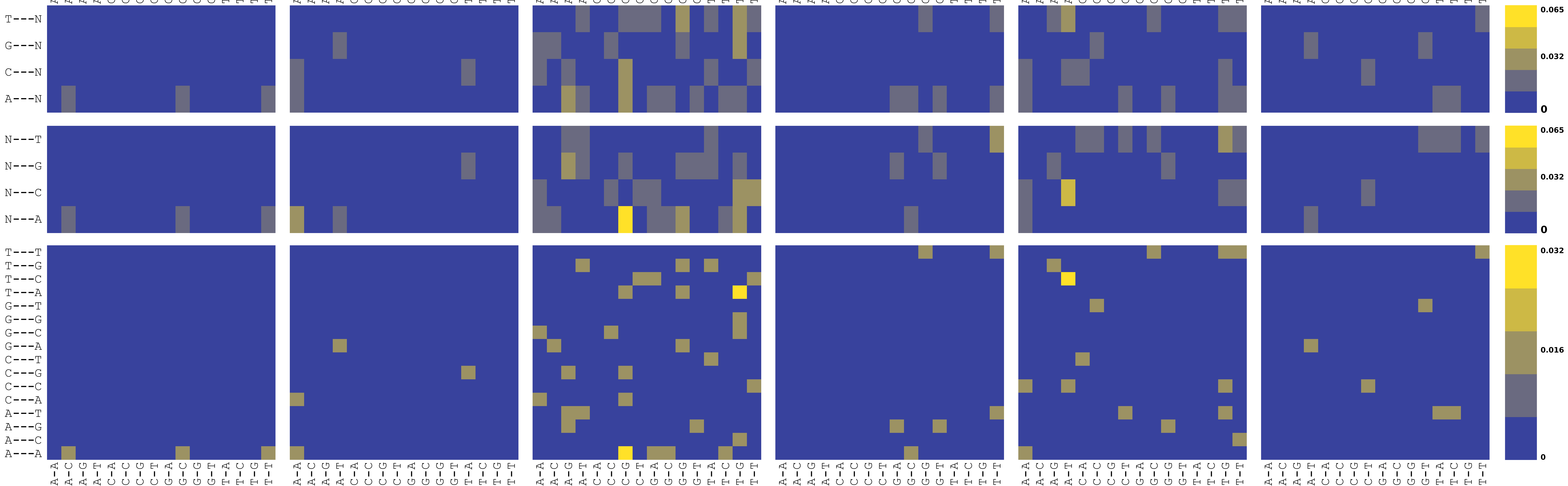
T>G

Number of Single Base Substitutions



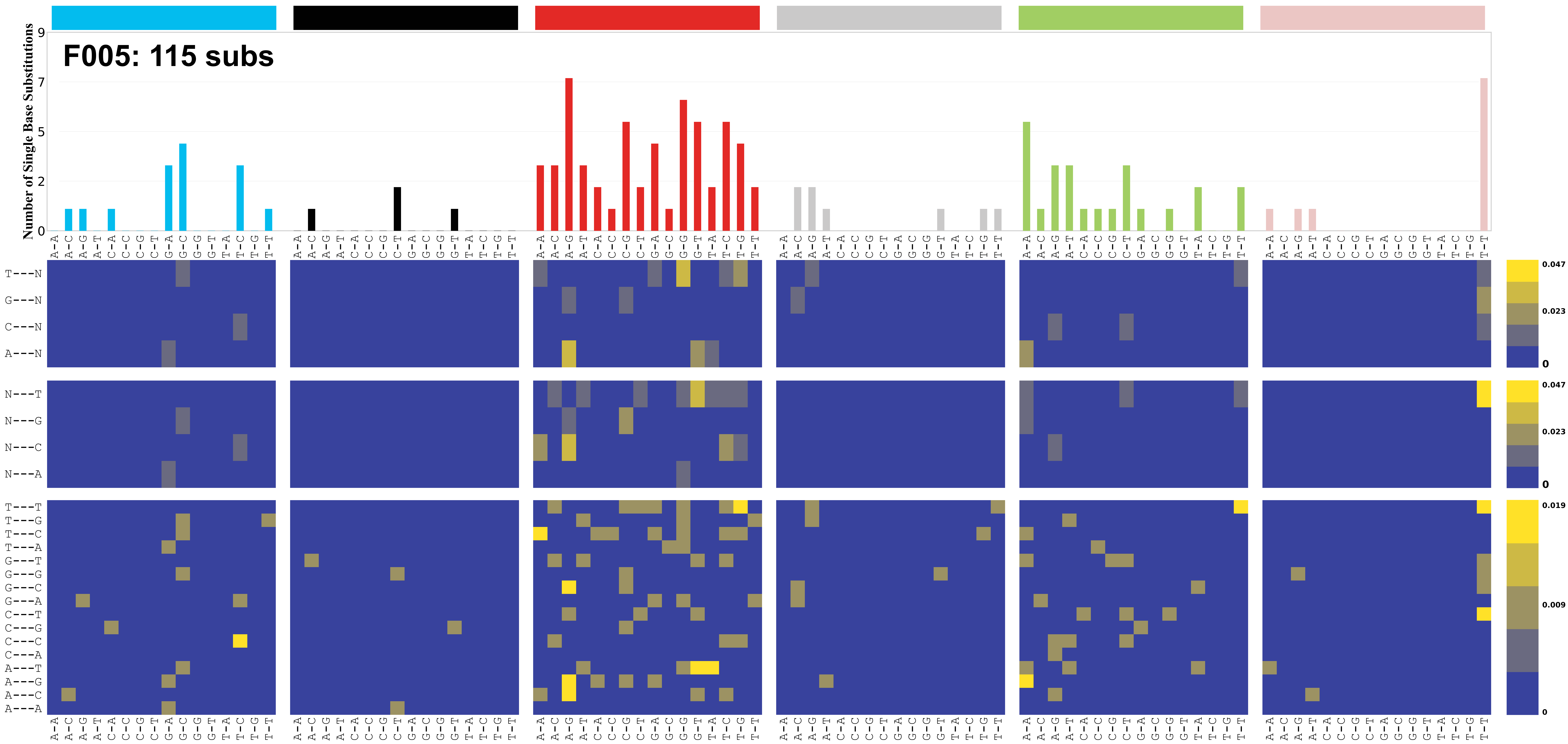
T>G

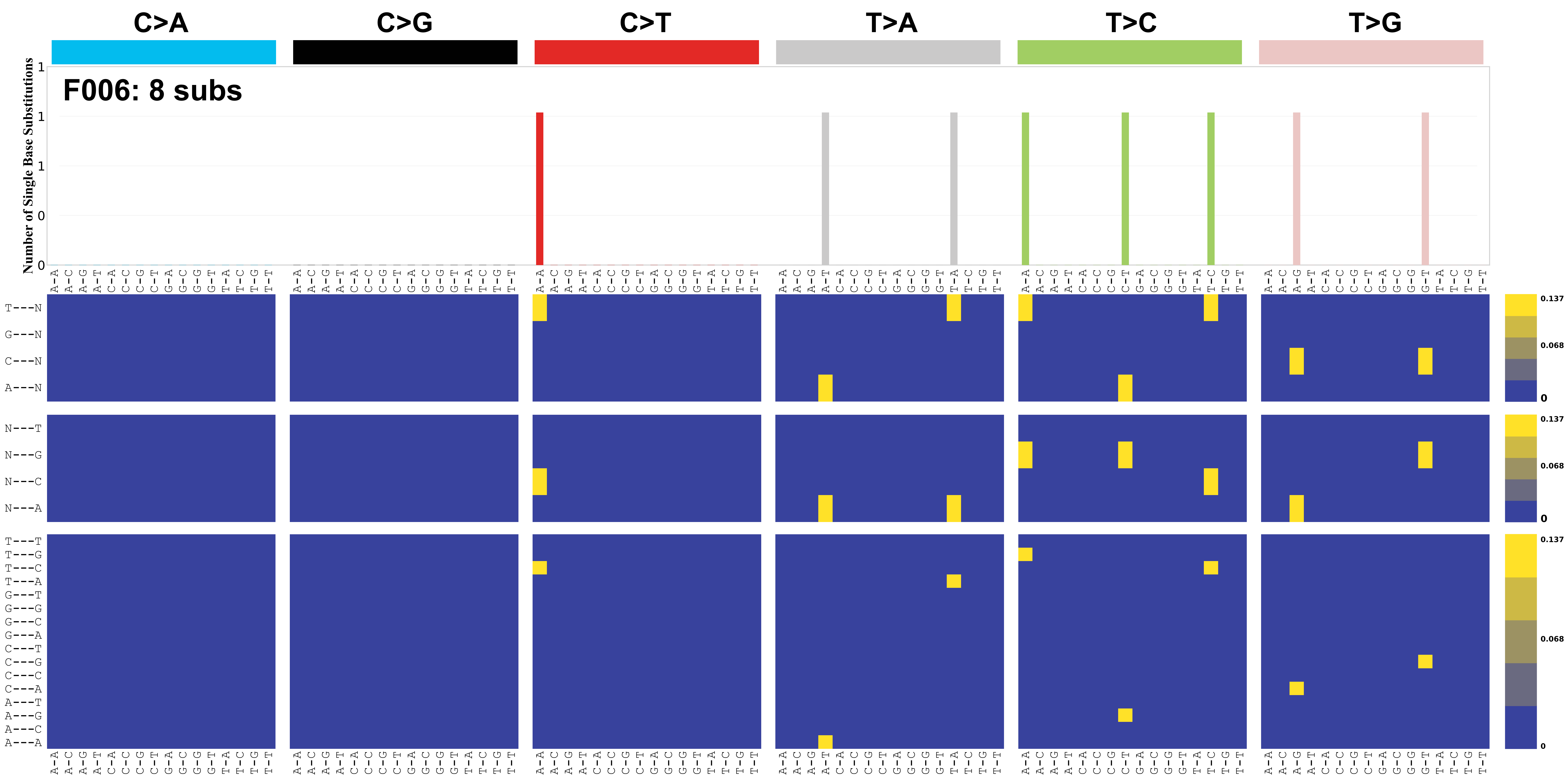
Number of Single Base Substitutions



T>G

F005: 115 subs





T>G

F007: 2,059 subs

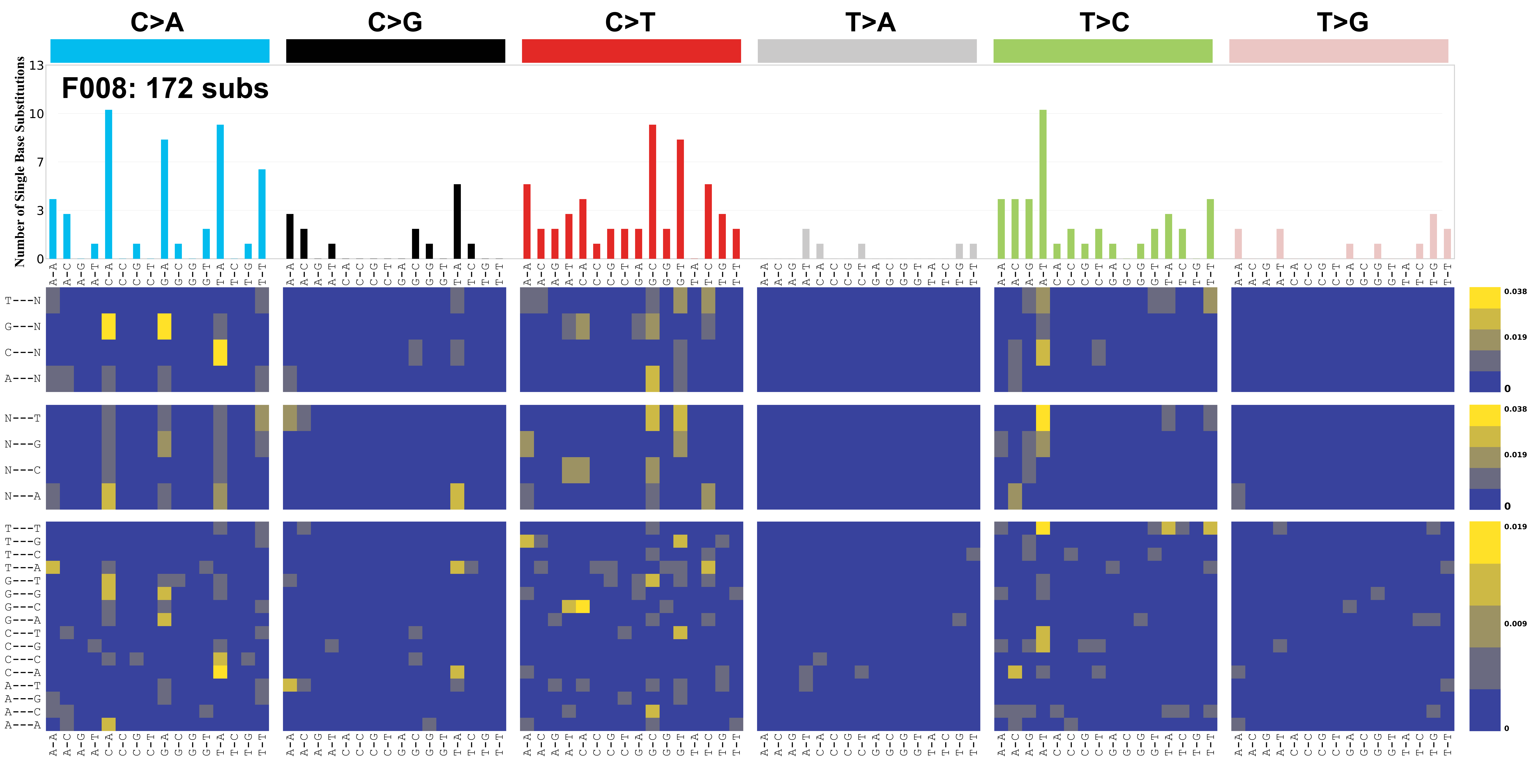
The figure displays the distribution of single and double base substitutions for the F007 dataset, which consists of 2,059 substitutions.

Single Base Substitutions (Bar Chart):

The top bar chart shows the number of single base substitutions for each of the 20 possible pairs (A-A, A-C, A-G, A-T, C-A, C-C, C-G, C-T, G-A, G-C, G-G, G-T, T-A, T-C, T-G, T-T). The y-axis represents the number of substitutions, ranging from 0 to 163. The bars are color-coded by the type of substitution: A-A (black), A-C (red), A-G (green), A-T (cyan), C-A (black), C-C (red), C-G (green), C-T (cyan), G-A (black), G-C (red), G-G (green), G-T (cyan), T-A (black), T-C (red), T-G (green), and T-T (cyan).

Double Base Substitutions (Heatmap):

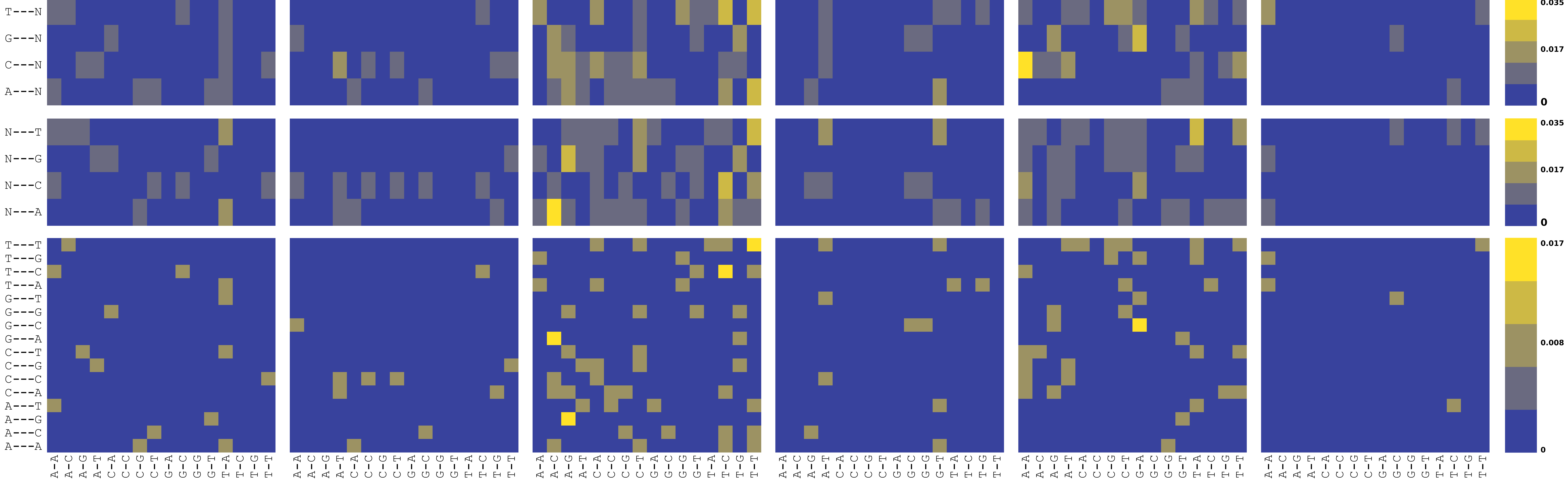
The bottom heatmap shows the number of double base substitutions for each of the 20x20 possible pairs. The color scale ranges from 0 (dark blue) to 0.044 (yellow). The heatmap is organized into four quadrants based on the first base (A, C, G, T) and the second base (A, C, G, T).



T>G

F010: 124 subs

Number of Single Base Substitutions



T>G

F011: 100 subs

Number of Single Base Substitutions

T-
G-
C-
A-

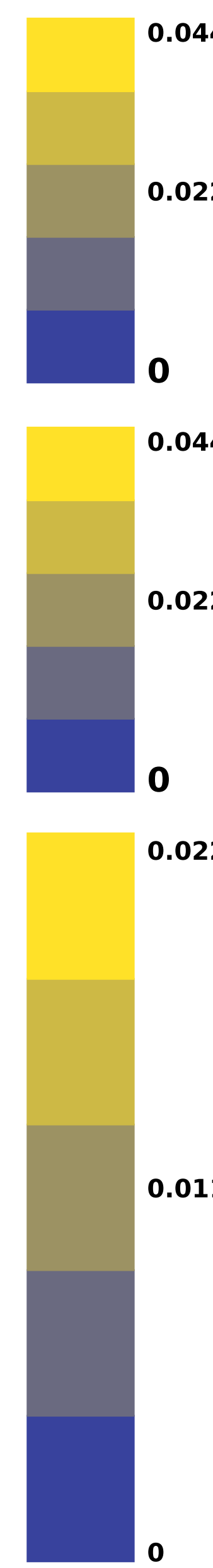
N -
N -
N -
N -

T-
T-
T-
T-
G-
G-
G-
G-
C-
C-
C-
C-
A-
A-
A-
A-

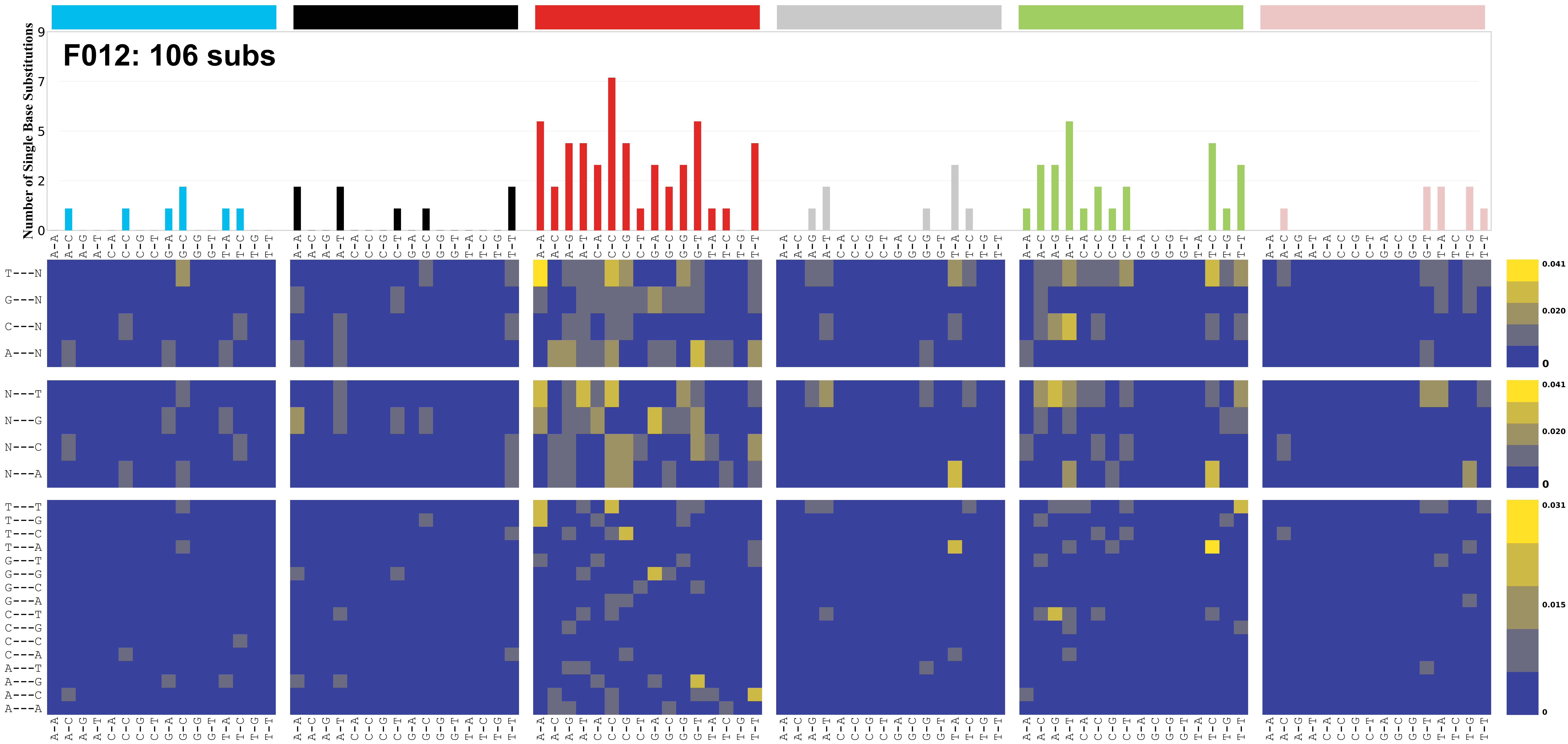
[illegible]

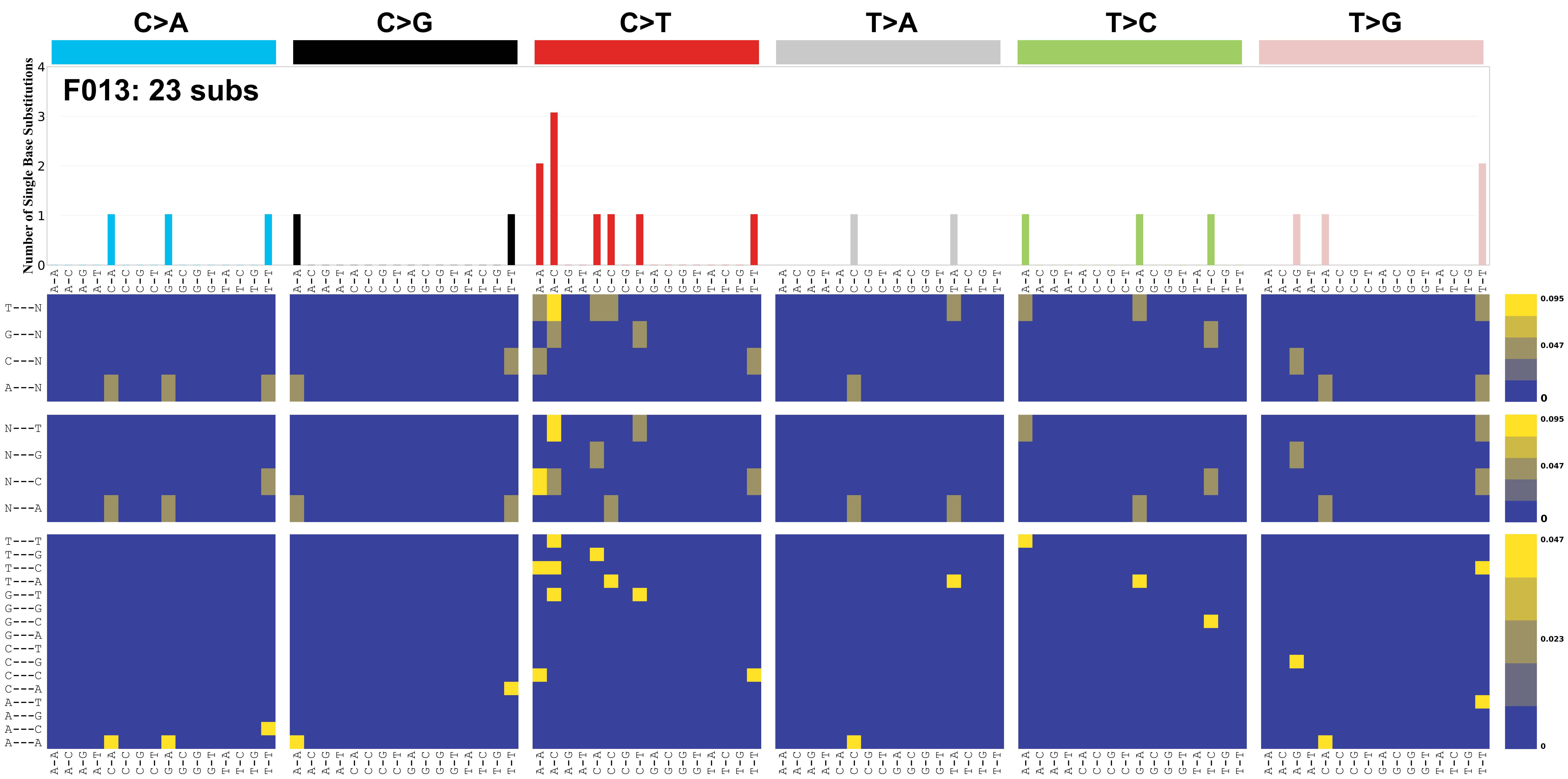
Figure 1 displays three heatmaps showing the distribution of amino acid pairs (A-A, A-C, A-G, A-T, C-A, C-C, C-G, C-T, G-A, G-C, G-G, G-T, T-A, T-C, T-G, T-T) across three different datasets. The color scale ranges from blue (low frequency) to yellow (high frequency).

The first heatmap (left) shows a high frequency of A-A, A-C, A-G, A-T, C-A, C-C, C-G, C-T, G-A, G-C, G-G, G-T, T-A, T-C, T-G, and T-T pairs. The second heatmap (middle) shows a high frequency of A-A, A-C, A-G, A-T, C-A, C-C, C-G, C-T, G-A, G-C, G-G, G-T, T-A, T-C, T-G, and T-T pairs. The third heatmap (right) shows a high frequency of A-A, A-C, A-G, A-T, C-A, C-C, C-G, C-T, G-A, G-C, G-G, G-T, T-A, T-C, T-G, and T-T pairs.



T>G



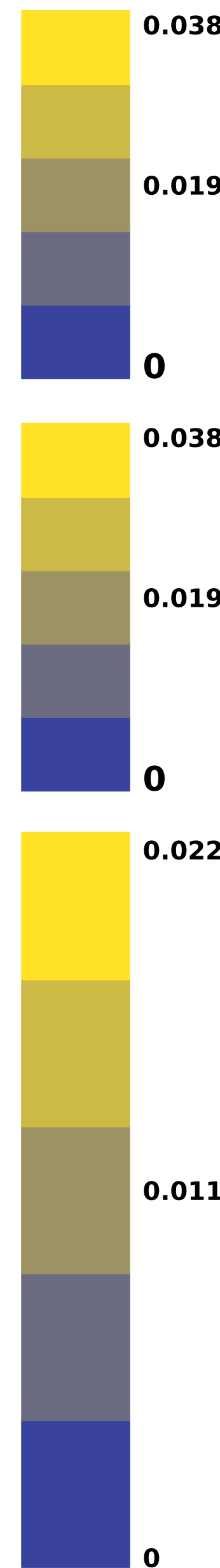
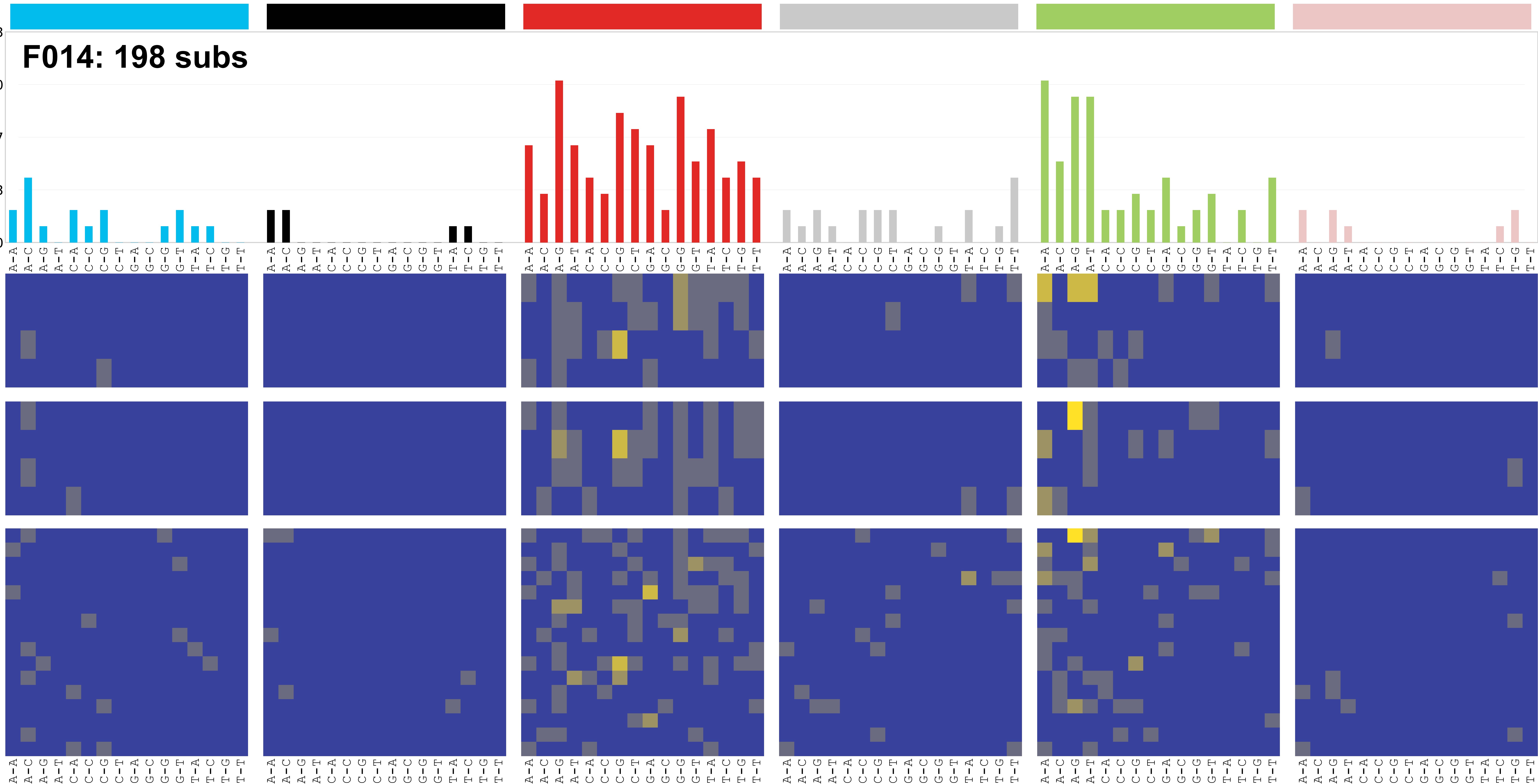


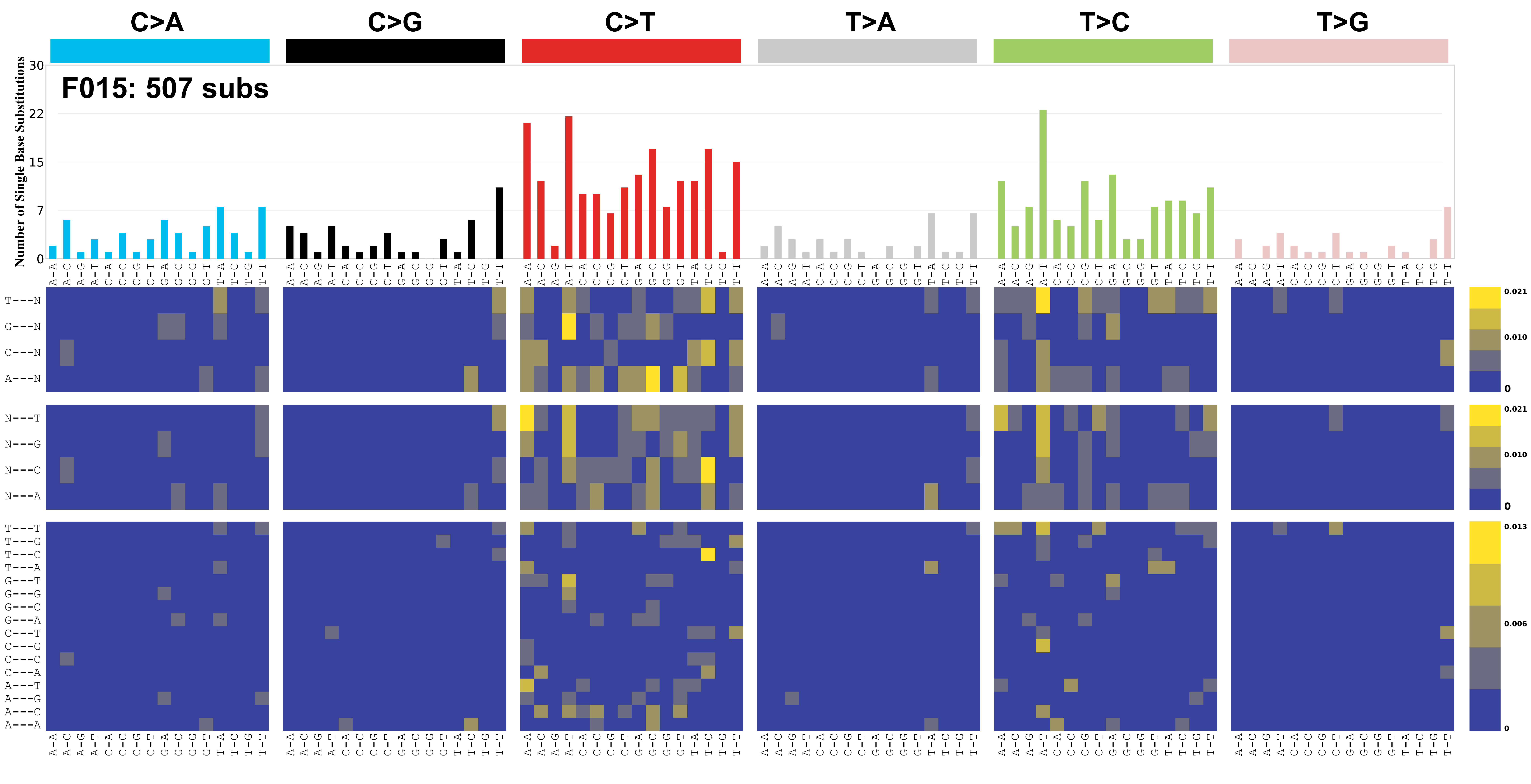
T>G

F014: 198 subs

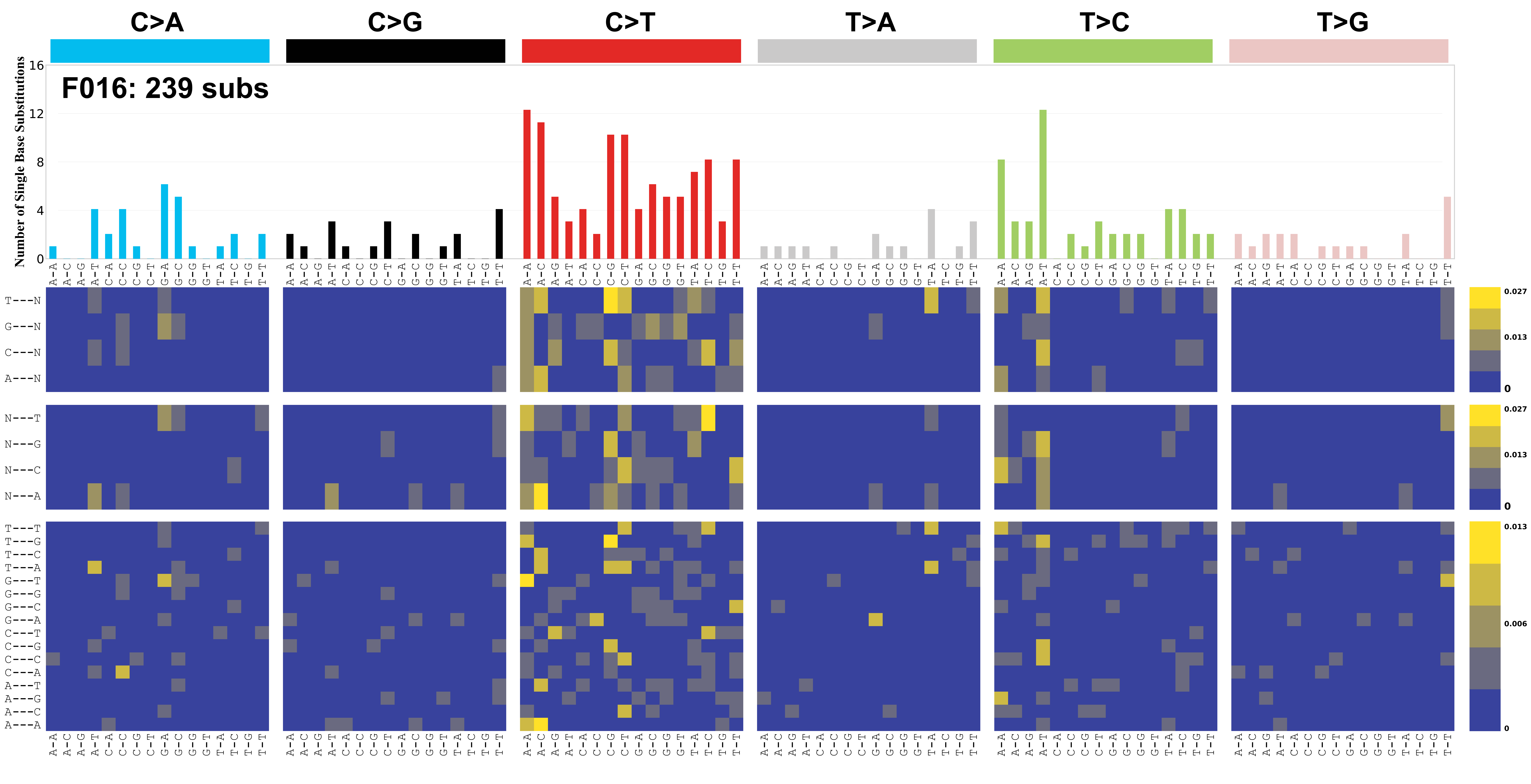
Number of Single Base Substitutions

T-
G-
C-
A-
N-
N-
N-
N-
T-
T-
T-
T-
G-
G-
G-
G-
C-
C-
C-
C-
A-
A-
A-
A-

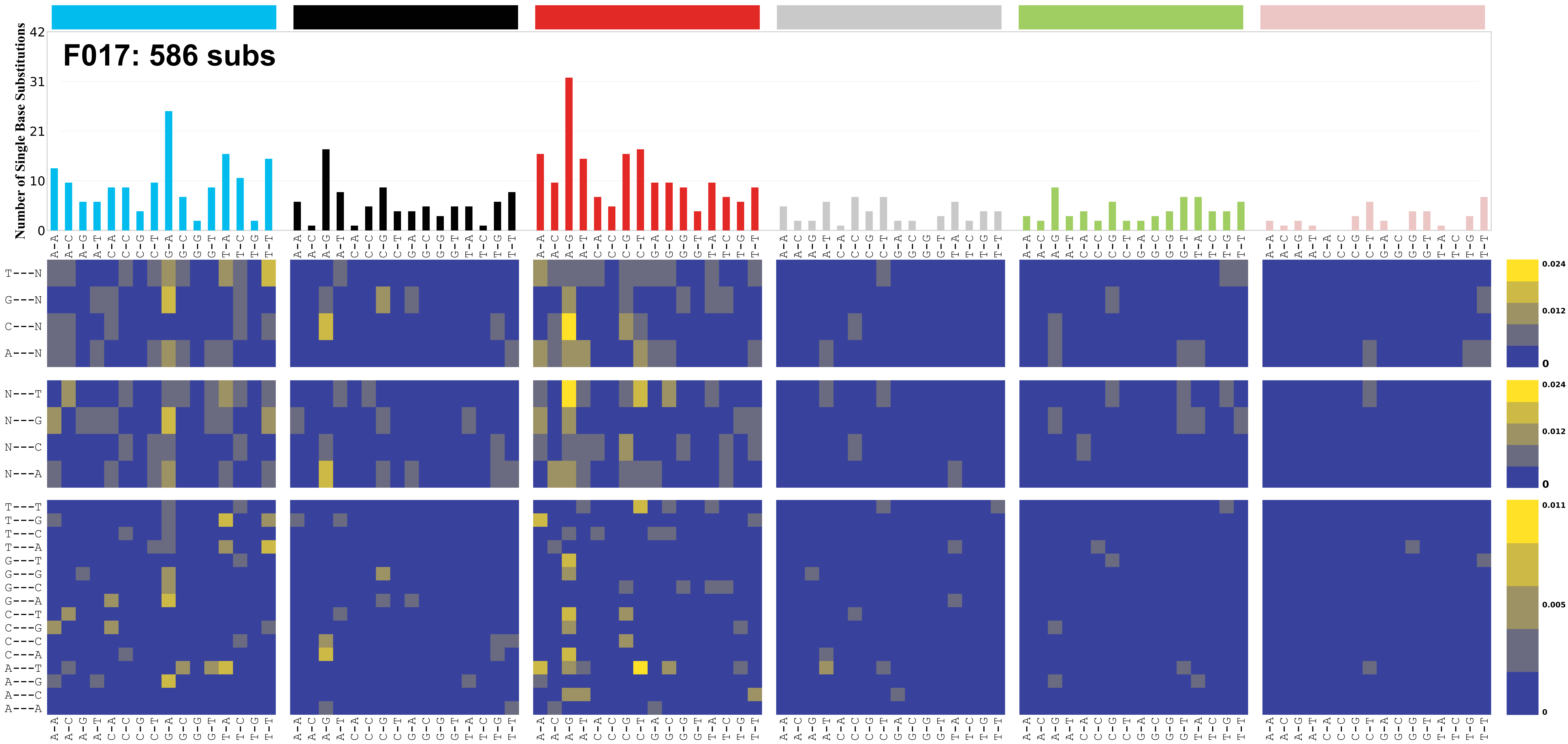


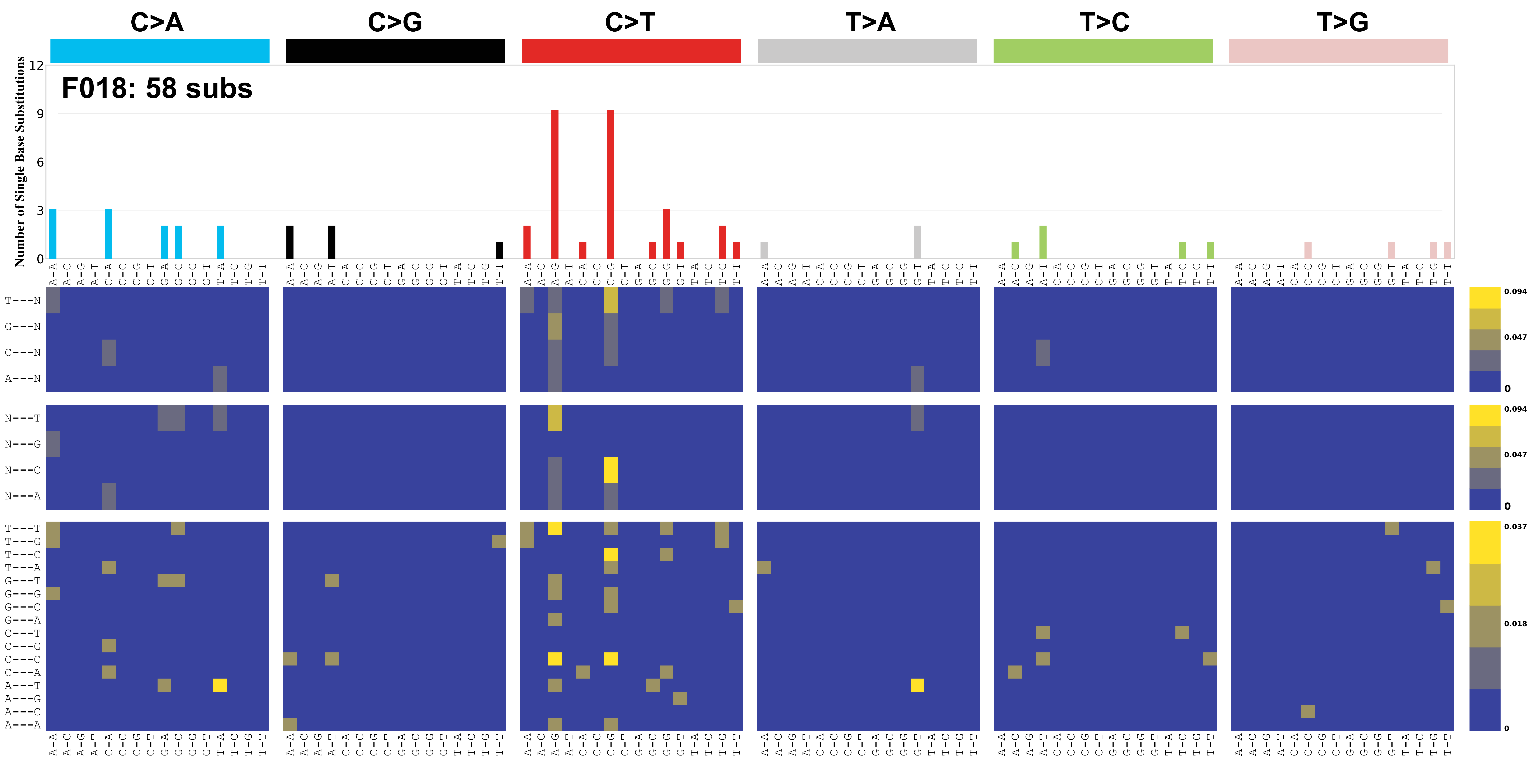


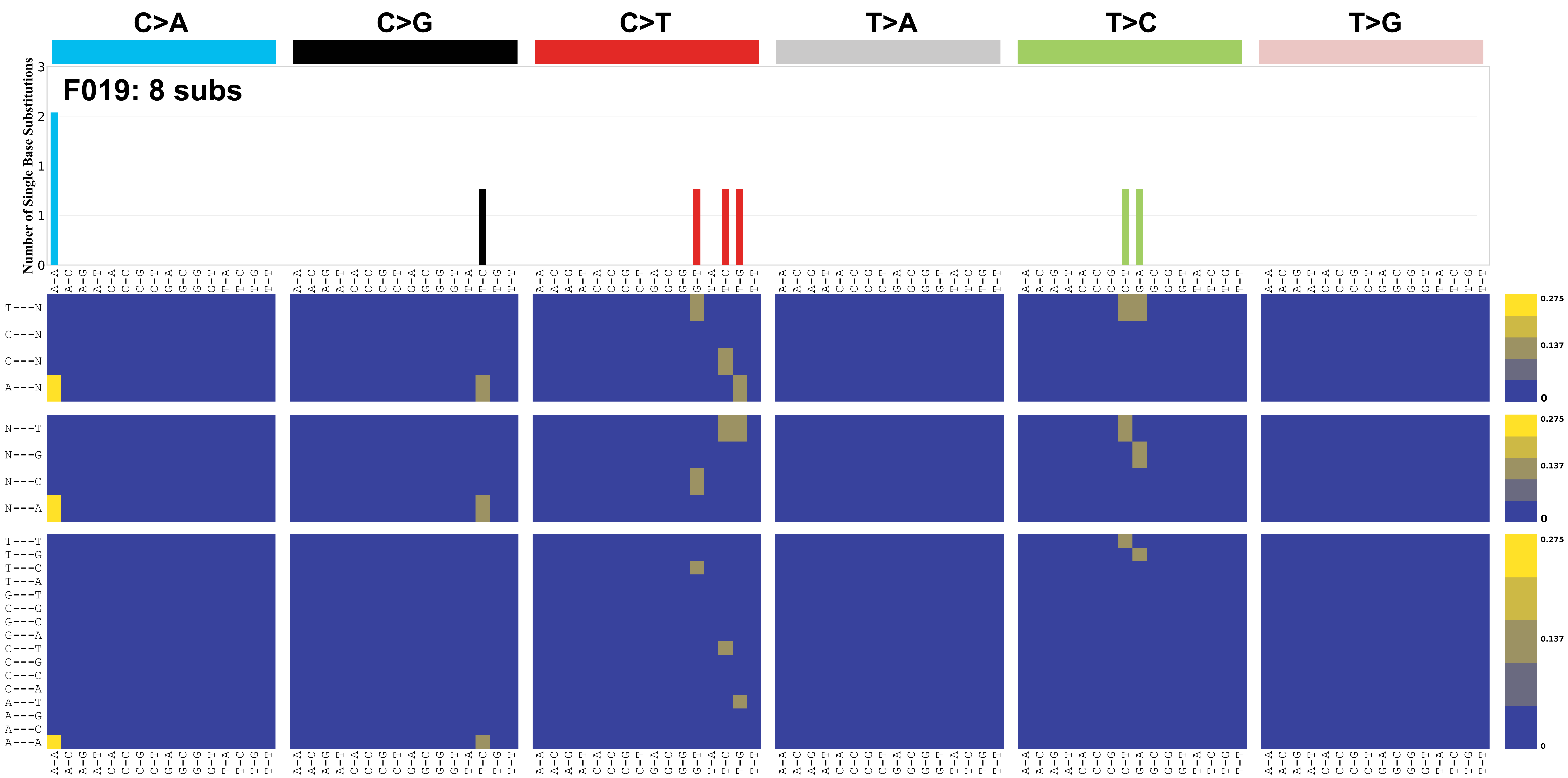
T>G

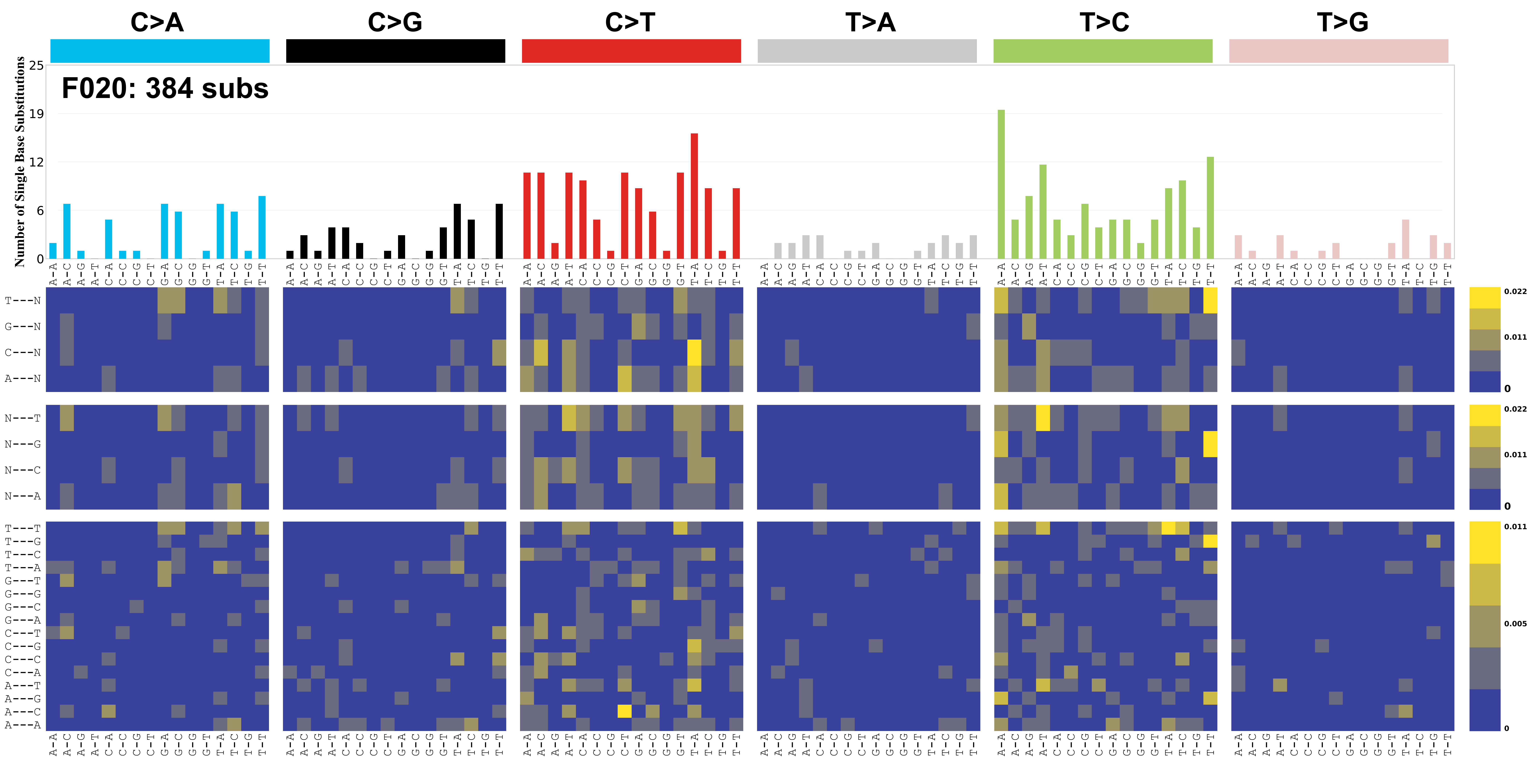


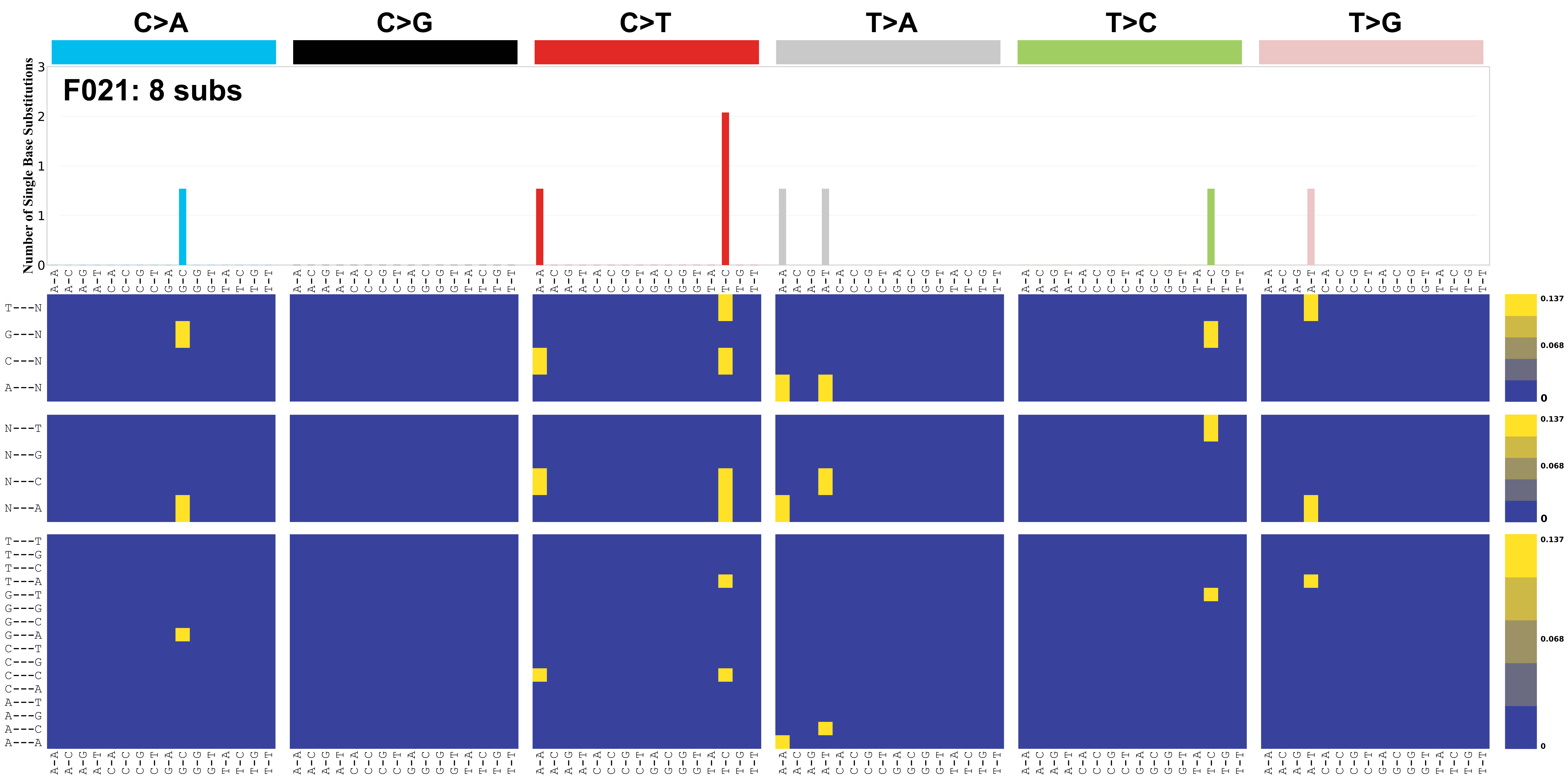
T>G





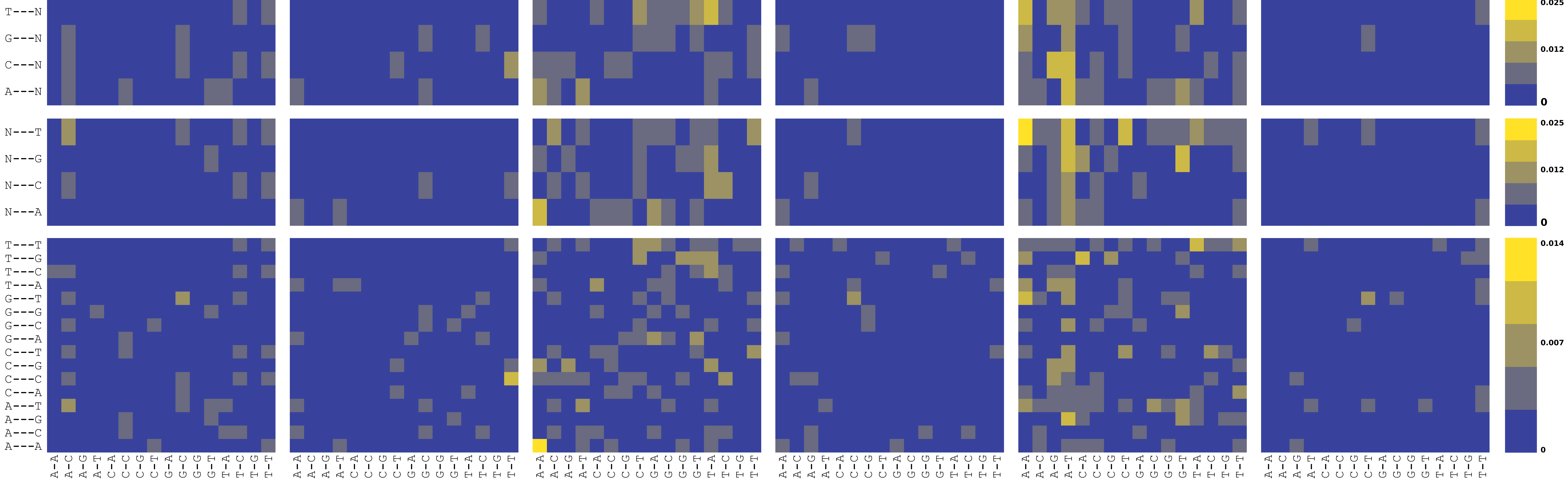






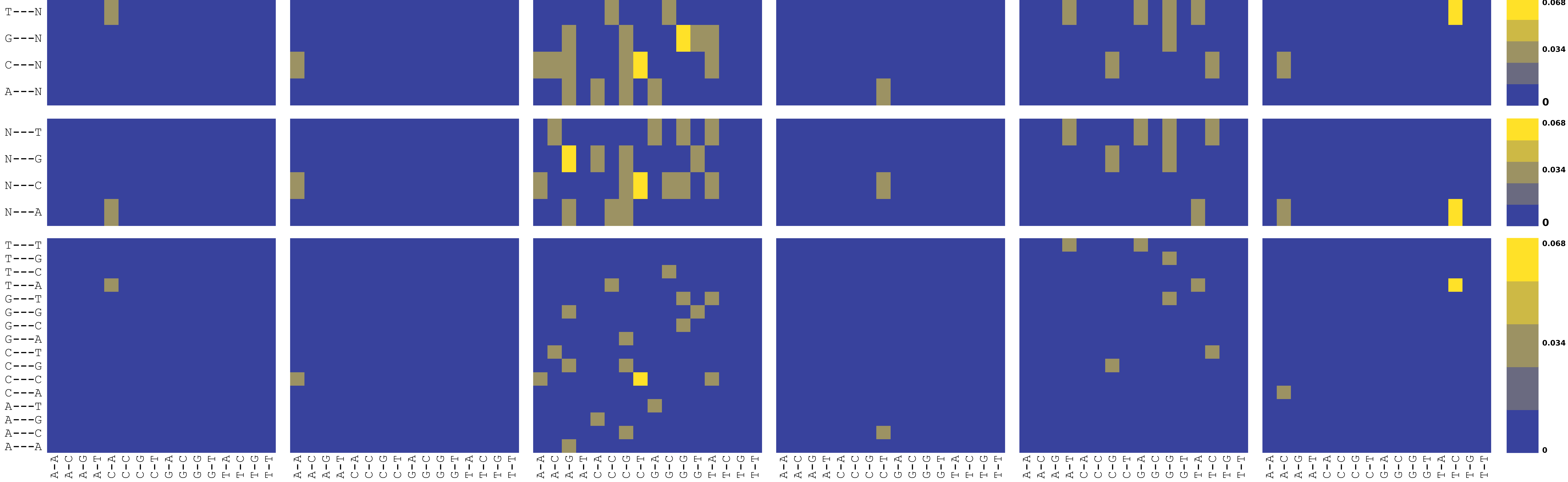
T>G

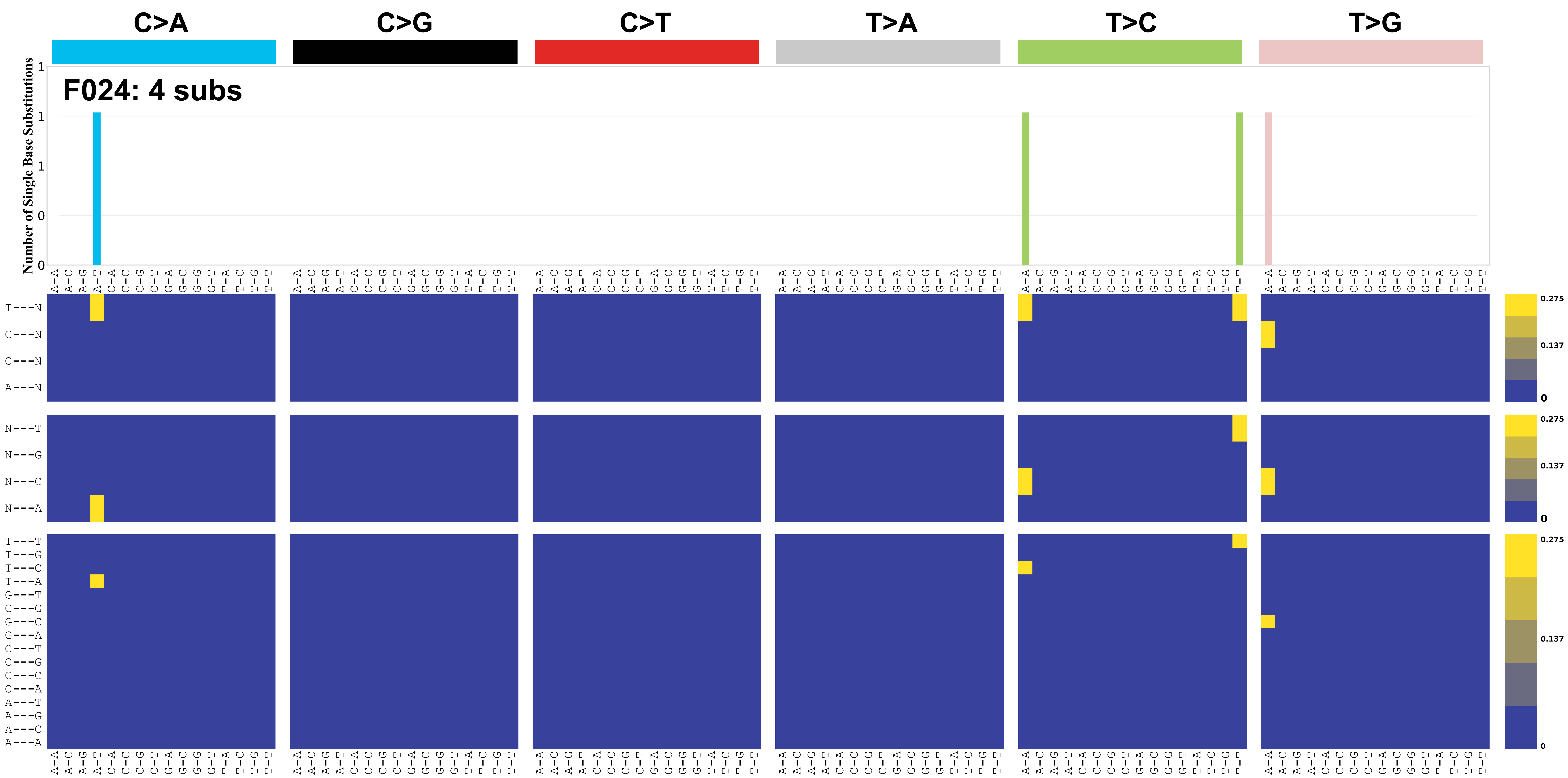
Number of Single Base Substitutions



T>G

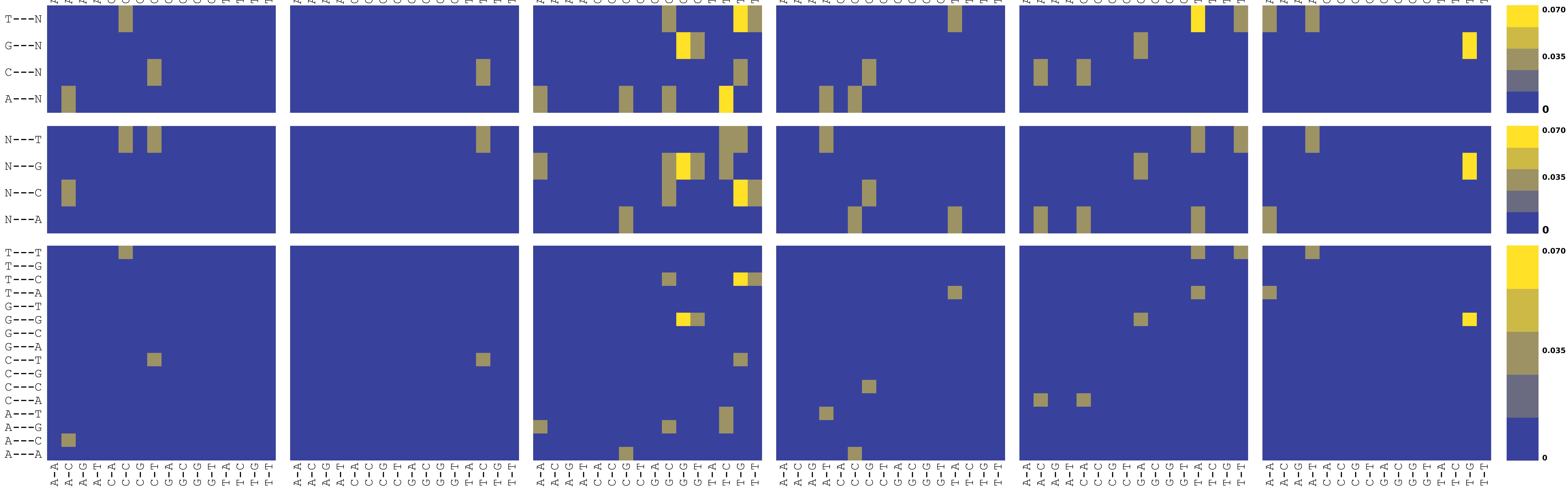
Number of Single Base Substitutions

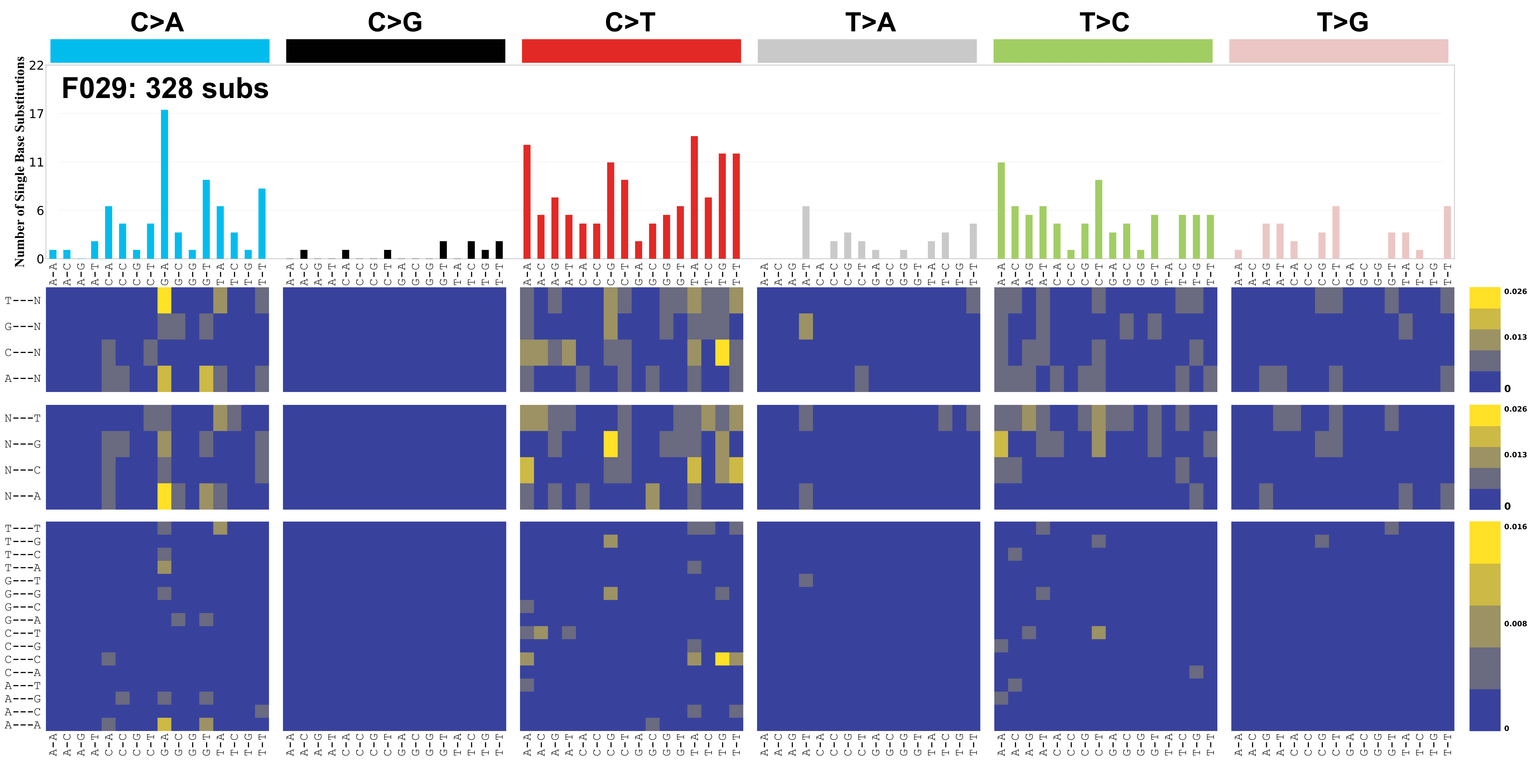




T>G

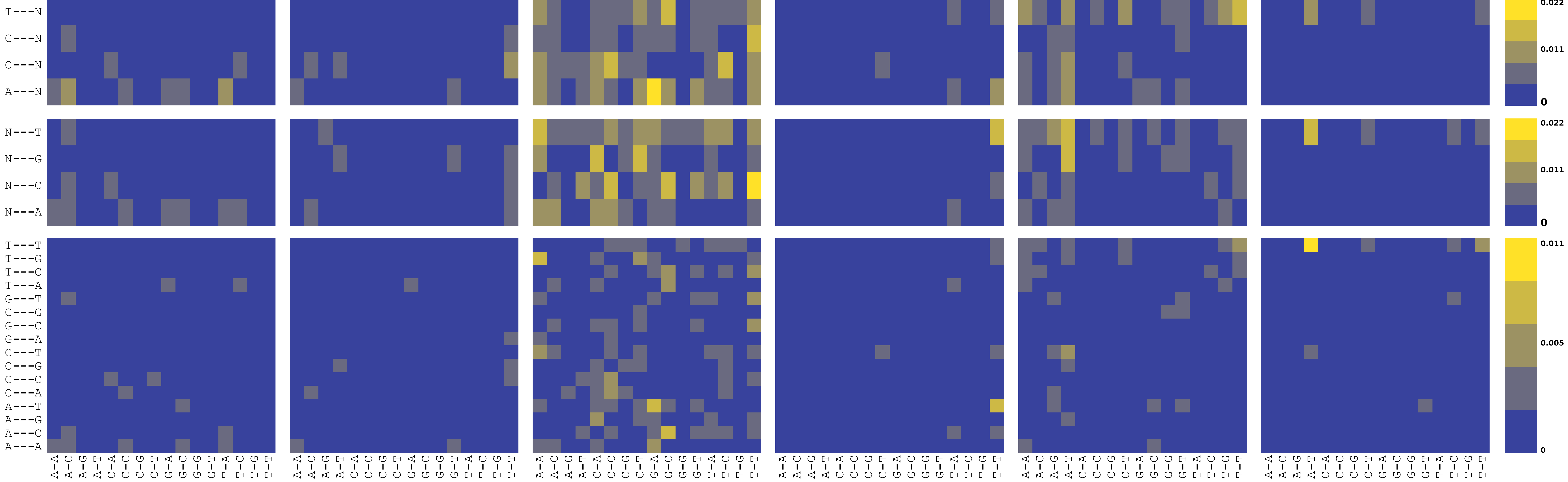
Number of Single Base Substitutions





T>G

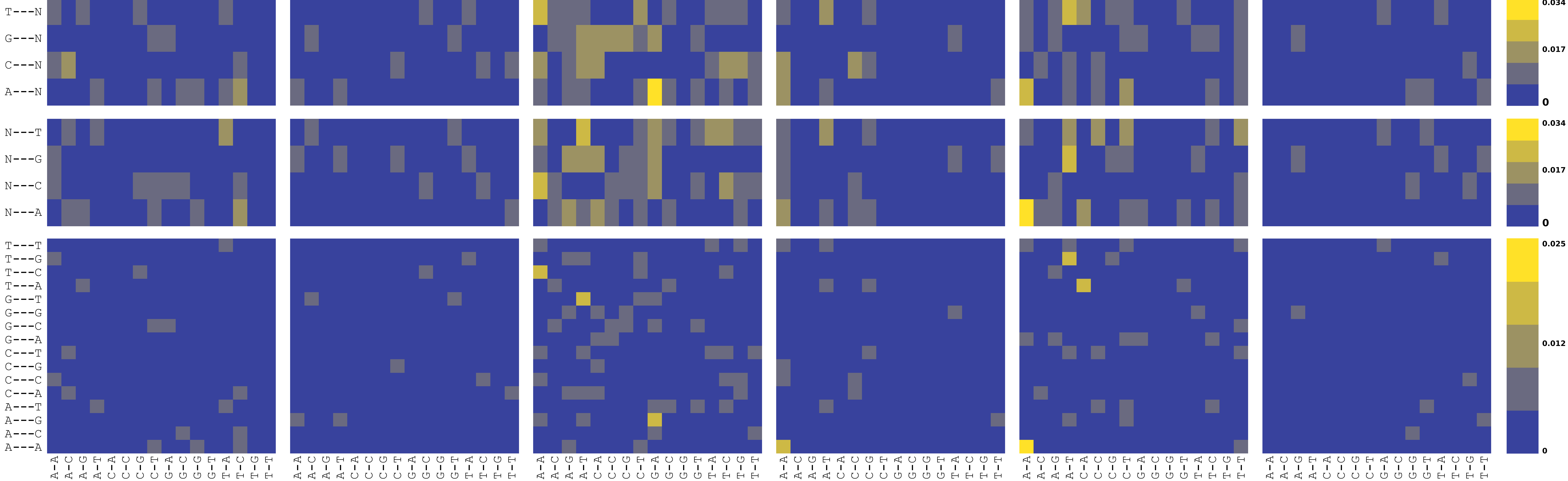
Number of Single Base Substitutions



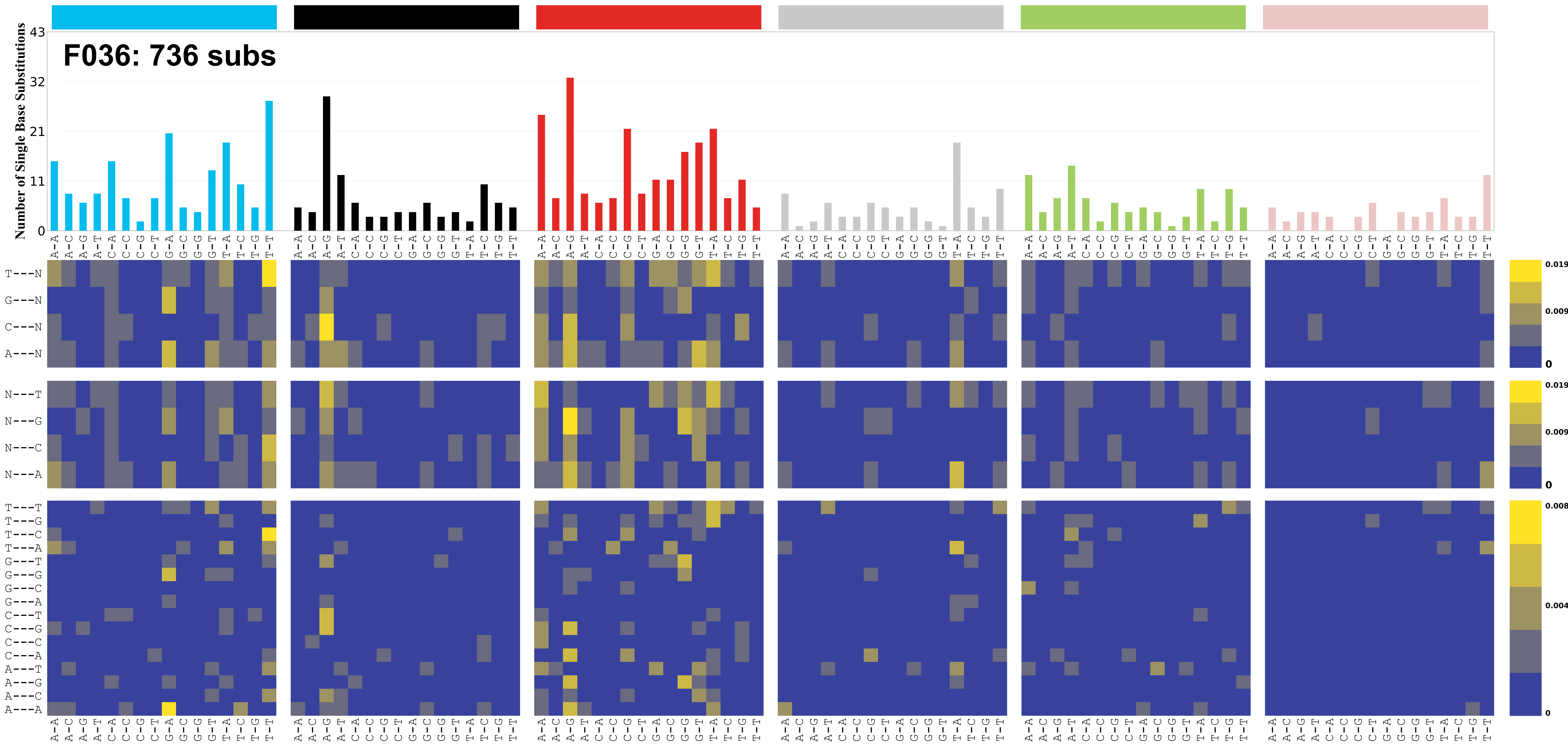
T>G

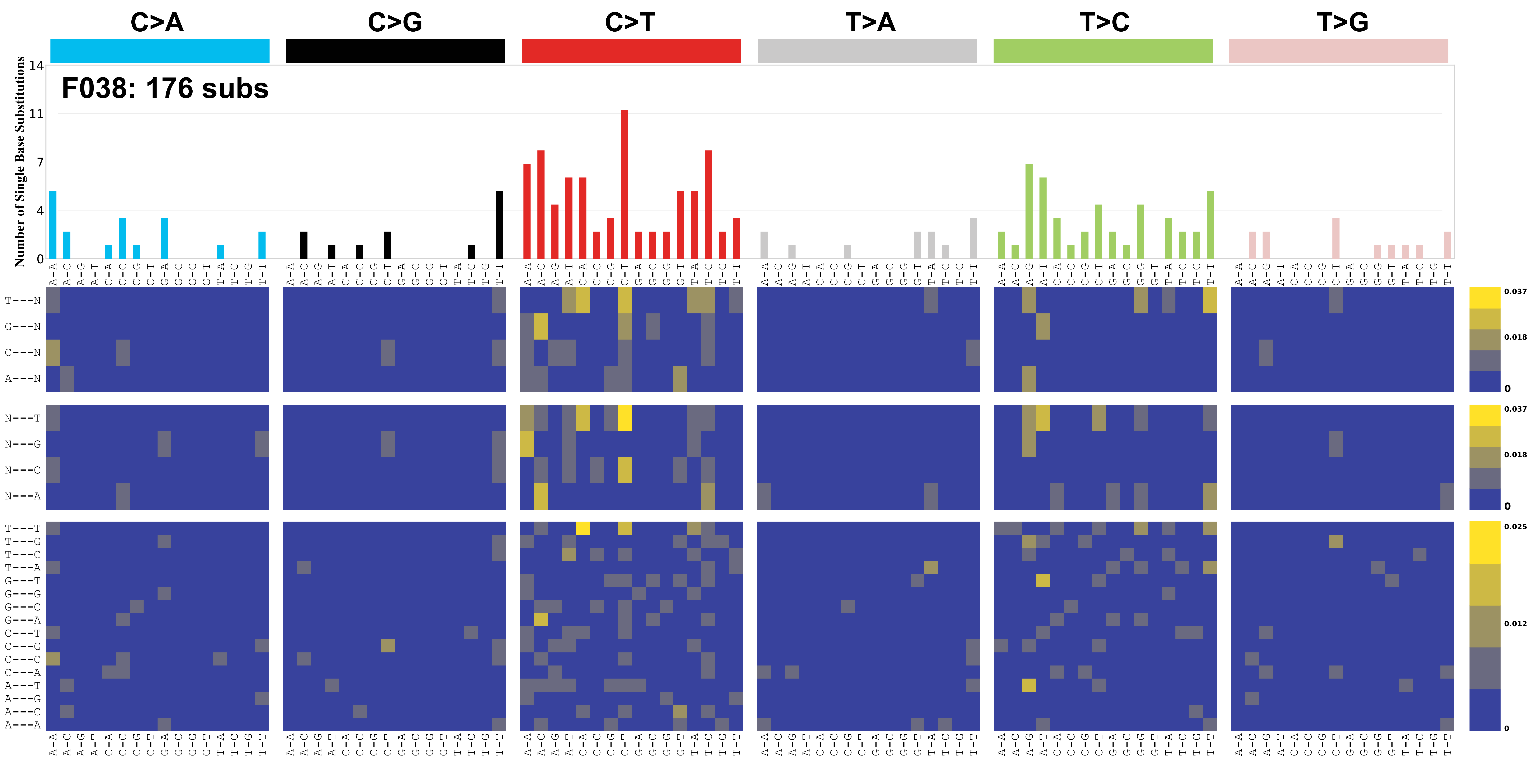
F034: 129 subs

Number of Single Base Substitutions

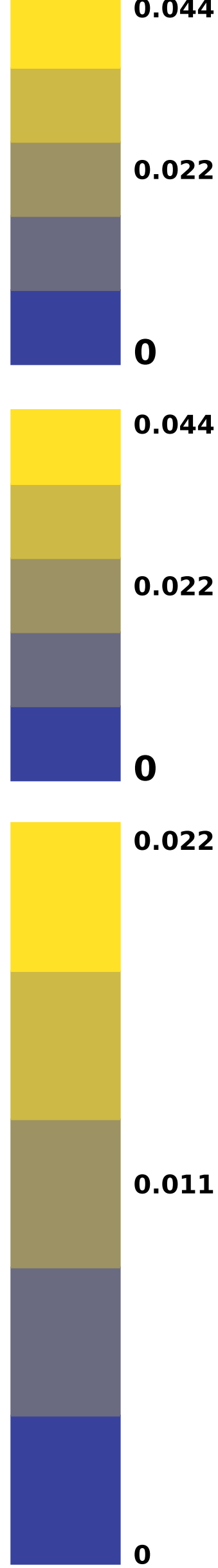


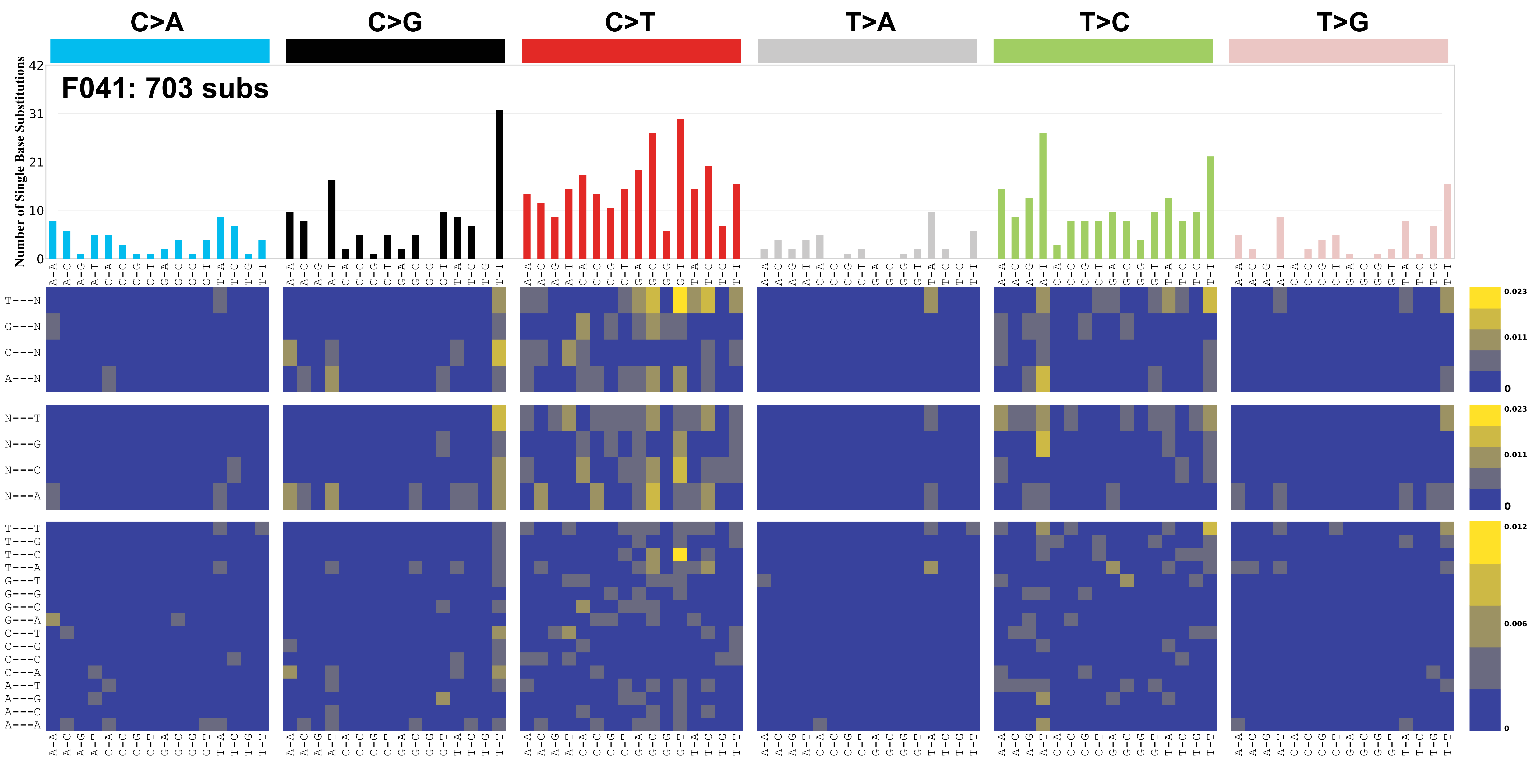
T>G

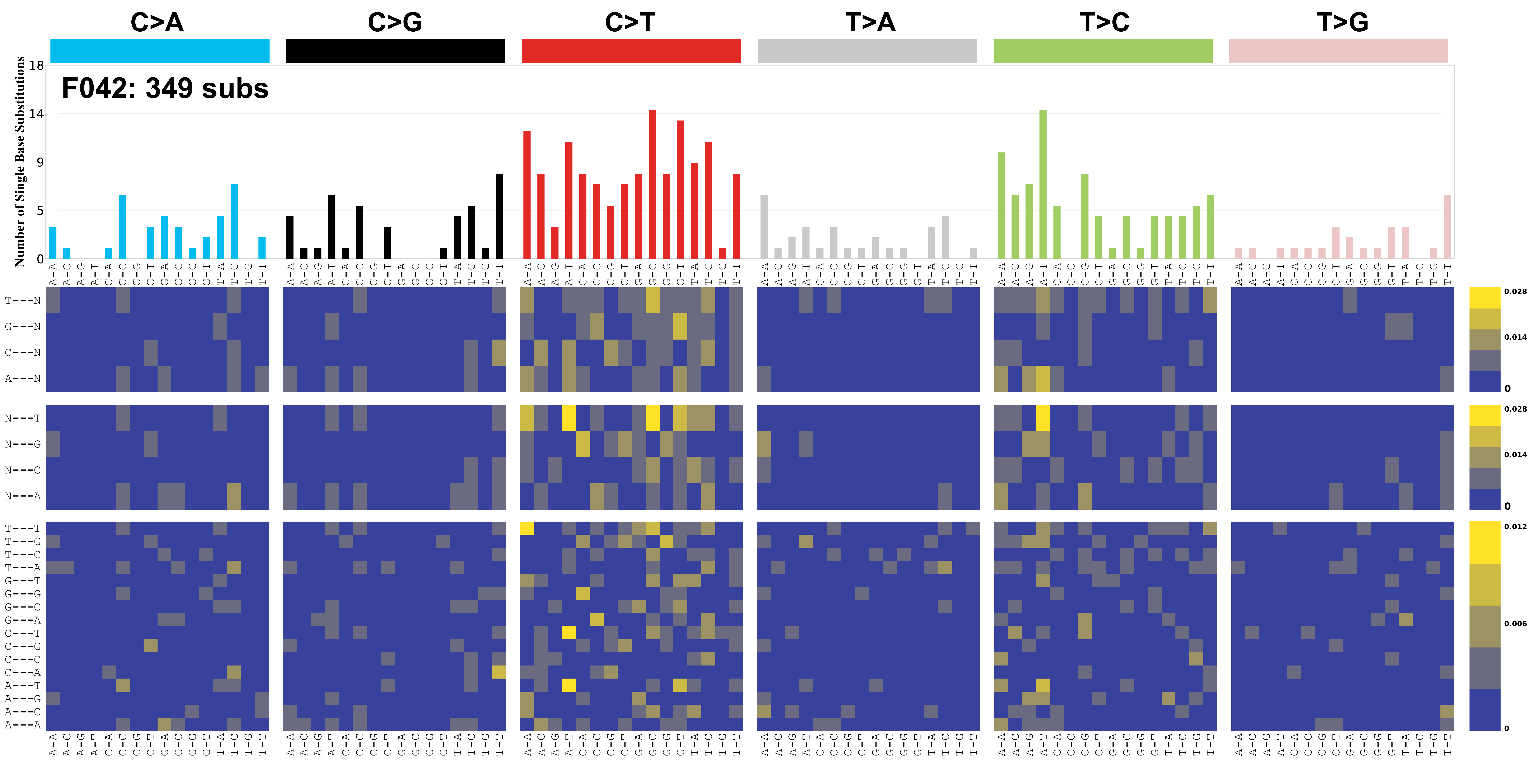


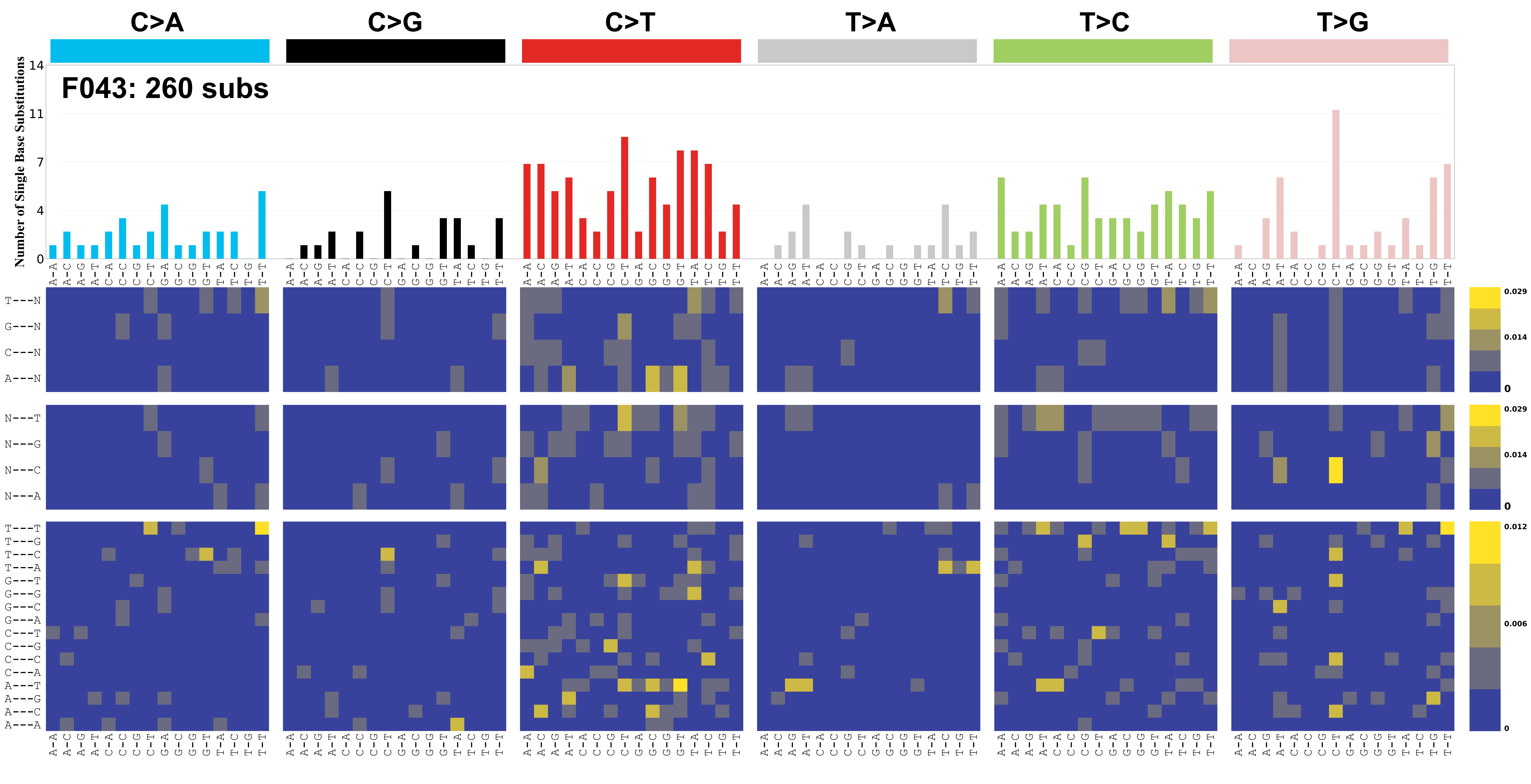


T>G



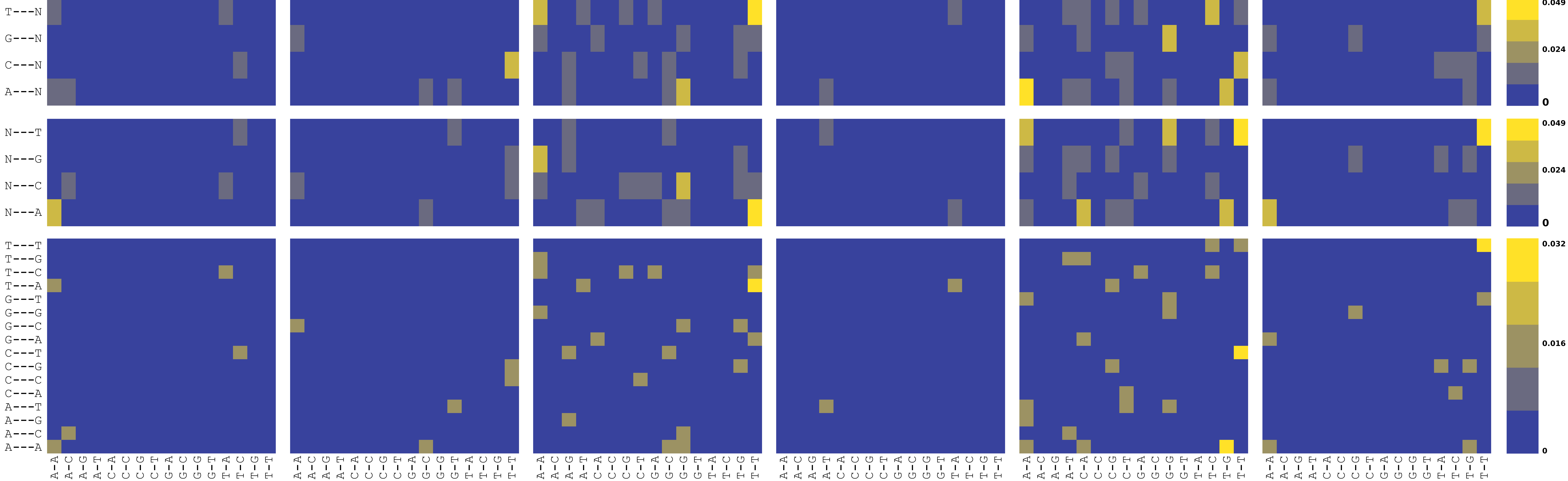


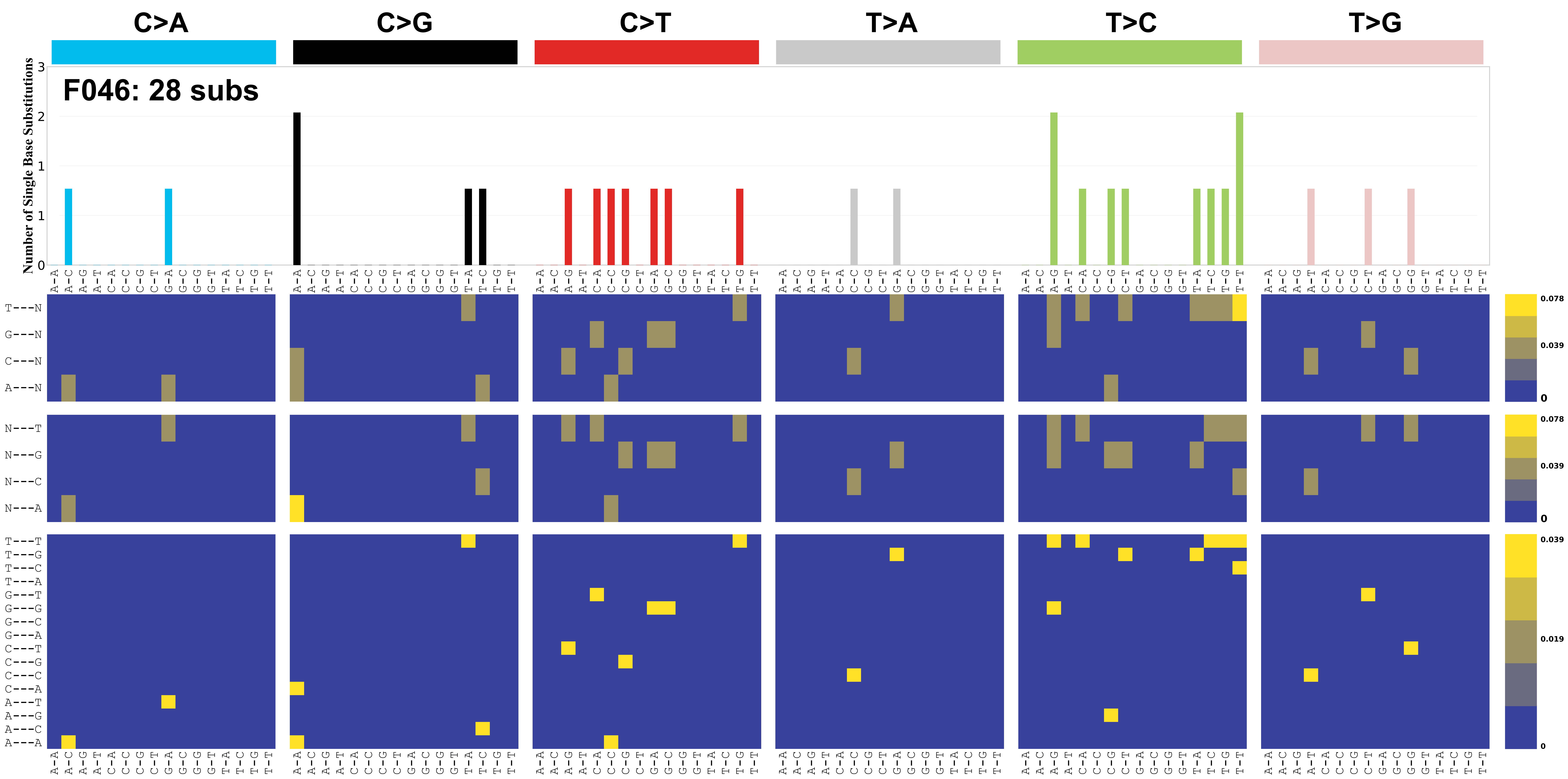


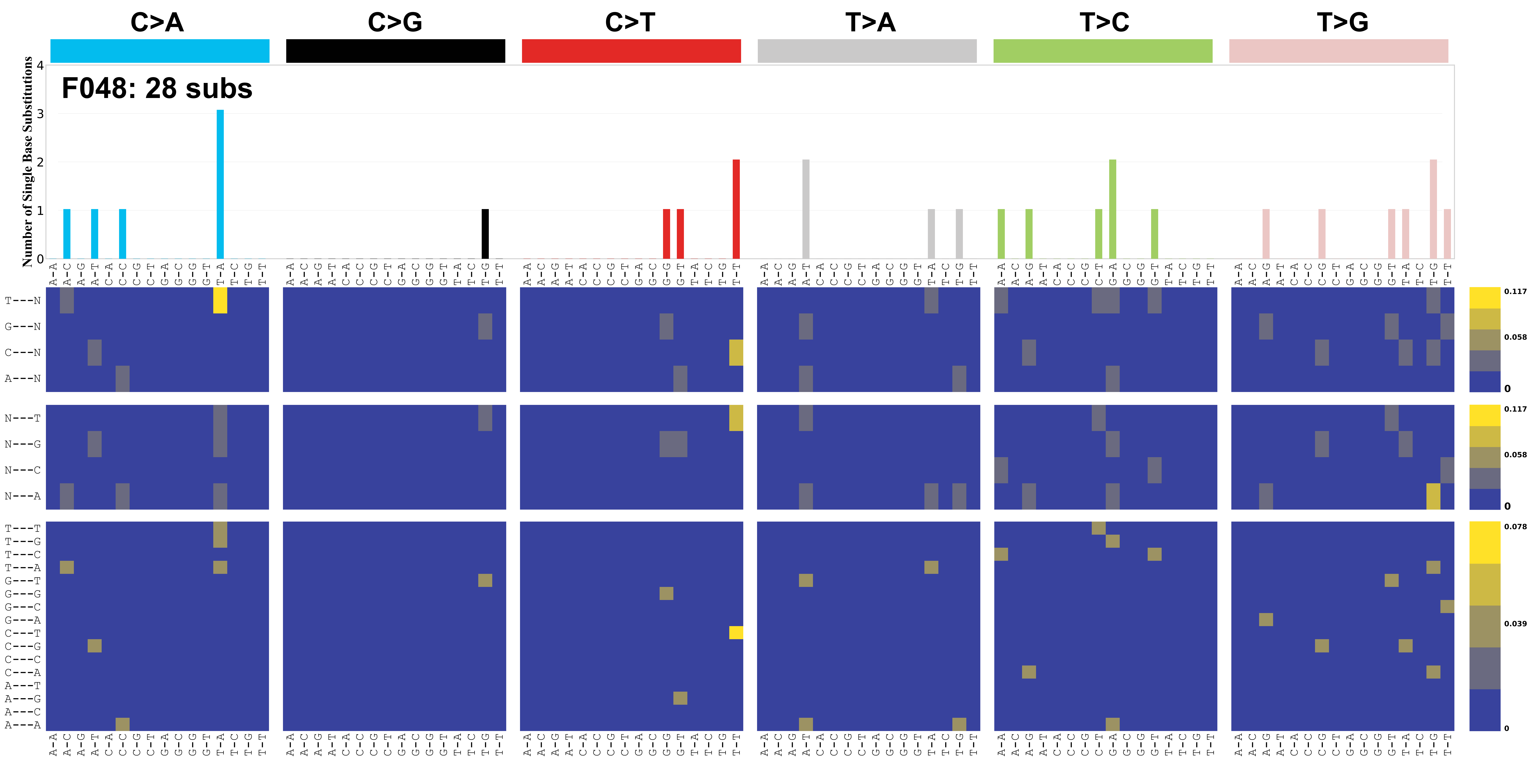


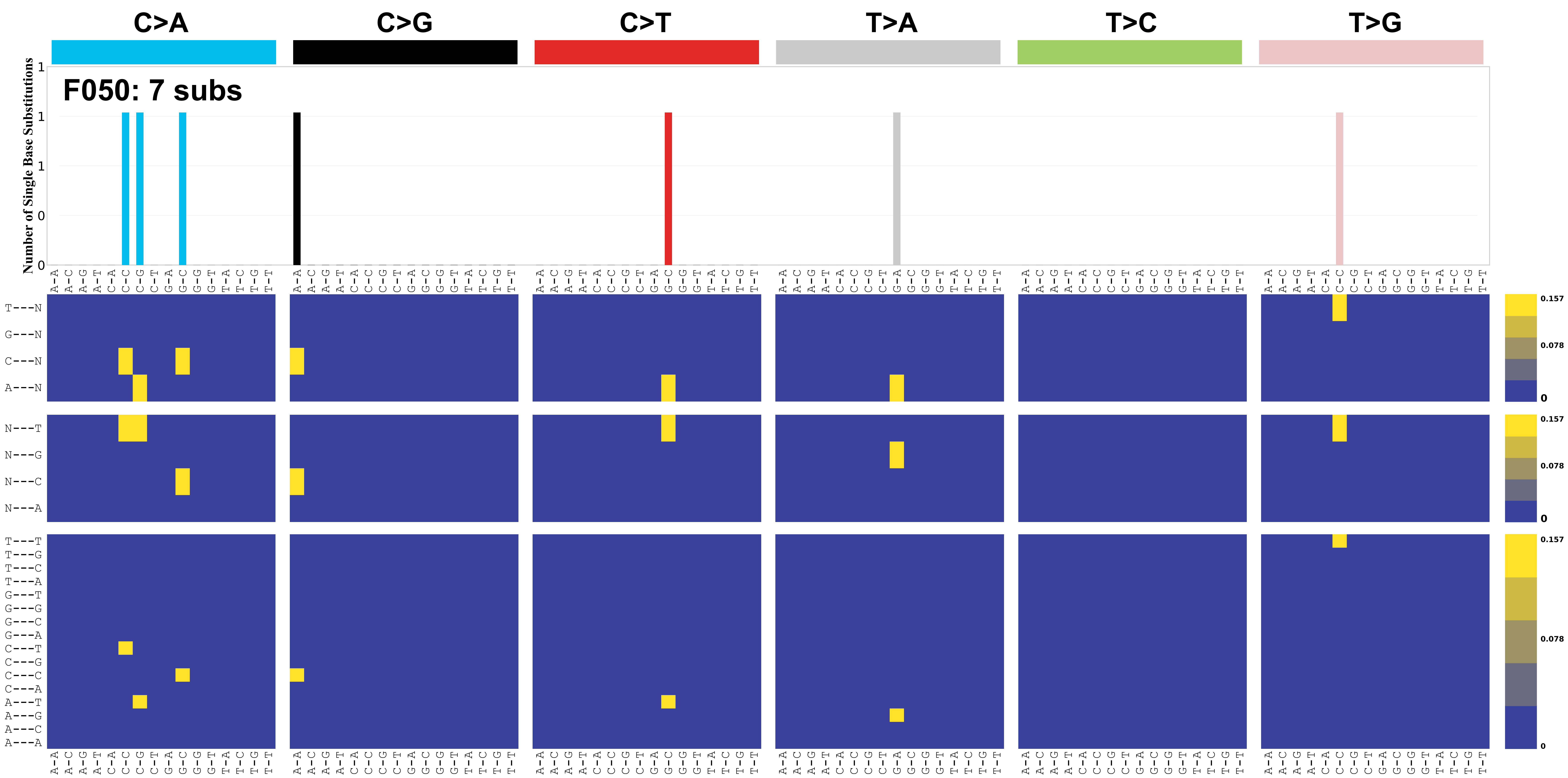
T>G

Number of Single Base Substitutions

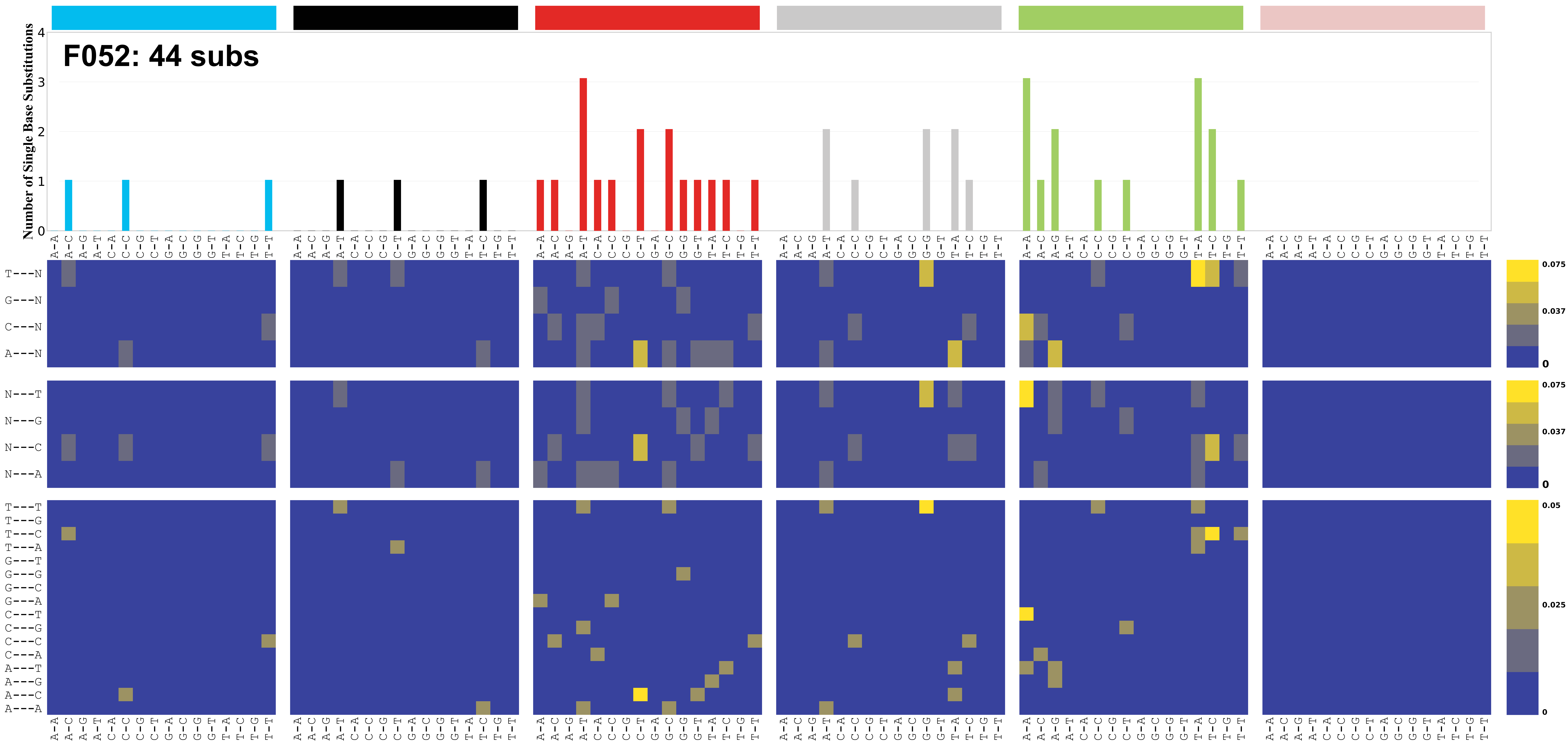








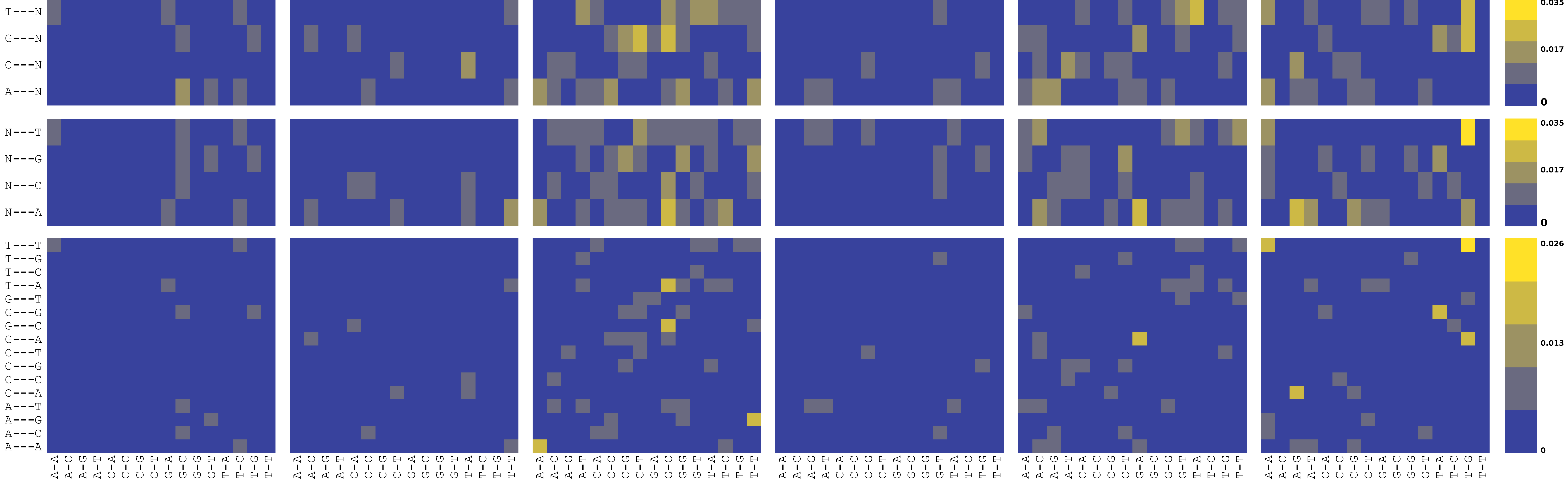
T>G



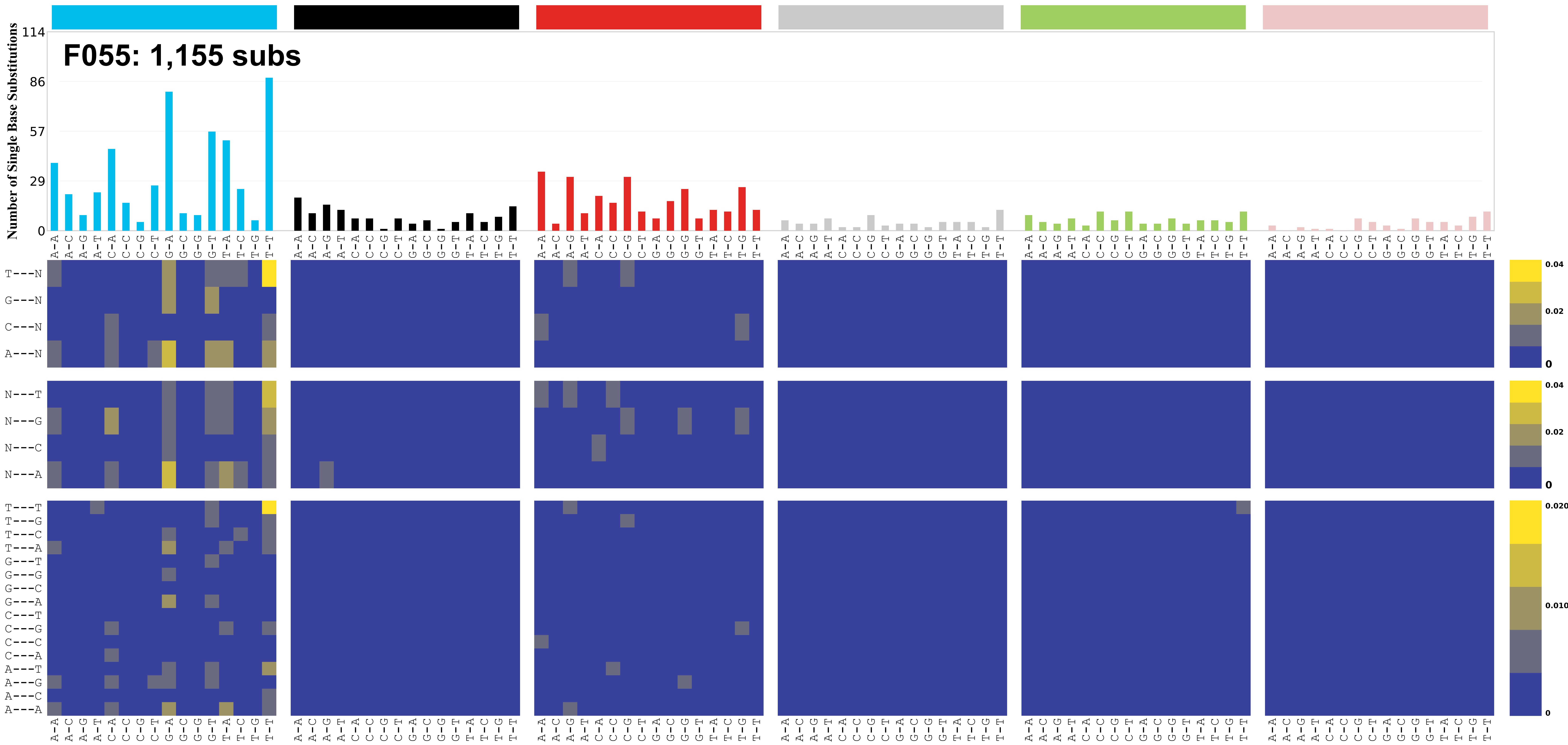
T>G

F053: 125 subs

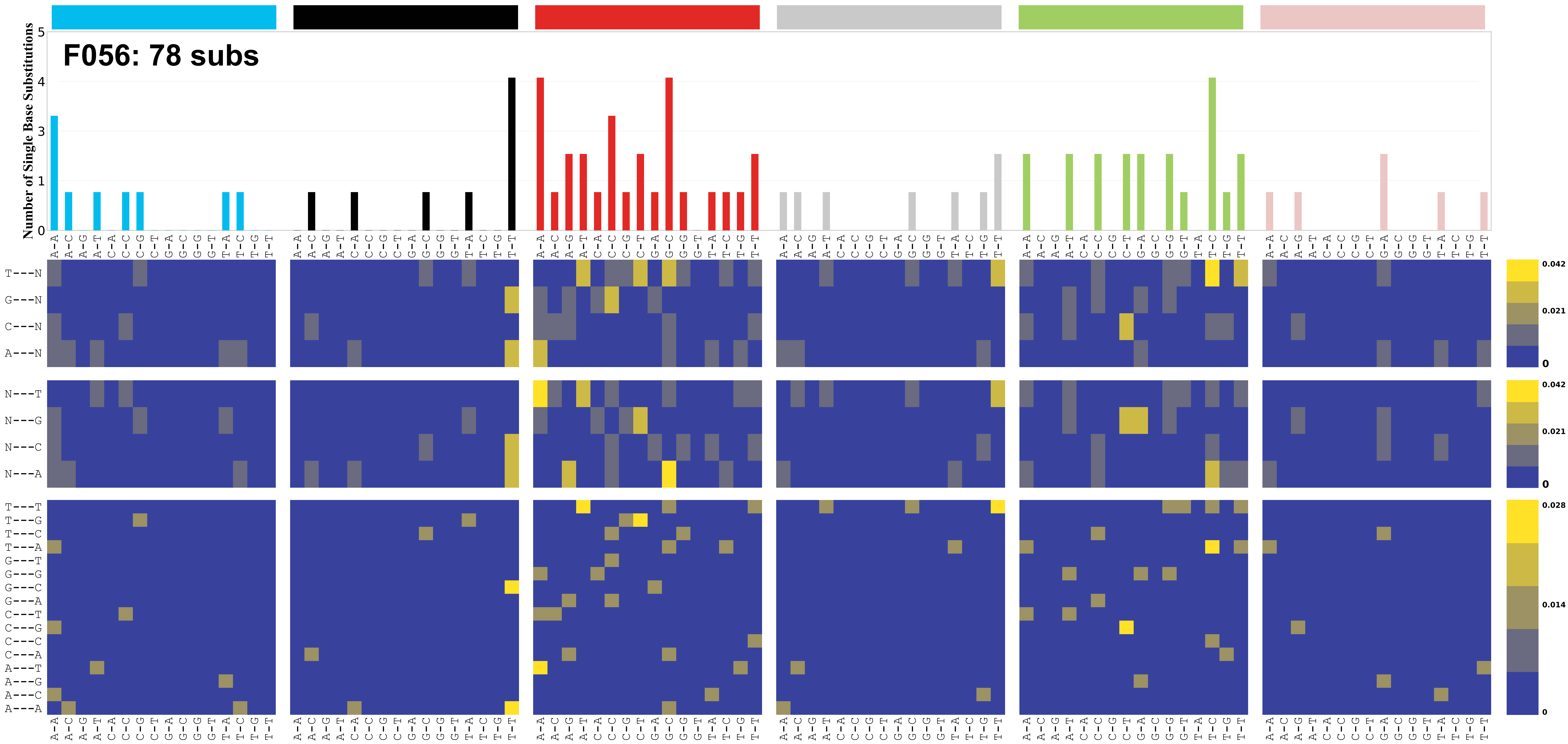
Number of Single Base Substitutions



T>G



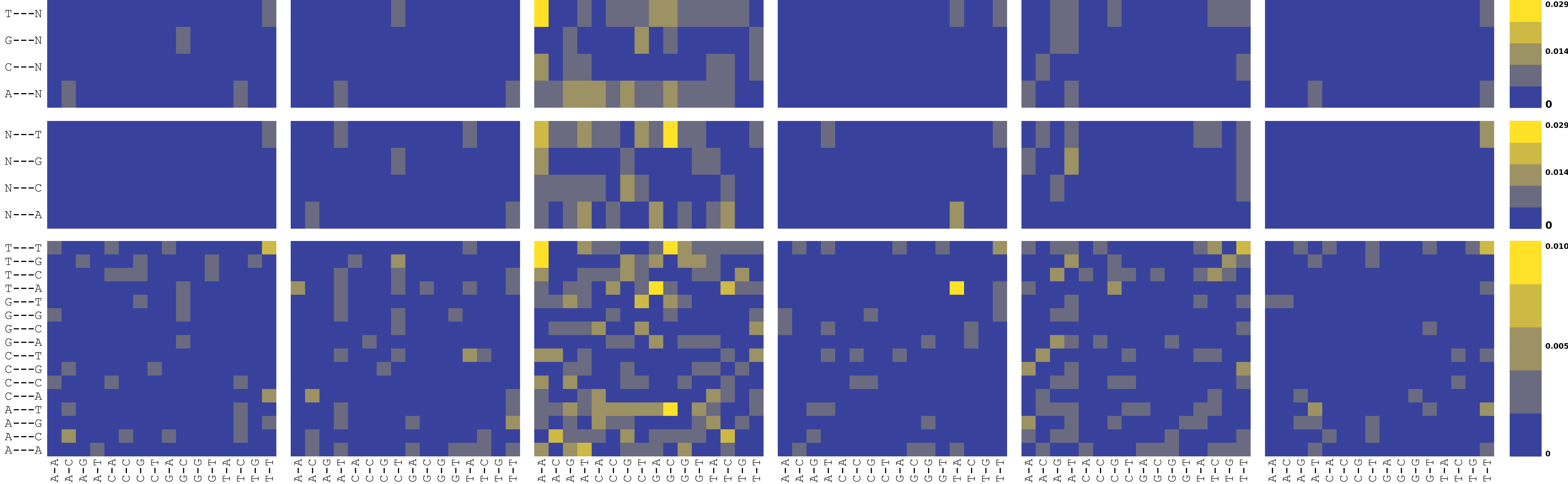
T>G



T>G

F057: 411 subs

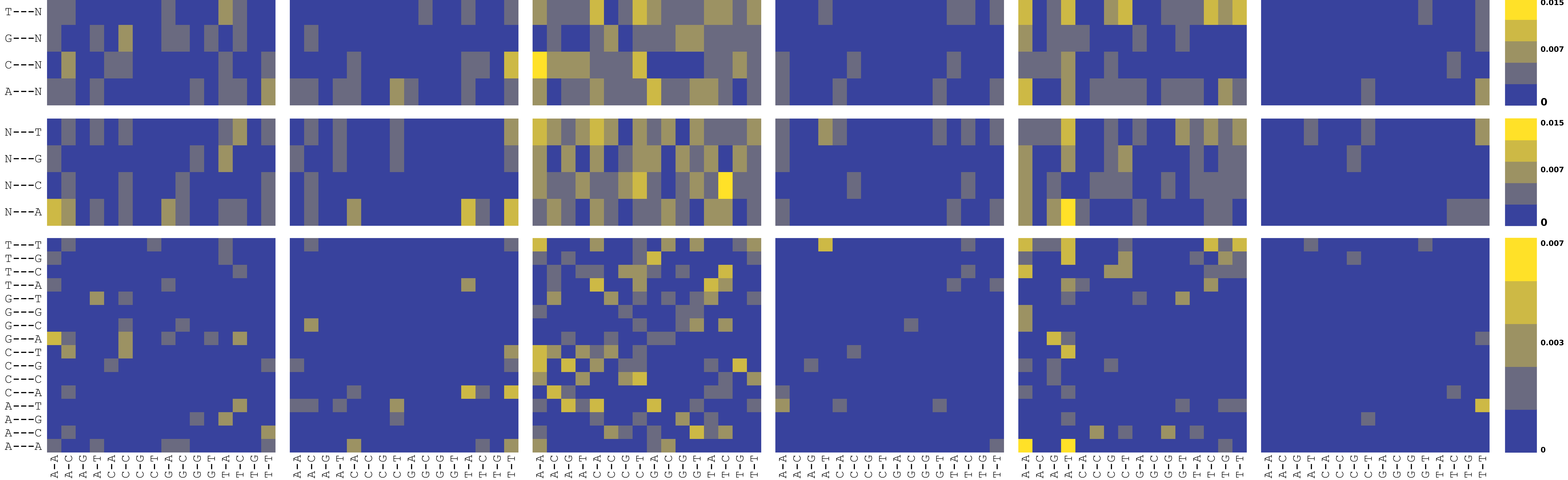
Number of Single Base Substitutions

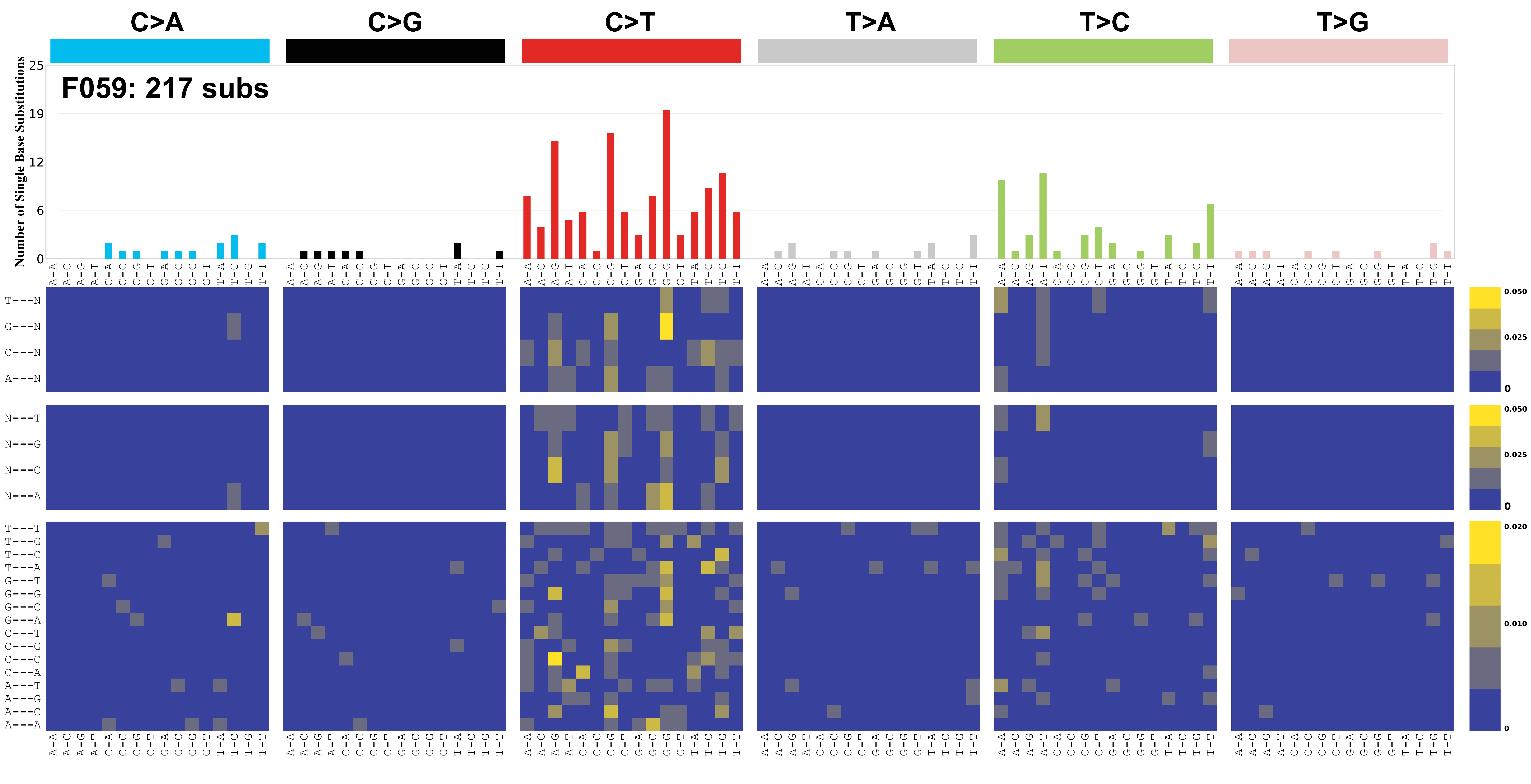


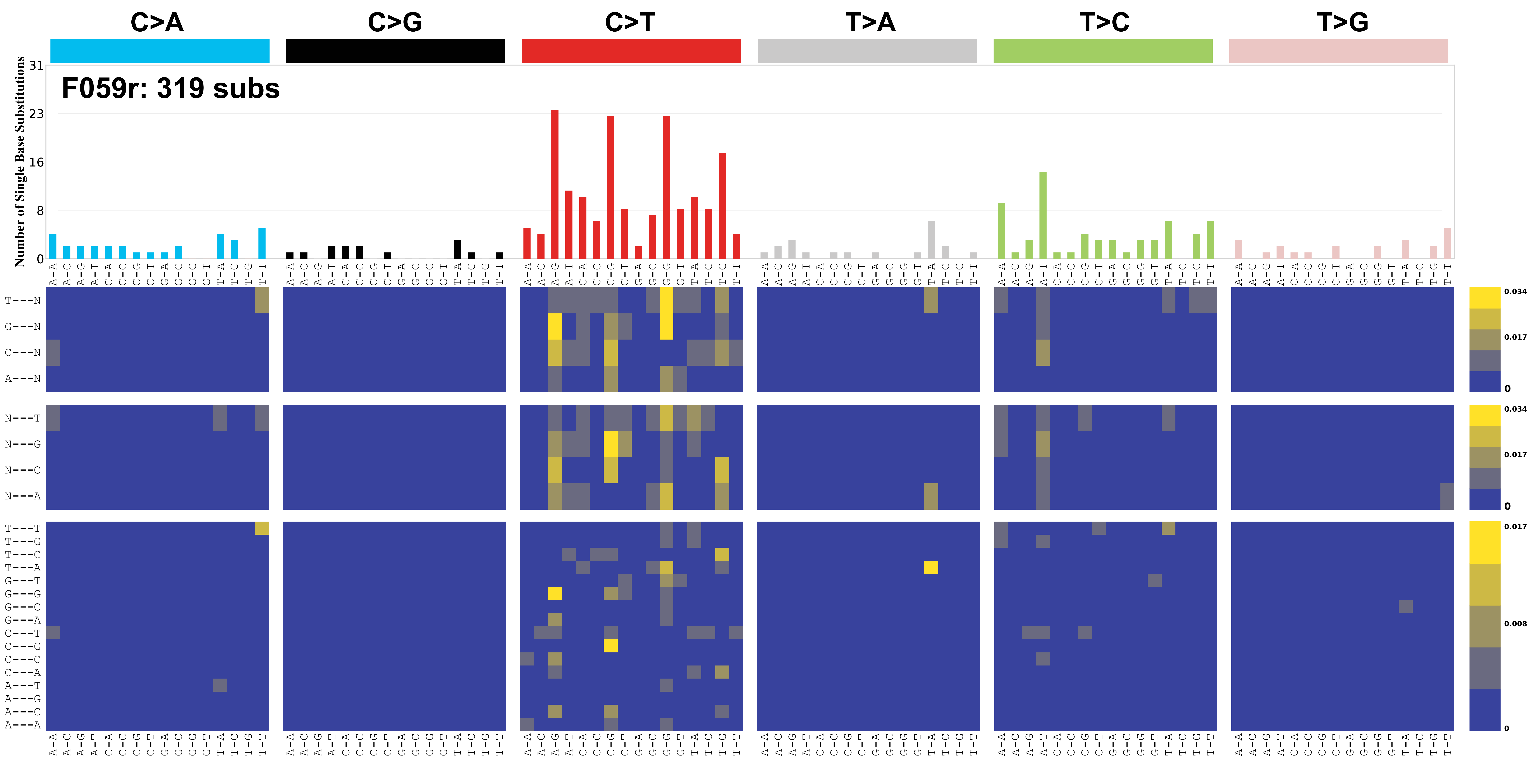
T>G

F058: 896 subs

Number of Single Base Substitutions

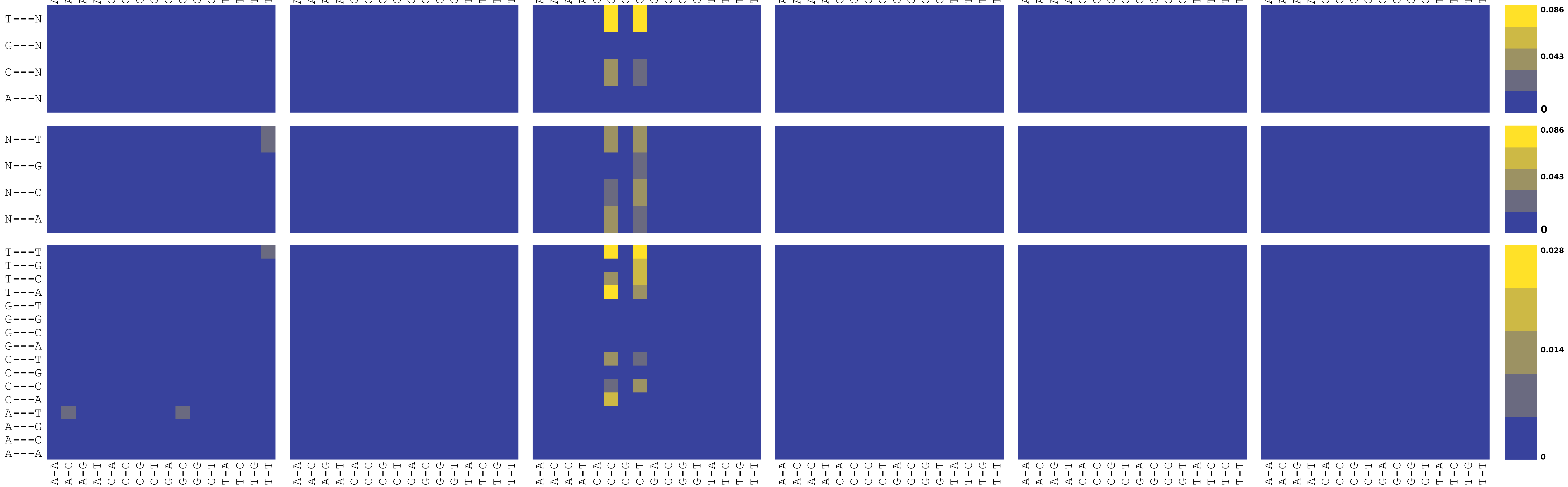




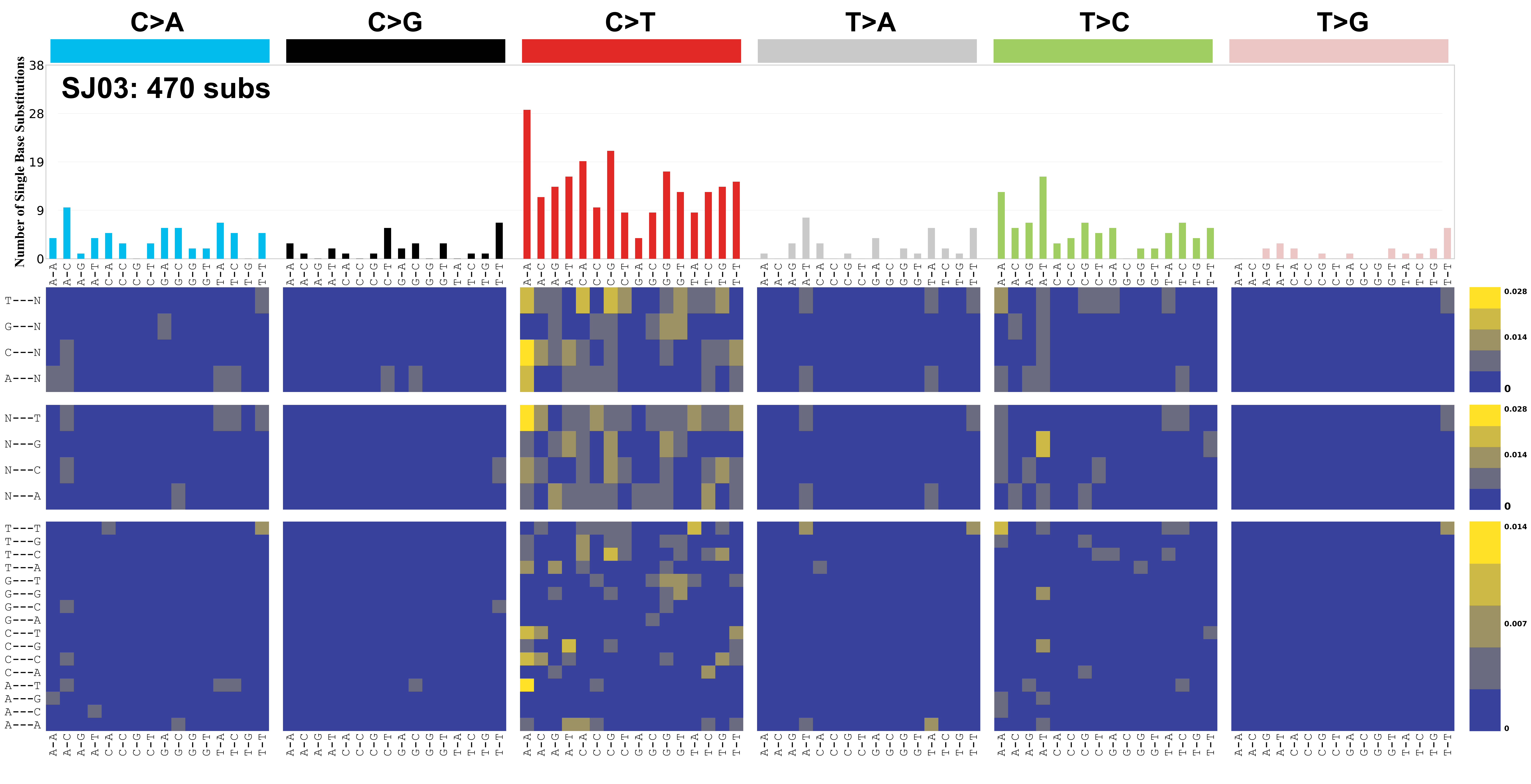


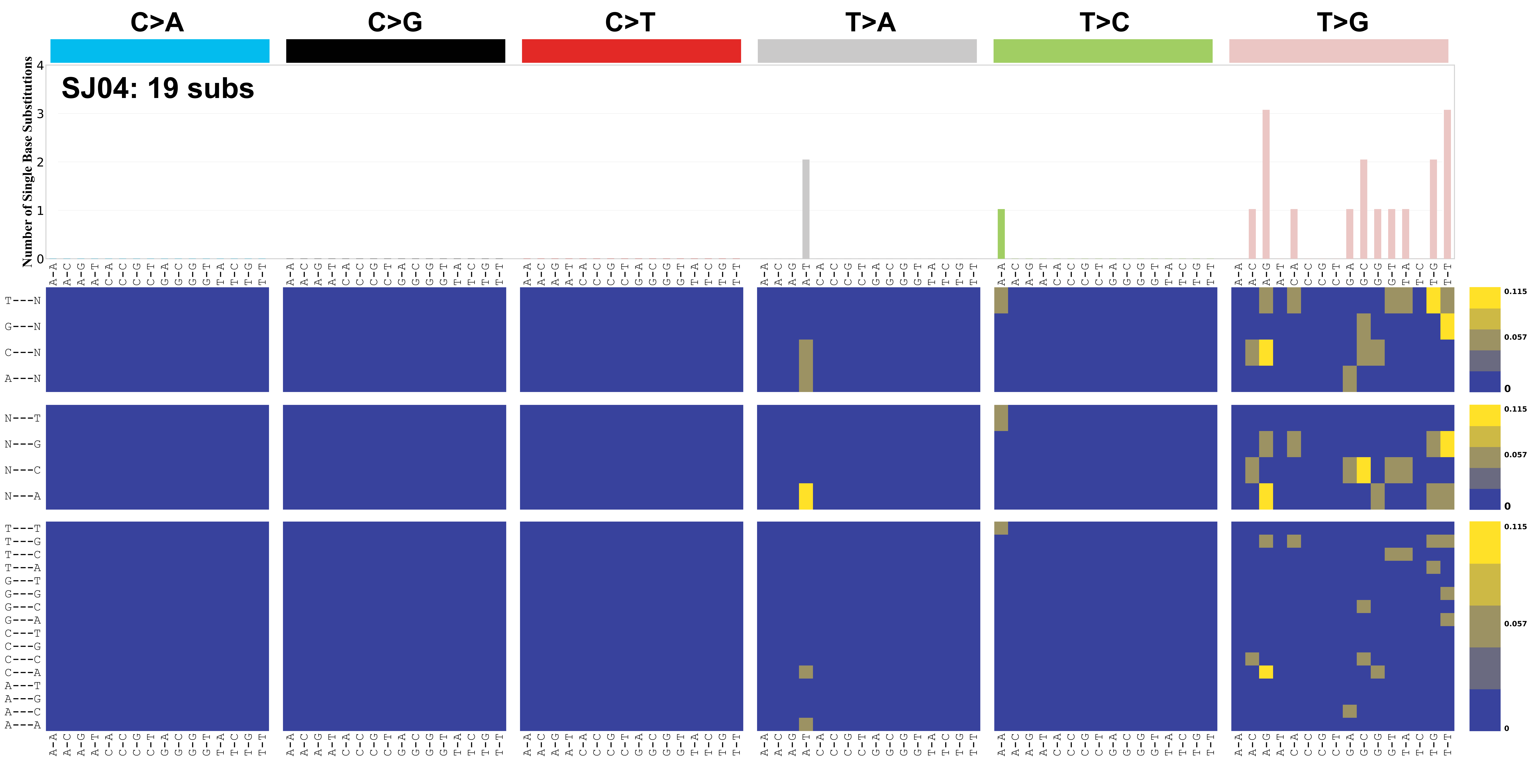
T>G

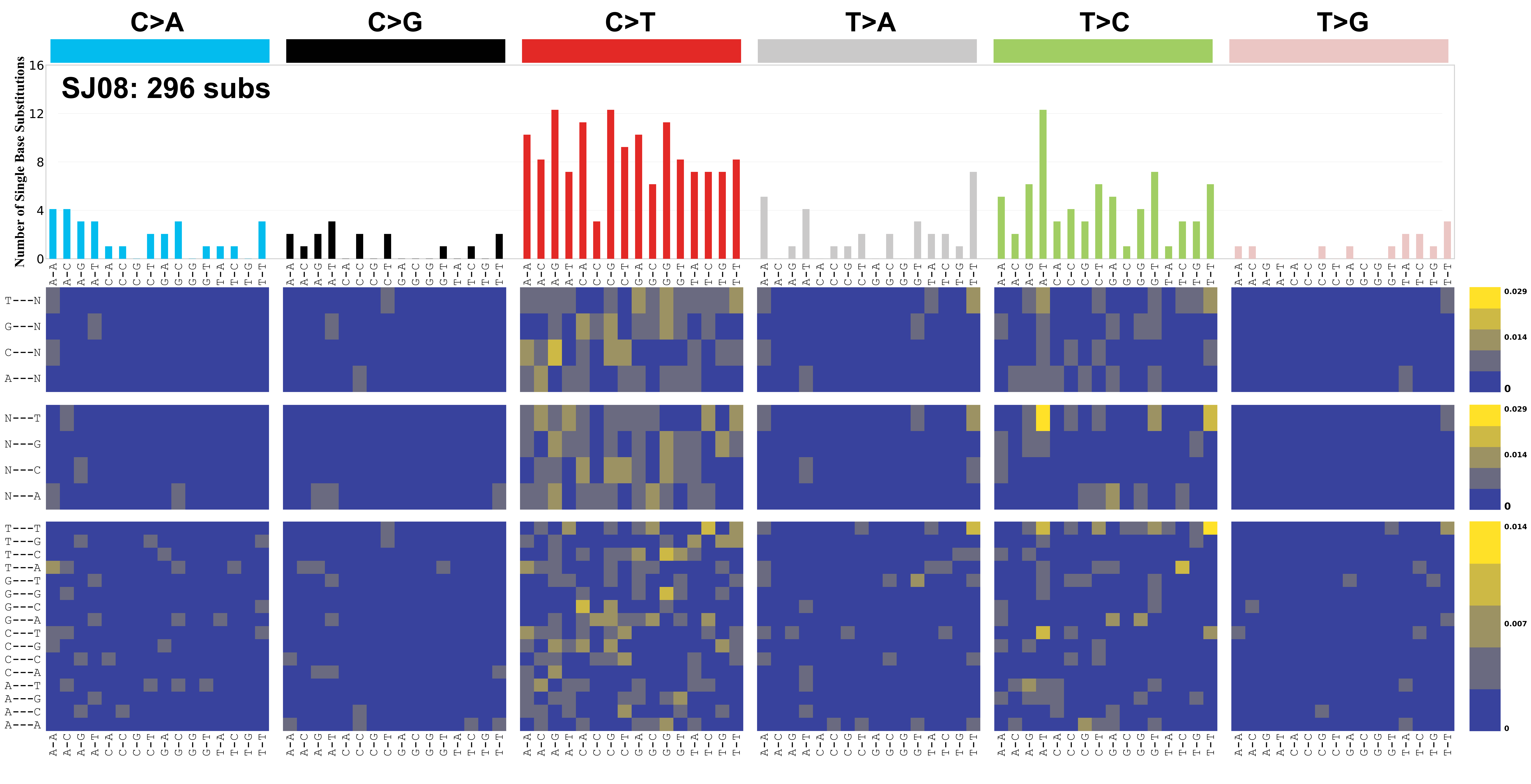
Number of Single Base Substitutions



SJ02: 304 subs





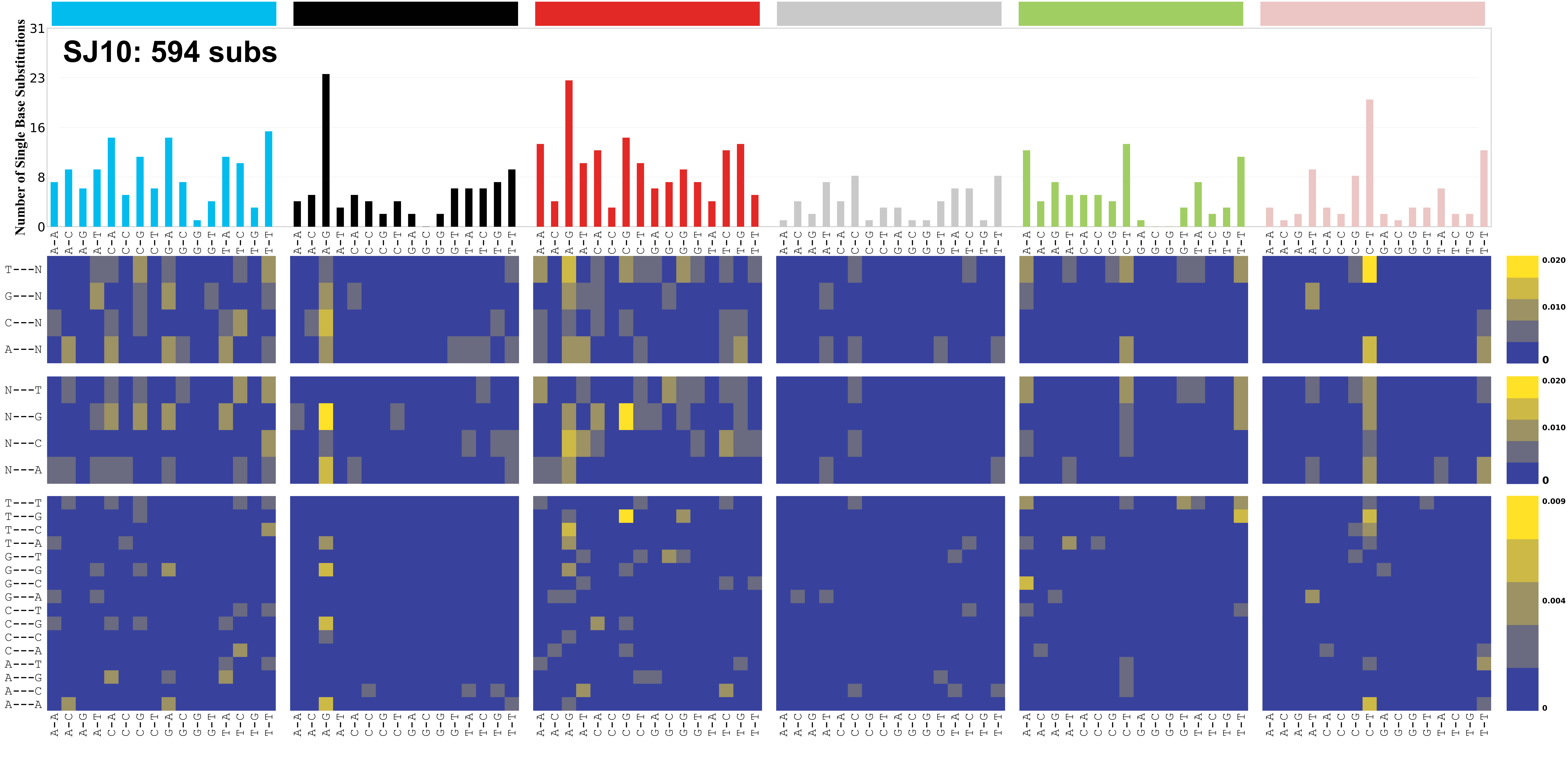


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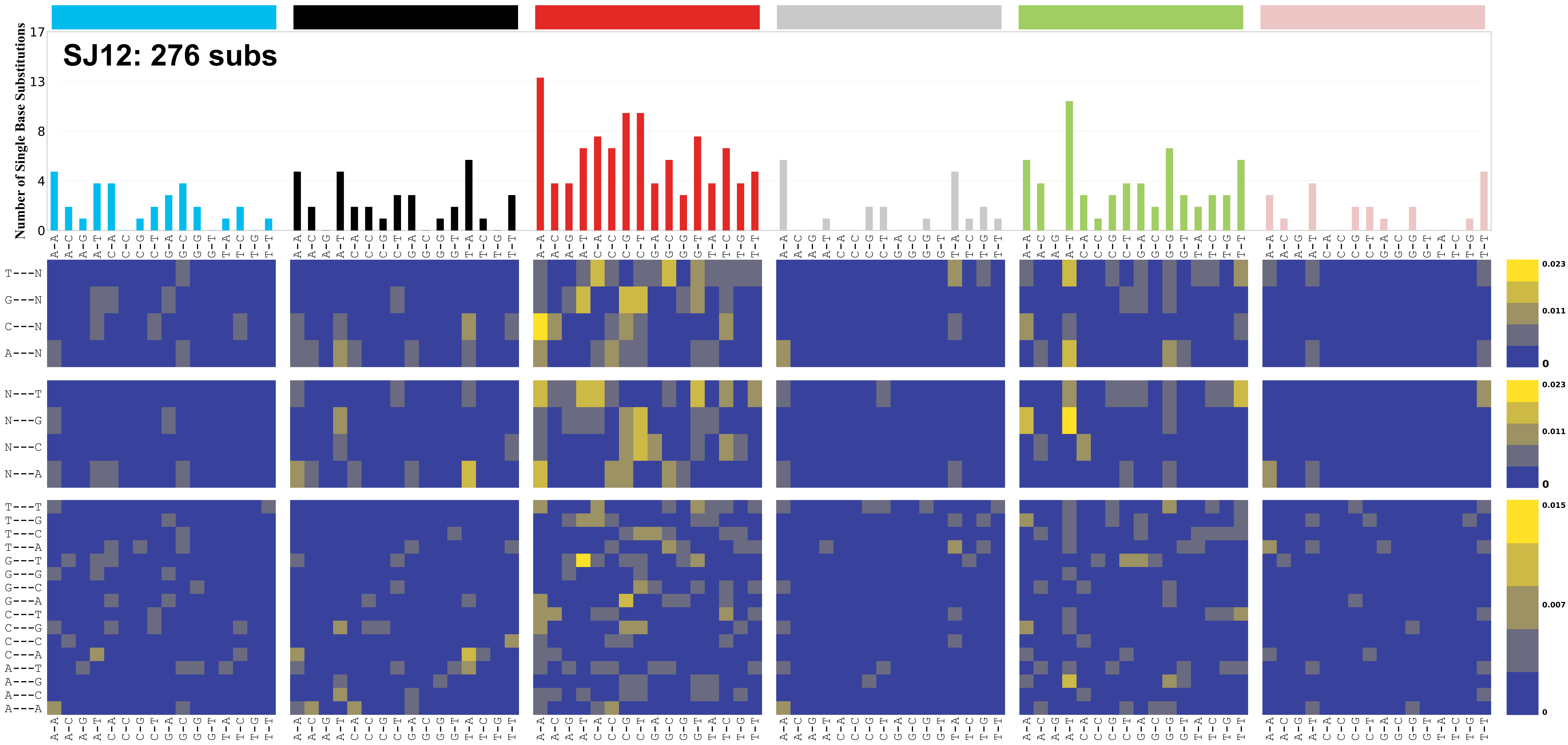
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Journal compilation © 2006 Blackwell Publishing Ltd

T--
G--
C--
A--

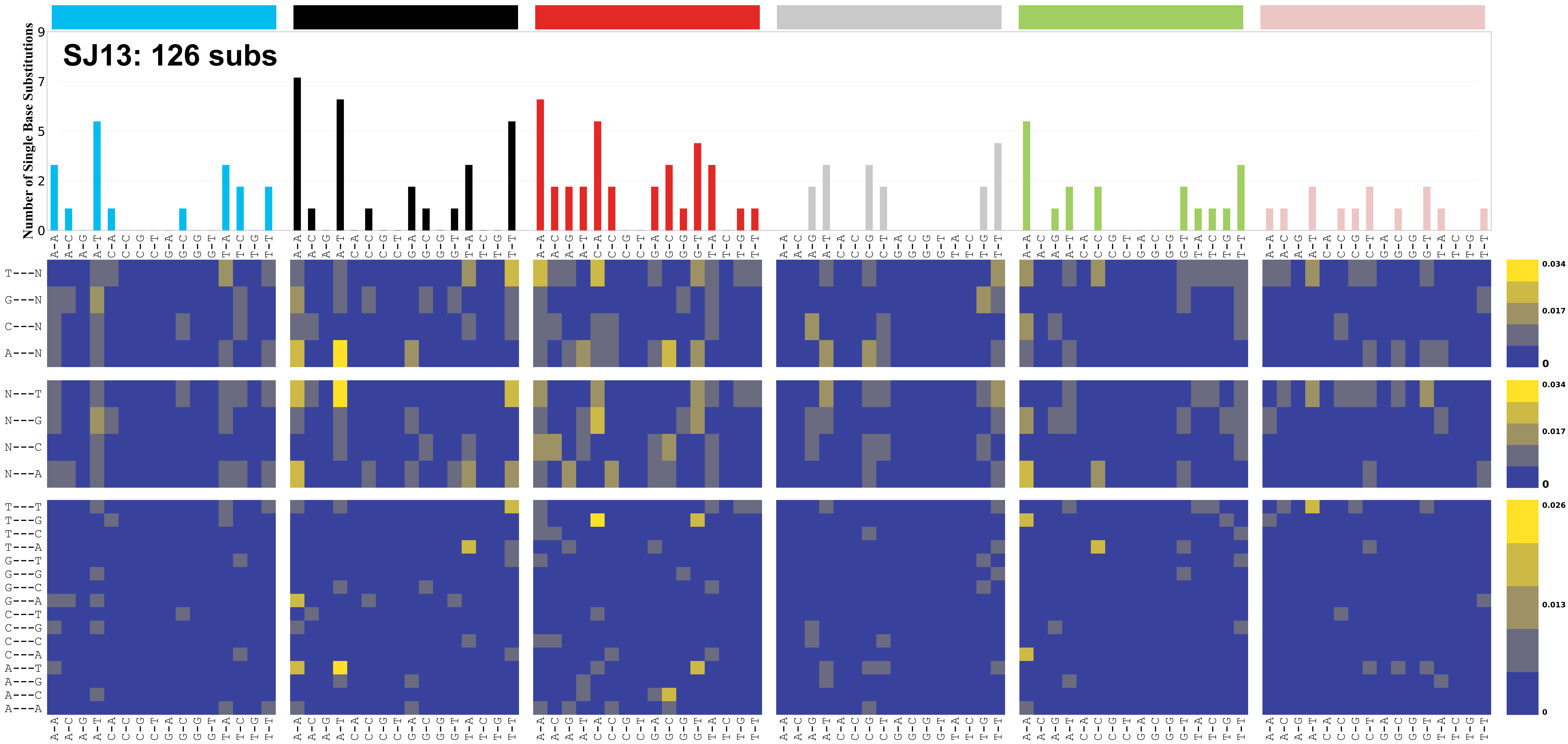
N--
N--
N--
N--



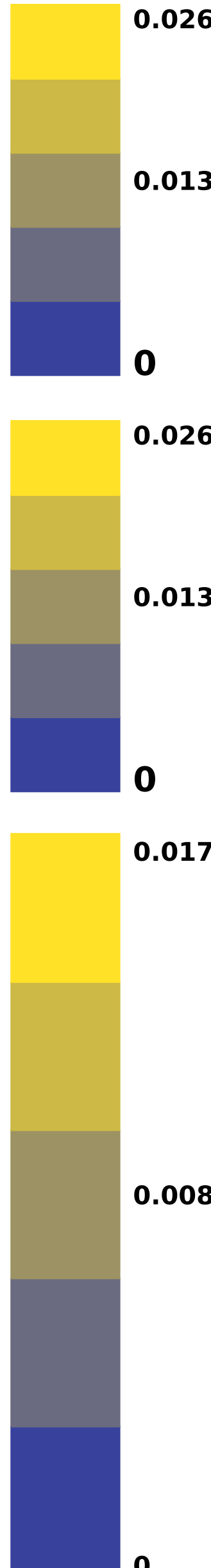
T>G

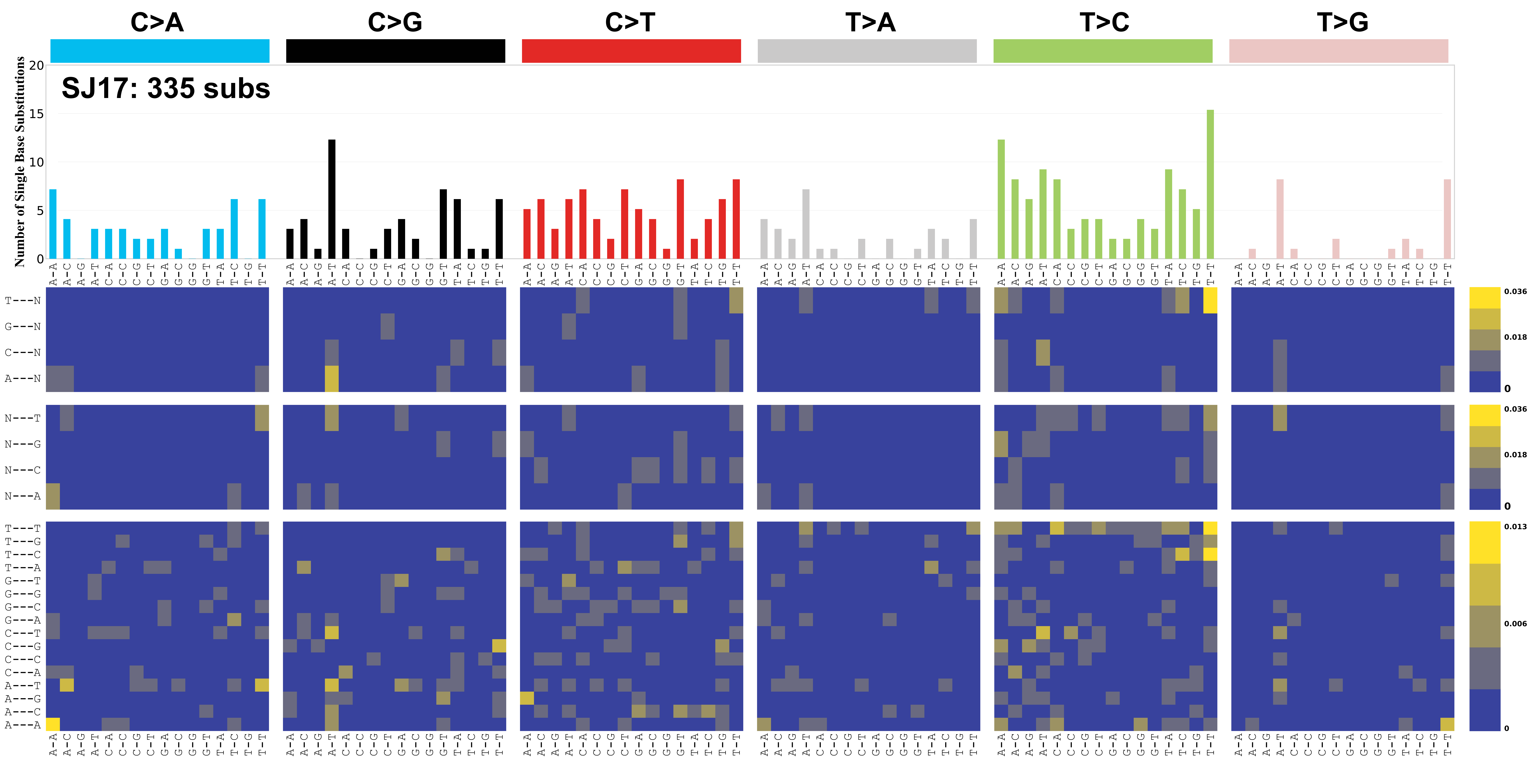


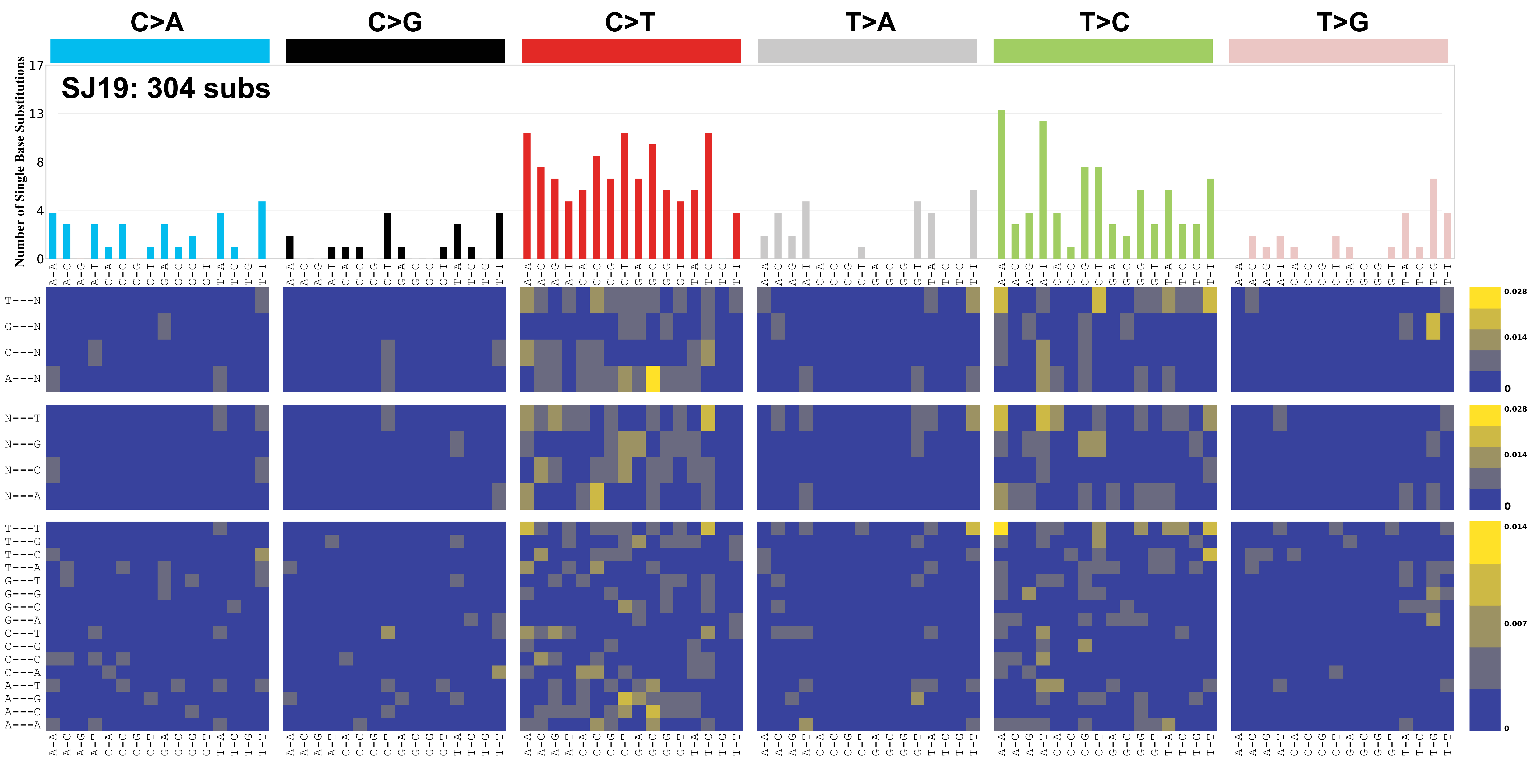
T>G



T>G







C > A

C>G

C>T

T > A

T>C

T>G

SJ20: 1,600 subs

Number of Single Base Substitutions

T---N
G---N
C---N
A---N

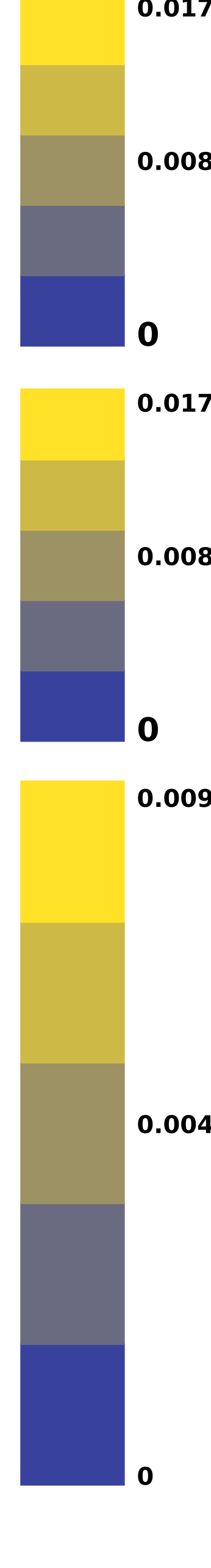
N---T
N---G
N---C
N---A

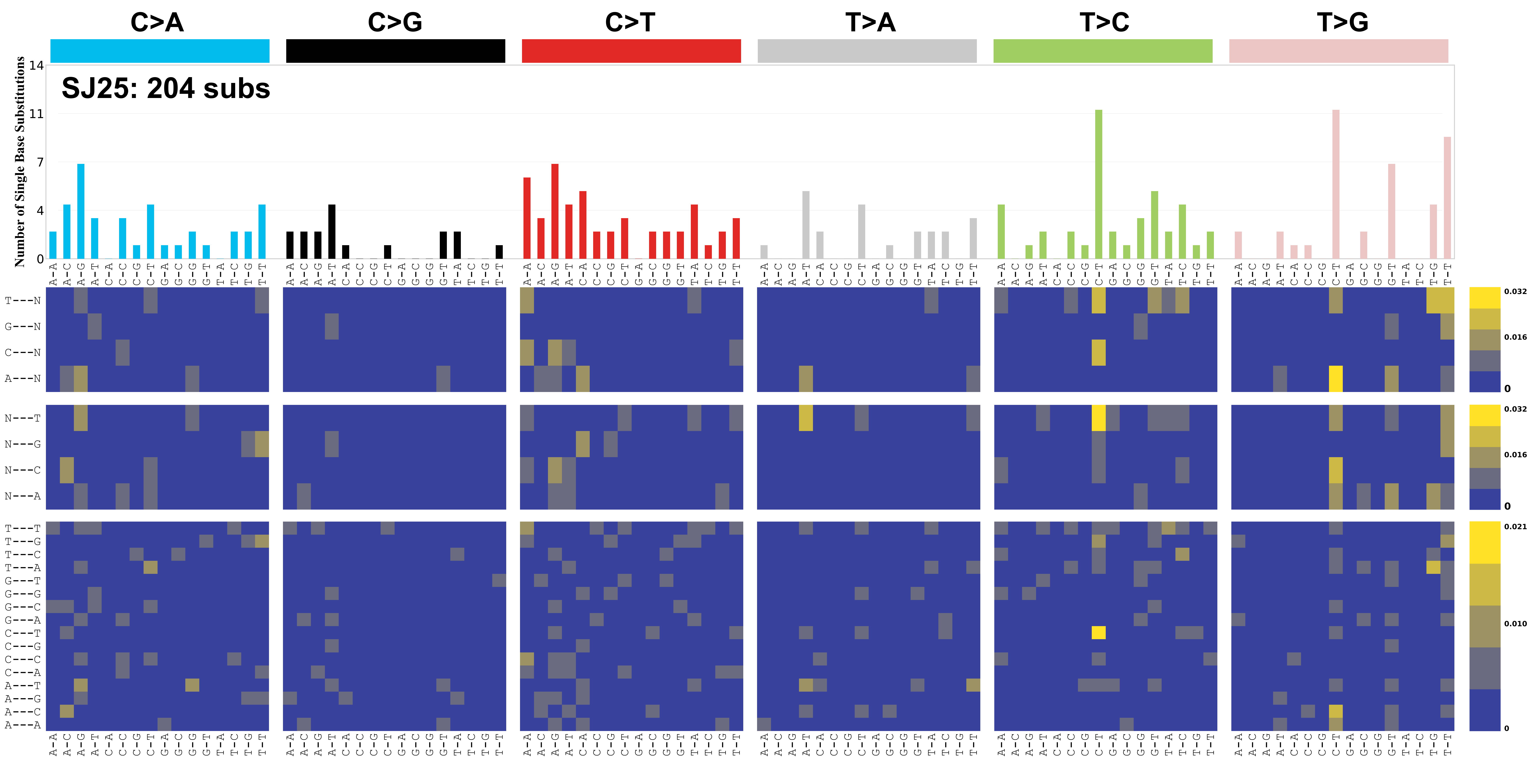
T---T
T---G
T---C
T---A
G---T
G---G
G---C
G---A
C---T
C---G
C---C
C---A
A---T
A---G
A---C
A---A

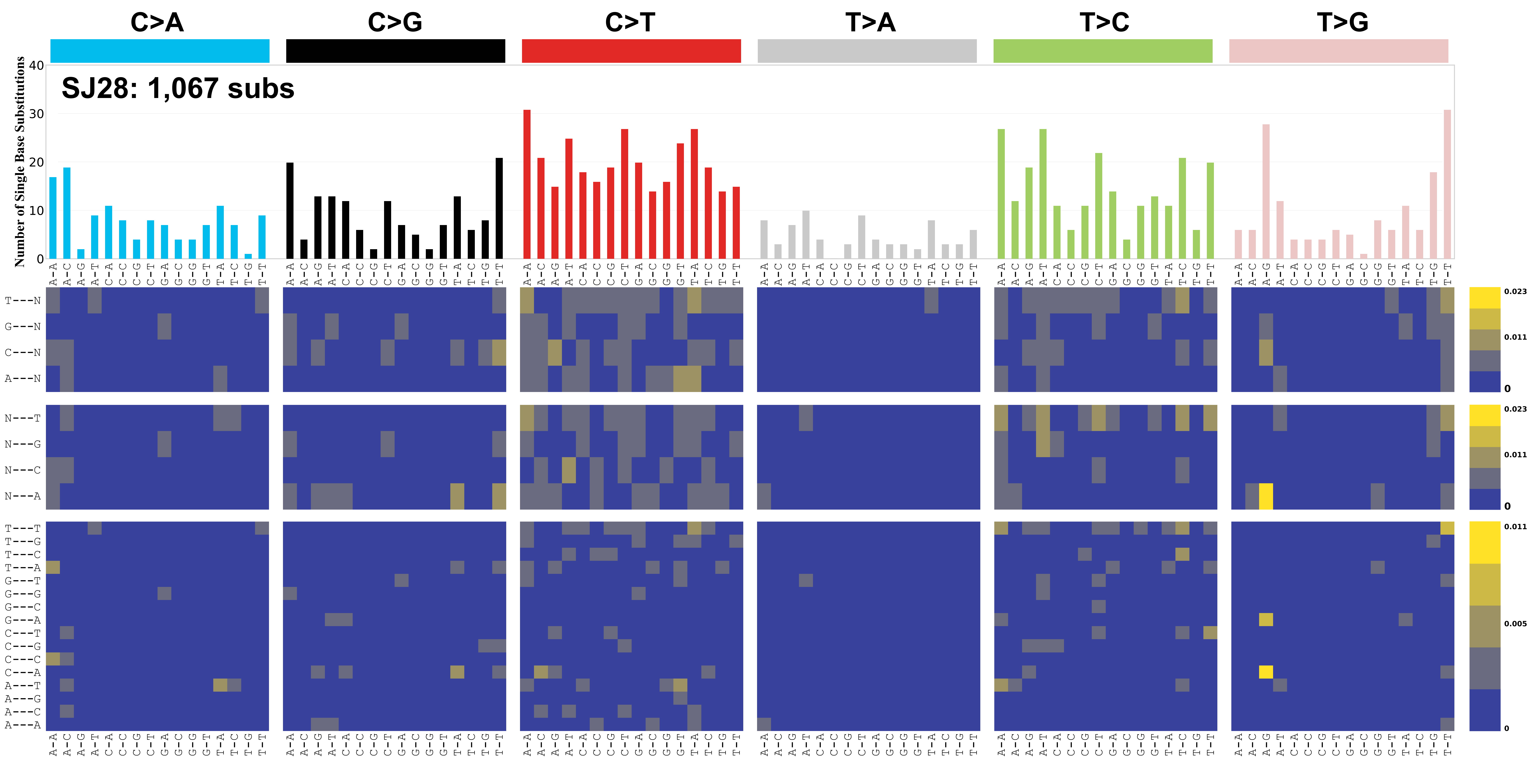
Figure 1 consists of three stacked bar charts, labeled (a), (b), and (c), each representing the distribution of 1000 simulated sequences across 16 categories: A-A, A-C, A-G, A-T, C-A, C-C, C-G, C-T, G-A, G-C, G-G, G-T, T-A, T-C, T-G, and T-T. The y-axis for all charts ranges from 0 to 1000. The x-axis labels are repeated for each chart. The bars are colored in shades of blue and grey. In all three models, the 'A-A' category is the most frequent, followed by 'A-T' and 'C-G'. The 3D CNN (c) shows a significantly higher frequency for the 'T-T' category compared to the other two models.

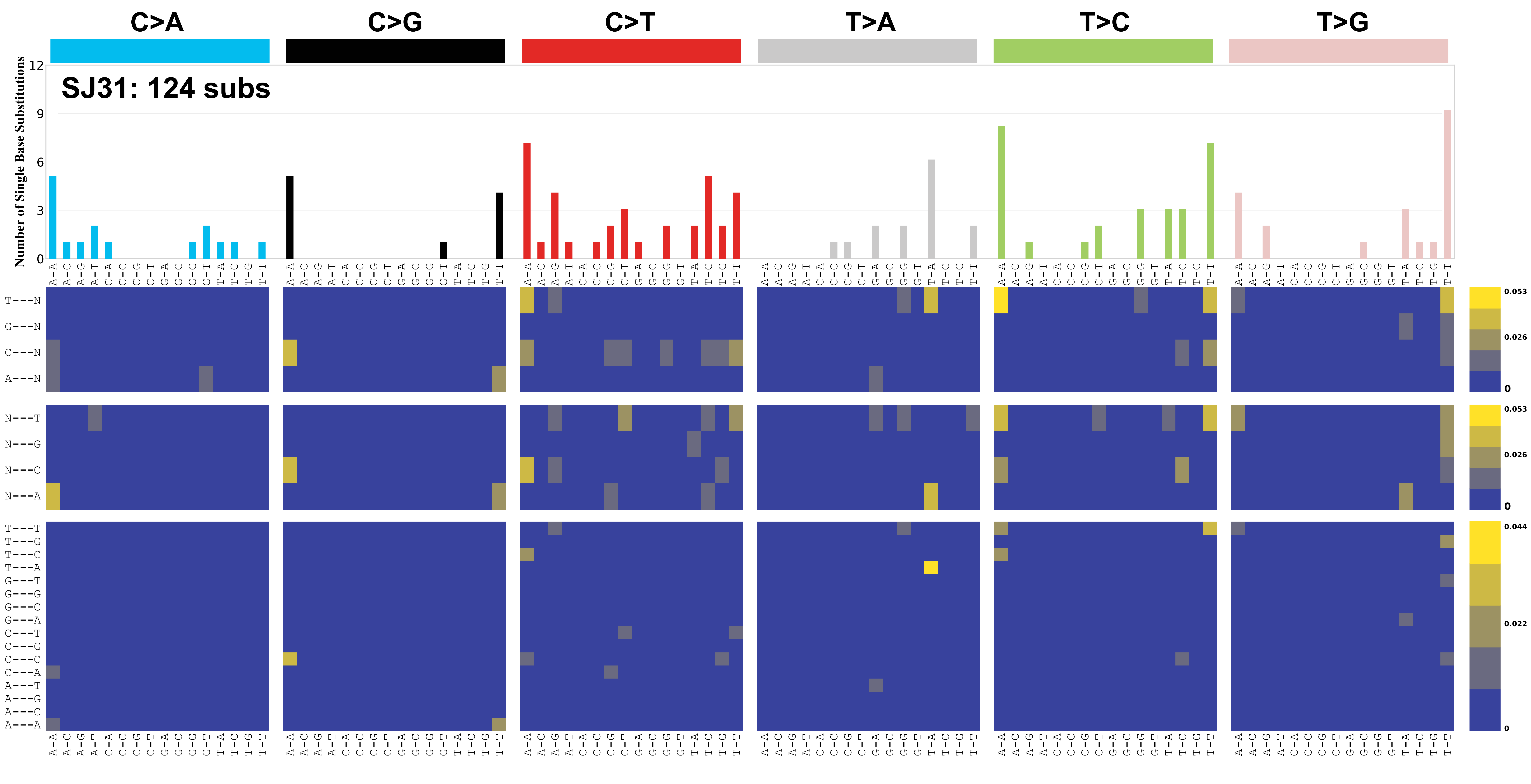
Figure 1 displays three heatmaps showing the distribution of 16 amino acid pairs (A-A, A-C, A-G, A-T, C-A, C-C, C-G, C-T, G-A, G-C, G-G, G-T, T-A, T-C, T-G, T-T) across three categories: I (top), II (middle), and III (bottom). The color scale ranges from 0 (dark blue) to 1 (yellow).

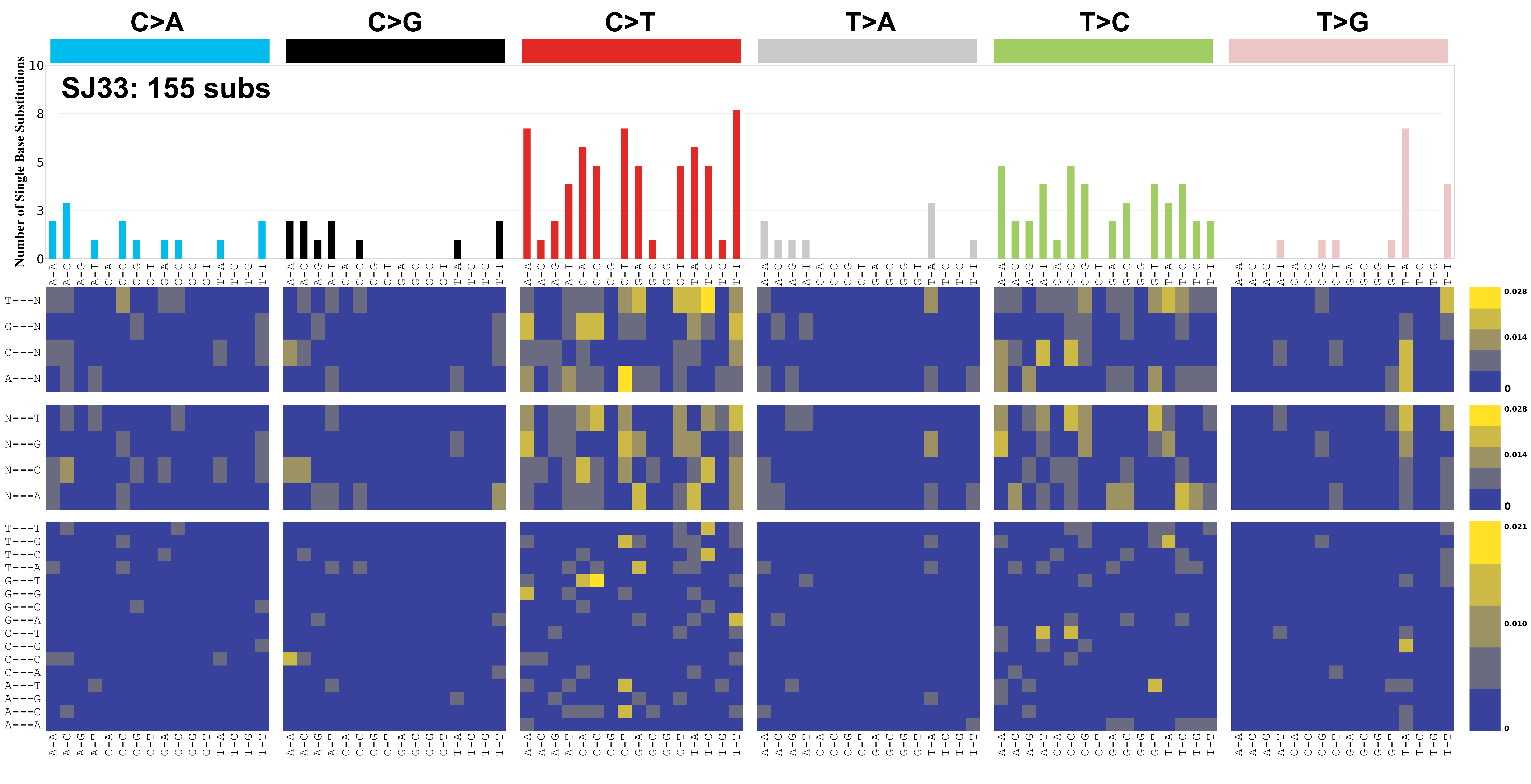
The heatmaps show that the distribution of amino acid pairs is highly specific to the category. For example, the A-T pair is highly concentrated in category II, while the G-T pair is highly concentrated in category III. The A-A pair is highly concentrated in category I.











T>G

T135: 4,916 subs

